

Trinity College Dublin
Coláiste na Tríonóide, Baile Átha Cliath
The University of Dublin

Homotopical and effective methods for associative algebras

Pedro Tamaroff

School of Mathematics, Trinity College Dublin

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This dissertation is submitted for the degree of *Doctor of Philosophy*. It was supervised by *Vladimir Dotsenko* (Université de Strasbourg) and *Sergey Mozgovoy*, and the two appointed external examiners were *Bernhard Keller* (Université Paris Diderot) and *Steffen Koenig* (Universität Stuttgart).

Declaration and coauthorship

I declare that this thesis has not been submitted as an exercise for a degree at this or any other university and it is entirely my own work. I agree to deposit this thesis in the open access institutional repository of the University, or allow the library to do so on my behalf, subject to Irish Copyright Legislation and Trinity College Library conditions of use and acknowledgement. I consent to the examiners retaining a copy of the thesis beyond the examining period, should they so wish.

This thesis contains, among others, results of two coauthored research papers, one with Vladimir Dotsenko and Vincent Gélinas [51], and one with Vladimir Dotsenko [55].

Pedro Tamaroff
January 2021

For Olga, my loving parents and sisters, and my dear brother.

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Foreword

The work in this thesis has two general goals. First, it intends to further the use of methods coming from algebraic homotopy and homotopical algebra in the study of higher structures appearing in the context of algebraic operads and their resolutions —such as the celebrated A_∞-algebra structures of J. Stasheff— and showcase their usefulness when studying non-linear categories of objects or other non-linear phenomena, such as the deformation theory of associative algebras. It also intends to strengthen the idea that certain domains of algebra live considerably close to the domain of term rewriting theory, by arguing that the rather alluring areas of homotopical algebra concerned with deformation theory, Maurer–Cartan moduli, and structures 'up to homotopy', are indelibly linked to the hands-on and effective methods of term rewriting theory and the elegant insight of combinatorics: even though homotopical algebra is very technical and computationally heavy, such computations can be approached, almost universally, through the judicious use of interesting methods coming from combinatorics, and through the well-developed theory of term rewriting. The latter brings new insight to an already rich field, makes proofs effective —so that, for example, homological invariants are computable not just in principle—, and allows to use the elegant machinery of combinatorics to study algebraic problems. Pursuing these kinds of projects aims to bring together three rich fields of algebra: homotopical algebra, term rewriting and algebraic combinatorics. These three fields are in constant development and the potential to produce a positive feedback loop between the three cannot be missed. Naturally, this is not a novel statement, and each of [31, 56, 63, 74, 128, 146, 148] could serve as a good example: we just intend to stress the statement here.

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Several mathematicians have in one way or another influenced the contents of the articles this thesis is based on, either by their direct comments and suggestions or through valuable conversations: for this I sincerely thank Joan Bellier-Millès, Claude Cibils, Alexander Efimov, Eric Hoffbeck, Bernhard Keller, Andrey Lazarev, Markus Linckelmann, Muriel Livernet, María Julia Redondo, Daniel Robert-Nicoud, Luis Scoccola, Andrea Solotar, Jim Stasheff, Mariano Suárez-Álvarez, Guillermo Tochi, Christine Vespa, Sarah Witherspoon, and Guodong Zhou.

I sincerely thank Joe Chuang and Alistair King who, in an unpublished manuscript, independently obtained a formula for the minimal model of a monomial algebra around 2005, and shared it with me shortly before a first draft of [171] was circulated. They also shared with me their unpublished manuscript from fifteen years ago that highlights the importance of previous work of Gruenberg [79] for computing higher structures on Ext algebras of monomial algebras; the beautiful example of Section 3.3.3 is also coming from their unpublished note, which was reproduced with their permission.

Finally, I thank Bernhard Keller and Steffen Koenig kindly for accepting the task of examining this thesis, and thank Bernhard Keller once more for his very careful and precise feedback on the manuscript [172].

Introduction

This thesis is based on four papers [51, 55, 171, 172]. The first chapter gives individual summaries of each of these four chapters, while the second one gives useful recollections and conventions we will use. In particular, it contains recollections on term rewriting theory, which is essential for chapter six, on the homological algebra of associative algebras and, particularly, the machinery of algebraic discrete Morse theory, which is indispensable for chapter three. This chapter also contains a short reminder on the elements of algebraic operads, which we use in chapter four. The remaining four chapters present the results of the papers we just mentioned.

In the third chapter, which is based on [171], we solve the problem of computing the minimal model of an arbitrary associative monomial algebra. We accomplish this by computing what is known as a 'canonical' A_∞-coalgebra structure on its Yoneda coalgebra: it is well known that such a structure is tantamount to producing a quasi-free model of the algebra [150]. Our methods are combinatorial and depend on a detailed analysis of a homotopy retract datum on the bar resolution of the trivial module obtained from the algebraic discrete Morse theory of Jöllenbeck–Welker [102]. In it, we also explain how the bimodule resolution of Bardzell [12] can be obtained as a by-product of our resolution through a twisting cochain argument, and give some interesting examples of vanishing and non-vanishing patterns for higher multiplications of homotopy associative algebras.

The importance of using such higher structures to study associative algebras from a homotopical point of view and in particular obtain models of them —as opposed to using the dual structures, known as an A_∞-algebras— was highlighted by the work of B. Keller. The third, fourth and fifth chapters in this thesis are based on papers that have been strongly influenced by this point of view.

The fourth chapter, based on [172], centers on a homotopically invariant effective computation of the Tamarkin–Tsygan calculus of an associative algebra, which was shown to be a derived invariant —and hence homotopy invariant— after a series of papers of B. Keller and two joint papers of M. Armenta and himself. This result motivates the problem to compute this derived invariant; since part of this computation involves computing the Hochschild cohomology groups of an associative algebra along with its Gerstenhaber bracket, which are known to be very hard to compute in practice, effective and streamlined homotopical methods of computations are desirable. To begin moving towards such a result, we construct out of each cofibrant resolution of an associative algebra a pair of complexes and show they are ∞-quasi-isomorphic to the standard pair of Hochschild complexes defining the calculus structure on this algebra, and give computational examples at the

end of the chapter. At the same time, we prove that the colored operad defining Tamarkin–Tsygan calculi is inhomogenous Koszul, which is essential for our proof to go through.

The fifth chapter is based on joint work with Vladimir Dotsenko and Vincent Gélinas [51], and we feel like it is useful as a computationally explicit work where we follow the philosophy that canonical higher structures are a pivotal invariant that allows to obtain information on other homological invariants. In it, we construct, through the use of Anick chains and perfect paths in a Gorenstein monomial algebra Λ, a monogenic polynomial subalgebra $\Bbbk[\chi]$ of the A_∞-centre of the Yoneda algebra $\mathrm{Ext}_\Lambda(\Bbbk,\Bbbk)$ exhibiting this higher centre and the Yoneda algebra itself as module finite over $\Bbbk[\chi]$, ultimately showing that Λ satisfies the **FG** conditions of Snashall–Solberg through the methods of [32]. The ideas developed in this last paper follow, once more, deep insight of B. Keller in communications during 2003 regarding the image in the Yoneda algebra of the characteristic morphism coming from Hochschild cohomology, and its description through higher structures.

The sixth and last chapter, based on joint work with Vladimir Dotsenko [55], presents a new way to interpret and prove the celebrated Diamond Lemma of Bergman from the viewpoint of homotopical algebra, while at the same time providing a general framework to state and prove the respective result for other algebraic structures. Our main result states that every multiplicative free resolution of an algebra with monomial relations gives rise to its own Diamond Lemma, so that Bergman's condition of "resolvable ambiguities" becomes the first non-trivial component of the Maurer–Cartan equation in the corresponding tangent complex. We also show how our methods immediately go through to recover the Diamond Lemma for non-symmetric and shuffle operads [31] and how it can be extended for other 'multiplicative algebraic structures', our main example being colored operads coming from linearisations of set theoretical ones.

It is fair to say that only the main results of [51] depend crucially on the explicit computation of the minimal model of a monomial algebra: the two remaining papers use it for computations and to obtain interesting connections to rewriting theory, but the main results there are independent of the results obtained in [171].

Chapter 1

Summary

1 Models for monomial algebras

The problem solved in the first chapter is readily motivated by the need to understand the homotopy category of associative algebras and, more generally, the problem of developing effective and useful computational methods for the homotopical algebra of such objects. It is known quasi-free models determine the homotopy type of an algebra, and may be used to compute its Hochschild cohomology —the cohomology theory controlling the deformation theory of the algebra— and its non-commutative Cartan calculus, due to Tamarkin–Tsygan [169, 170]; also see Subsection 4.2.1. A quasi-free model provides a representative of the homotopy type of the algebra in the homotopy category, and hence can be used, *a priori*, to compute any homotopy invariant of the algebra.

Our specification of the problem above to the class of monomial algebras follows the philosophy that monomial algebras provide good approximations, in the light of term rewriting theory and homological perturbation theory, for general associative algebras. This philosophy has been already pursued in the literature [3, 12, 18, 73], and made concrete in the article [74]: associative algebras with a fixed set of leading terms can be made into a variety, and the monomial algebra defined by these leading terms controls many generic properties of the points in this variety, such as: dimension, the Cartan matrix, the global dimension, the N-Koszul property, the Hilbert series and the property of being quasi-hereditary.

Main result. Our solution is presented in the language of higher structures, and we couple it with the streamlined combinatorial approach to the computation of homological invariants through chains, also known as overlappings, ambiguities, and overlapping ambiguities. As we just remarked, it is well known that a quasi-free model of an algebra is generated, as a free algebra, by the homology groups $\mathrm{Tor}_A(\Bbbk,\Bbbk)$ underlying the Yoneda coalgebra of A, and these homology groups admit a combinatorial basis determined by chains: these are simply monomials that can be, quite literally, obtained by gluing defining relations of A into a chain, with suitable technical conditions

that remove superfluous information. The novel concept introduced in our first paper is the notion of a decomposition of a chain, which suggests itself immediately during computations, and allows for the description of the differential of the model.

A deconcatenation [171, pp. 1-2] of a chain γ (which is simply a monomial in the defining variables of A) is a factorization of this monomial into a product $\gamma_1 \cdots \gamma_n$ where each of the factors is itself a chain. The only technical requirement is the following: the chain γ carries a homological degree $|\gamma|$, being an element of the homology group $\mathrm{Tor}_A(\Bbbk,\Bbbk)$, and we require that $|\gamma| = |\gamma_1| + \cdots + |\gamma_n| + 1$. One of the striking features of this requirement, which seems rather unconstrained at first, is that it is the only condition required for a term $\gamma_1 \cdots \gamma_n$ to appear in the differential $d\gamma$, as the main result of [171] guarantees.

Theorem (Theorem 3.2.6)**.** *For each monomial algebra A there is a minimal model $(TV,d) \longrightarrow A$ where V is the desuspension of $\mathrm{Tor}_A(\Bbbk,\Bbbk)$. The differential d is such that for a chain $\gamma \in \mathrm{Tor}_A(\Bbbk,\Bbbk)$,*

$$d\gamma = -\sum_{n\geqslant 2} (-1)^{\binom{n+1}{2}+|\gamma_1|} \gamma_1 \cdots \gamma_n,$$

where the sum ranges through all possible decompositions of γ. □

This result is attractive for various reasons: first, it gives a completely combinatorial answer to a purely homological question and, second, the form of it is presented in —in terms of factorizations of words in a free monoid— allows for the implementation of this result in a computer; this has already been done in case of the "Anick resolution" (which this theorem obtains as a by product) in the computer software `bergman` [10]. Third, it recovers some of the results in [72] describing cup products in $\mathrm{Ext}_A(\Bbbk,\Bbbk)$ for a monomial quiver algebra A using a multiplicative basis of chains, and some of the results in [86] describing the A_∞-algebra structure of Ext_A for monomial algebras which are p-Koszul. Finally, we have been able to use it in three subsequent projects to compute derived invariants of algebras [172], study the support variety theory of monomial algebras [51] and to provide a concrete link between effective homotopical methods and the Diamond Lemma of rewriting theory [55], which are the subjects of the three other chapters.

Further directions. At the end of the chapter, we suggest a conjectural algorithm to proceed and compute a possibly non-minimal model of an algebra with a chosen Gröbner basis. The complexity of the model depends directly on the associated monomial algebra of the chosen basis, and thus the basis should be chosen judiciously: the Anick chains for the chosen basis produce the requisite generators. We hope that the main result of this chapter, coupled with this conjectural algorithm, may be paired with homotopical methods —as detailed at the end of the introduction for the next chapter— to allow for the constructions of small models of associative algebras, a problem which, in the case of commutative algebras arising as cohomology algebras of topological spaces, among others, has received its due attention.

2 Non-commutative calculus

To every associative algebra we may associate its Hochschild homology and cohomology groups. As originally observed in [48], for every associative algebra A the pair

$$\mathsf{Calc}_A = (\mathrm{HH}^*(A), \mathrm{HH}_*(A)),$$

supports operations making it into a non-commutative analog of the Cartan calculus on a smooth manifold, which is called the *Tamarkin–Tsygan calculus of* A. In this chapter we focus our attention on giving formulas for these operations in terms of cofibrant resolutions in Alg, the category of dga algebras with the projective model structure or, what is the same, a *homotopy invariant* description of the action on the chain level. Initially, we focused our attention on the Gerstenhaber bracket on Hochschild cohomology, originally defined in [65], since computing the resulting Gerstenhaber algebra structure on Hochschild cohomology has been of interest [78, 142, 143, 167, 178], and is agreed to be a non-trivial task; the reason for this is nicely explained in [167].
D. Quillen [152, Part II, §3] identified the Hochschild cochain complex of an algebra with the space of coderivations of its bar construction. Later, in the same spirit, J. Stasheff [166] gave a definition of the Gerstenhaber bracket of an associative algebra as the Lie bracket of coderivations of its bar construction, which deserves to be thought of as intrinsic to the category of associative algebras. Interest for a description of the Tamarkin–Tsygan calculus of an associative algebra à la Stasheff appears in [183, Remark 7]. Our main result here is giving such an intrinsic description of the operations of defining the calculus that also lends itself to computations.

Main result. To state our result, consider a quasi-free associative algebra $B = (TV, d)$, and the following complexes of nc poly vector fields and of nc differential forms on B, respectively,

$$\mathscr{X}^*(B) = \mathrm{cone}(\mathrm{Ad} : B \longrightarrow \mathrm{Der}(B)), \quad \Theta_*(B) = \mathrm{cone}(C : V \otimes B \longrightarrow B).$$

where C is the commutator map $v \otimes b \longrightarrow [v,b]$ arising in cyclic homology. The space $\Theta_*(B)$ is spanned by nc-forms $\omega = b + b'dv$ where $b, b' \in B$ and $v \in V$, the domain of C being the nc space Ω^1_B of forms on B, while $\mathscr{X}^*(B)$ is spanned by nc-fields $X = \lambda + f$ where $\lambda \in B$ and f is a derivation.

Theorem (Theorem 4.2.7)**.** *The Tamarkin–Tsygan calculus* Calc_A *of an associative algebra A can be computed using the datum*

$$(\mathscr{X}^*(B), \Theta_*(B))$$

of nc poly vector fields and nc differential forms obtained from a model $B \longrightarrow A$*. In terms these two complexes, if we write a poly vector field by* $X = \lambda + f$ *and a differential form by* $\omega = b + b'dv$*, then*

(1) The cup product can be computed through a brace operation $\{X, Y; d\}$.

(2) The Lie bracket is the usual bracket of derivations
(3) The boundary of ω is computed as

$$d\omega = \sum_{i=1}^{n} (-1)^{\varepsilon} v_{i+1} \cdots v_n v_1 \cdots v_{i-1} dv_i.$$

To achieve this, we prove the seemingly new result that the colored operad Calc is inhomogeneous Koszul. The theory developed in [63] to obtain resolutions of inhomogeneous Koszul operads, generalizing the work of L. Positselski [147], then allows us to produce a cofibrant model Calc_∞ for Calc and endow the pair

$$(\mathscr{X}^*(B), \Theta_*(B))$$

with a Calc_∞-algebra structure.

Theorem (Theorem 4.1.4)**.** *The operad* Calc *is inhomogeneous Koszul. It admits a cofibrant model* Calc_∞ *of* Calc *with underlying symmetric sequence isomorphic to* $T(\delta) \circ \mathsf{PreCalc}^{\text{¡}}$.

Here $\mathsf{PreCalc}^{\text{¡}}$ is the Koszul dual cooperad of the Koszul operad PreCalc and $T(\delta)$ is a polynomial algebra with generator δ of degree two. Having this model for homotopy coherent calculi, we gather the following results from [49, 111]:

Theorem. *There is a dg colored operad* KS*, the Kontsevich–Soibelman operad, and a topological colored operad* Cyl *such that*

(1) The operad $C_*(\mathsf{Cyl})$ is formal and its homology is Calc*.*
(2) There is a quasi-isomorphism $\mathsf{KS} \longrightarrow C_*(\mathsf{Cyl})$ of dg-operads.
(3) KS *acts on the pair $(C^*(-), C_*(-))$ in such a way that*
(4) On homology we obtain the usual Calc*-algebra structure on $(\mathrm{HH}^*(-), \mathrm{HH}_*(-))$.*

In particular, for every cofibrant replacement Q of Calc*, the pair $(C^*(A), C_*(A))$ is a Q-algebra which on homology gives Calc_A.*

Specializing this to $Q = \mathsf{Calc}_\infty$, we obtain on the pair $(C^*(A), C_*(A))$ a Calc_∞-algebra which on homology gives Calc_A and which we write $\mathsf{Calc}_{\infty,A}$. With this at hand, the homotopy theory developed in [63] for inhomogeneous Koszul operads implies the following theorem.

Theorem (Theorem 4.2.6)**.** *For any quasi-free model B of A, the pair $(\mathscr{X}^*(B), \Theta_*(B))$ admits a Calc_∞-algebra structure which is ∞-quasi-isomorphic to the Calc_∞-algebra $\mathsf{Calc}_{\infty,A}$ so that the action of degree zero elements in Calc_∞ is given by the formulas above. In particular, this structure recovers Calc_A by taking homology.*

Towards a systematic and homotopical approach to computation of invariants. We remark that computing cofibrant resolutions in Alg is remarkably complicated. As explained in Chapter 3, we have solved the problem of computing models of *monomial* algebras, and using this and some

ideas coming from perturbation theory, it is possible to attempt to compute models of certain algebras with a chosen Gröbner basis, although the general description of the minimal model of such algebras is, at the moment, missing. At the same time —as the computations in Chapter 4 show— even in the case of algebras with few generators, the computation is highly non-trivial. Because of this, it is desirable to further develop a way to compute such invariants of algebras in the same way one computes homotopical invariants of topological spaces by working 'one cell at a time'. Such themes have already successfully explored in the literature, famously and systematically by D. Sullivan and H.-J. Baues, and in [42–45, 104] ideas in homological and homotopical algebra are used to approach the problem of computing Hochschild homology of an algebra, but to the best of our knowledge the full extent to which effective homotopical methods can be devised to compute such invariants has not been explored yet.

3 A_∞-algebras and support variety theory

The work in this chapter is intended as a contribution to the study of support varieties for finite dimensional algebras. Our main tool of choice for doing this is the use of 'canonical' A_∞-algebra structures on the Yoneda algebra of an associative algebra Λ. More precisely, the key ingredients for the main results in that paper are the contents of the first chapter, and the work of B. Briggs and V. Gélinas [32] on the notion of the A_∞-center of a minimal A_∞-algebra, following ideas of Buchweitz, Green, Snashall and Solberg in the case of Koszul algebras [36].
Support varieties were introduced in modular representation theory [38], relying on the finite generation theorems of Evens [61], Golod [68], and Venkov [177] for group cohomology. The existence of a good theory of support varieties for associative algebras —developed by Snashall and Solberg [163]— relies on the hypothesis that certain finite generation (**FG**) conditions hold for Hochschild cohomology; these guarantee that some semblance of a geometrical setting exists, and provide for us, in broad terms, with a geometric approach to study the representation theory of associative algebras. For example (see [58]), a module has trivial associated variety if and only if it has finite projective dimension, and it has a line as associated variety if and only if it is periodic for the syzygy operator Ω. Determining which finite dimensional algebras Λ satisfy **FG** is an important open problem of representation theory, and a foundational theorem of Erdmann, Holloway, Snashall, Solberg, and Taillefer [59, Theorem 1.5] states that any algebra satisfying **FG** is necessarily Gorenstein. The main result of this chapter is establishing the converse in the monomial case:

Theorem (Theorem 5.4.3)**.** *A monomial algebra satisfies the conditions **FG** if and only if it is Gorenstein.* □

Method: higher structures on monomial algebras. In [141], Nagase proved that a Nakayama algebra satisfies the conditions **FG** if and only if it is Gorenstein. This is a particular case of our result, since Nakayama algebras are monomial algebras for special type of quivers (cycles or type A quivers). Our methods are however very different, and rely on the work [32] of B. Briggs

and V. Gélinas who proposed, for a finite dimensional algebra Λ, an approach towards studying the conditions **FG** in terms of the canonical A_∞-structure on the Yoneda algebra $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ of the module $\Bbbk = \Lambda/\operatorname{rad}\Lambda$. Their work relies on earlier ideas of Green, Snashall, and Solberg [76] to study the ring structure on Hochschild cohomology in terms of its image under the characteristic homomorphism

$$\varphi_\Bbbk : \operatorname{HH}^*(\Lambda,\Lambda) \to \operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk).$$

We regard this as a useful concrete illustration of how one can apply the canonical A_∞-algebra structure on the Yoneda algebra towards determining more classical homological invariants, such as Hochschild cohomology. This philosophy was advocated in [131, 132], but since higher structures of the Yoneda algebras are generally quite hard to compute, not many applications emerged so far. In light of the results of Chapter 3, the class of monomial algebras is a natural starting point for exploring the philosophy of [131, 132]. Moreover, as one can approximate general algebras by monomial ones by means of Gröbner bases [74], a thorough understanding of the monomial case should serve as the basis for a more general line of investigation.

Invariants through combinatorics of chains. One of the main reasons that various invariants of monomial algebras can be computed explicitly is the extra grading that can be utilised. All important vector space and homology groups associated to a monomial algebra $\Lambda = \Bbbk Q/I$ have a grading by the category $\mathfrak{C}(Q)$ freely generated by the quiver Q. This means that some generally structureless vector space acquire distinguished bases, and thus become naturally identified with their linear duals, leading to elegant formulas that are not available otherwise; in particular, the Tor groups $\operatorname{Tor}^\Lambda_*(\underline{\Bbbk},\underline{\Bbbk})$ have combinatorial bases of the so called Anick chains. This was already noted and used in a recent preprint [88]. For us, this observation leads to intricate vanishing patterns for higher structures of Yoneda algebras that can be regarded as analogues of the formulas of He and Lu for higher structures associated to N-Koszul algebras [86], which we explained in Chapter 3 for convenience. Those vanishing patterns are at heart of some of our arguments, as they allow us to produce some explicit A_∞-central elements of Yoneda algebras, arising from what we call "stable relation cycles". A slightly weaker notion of stability for relation cycles was introduced by Green, Snashall and Solberg in [76]; our approach to stability is directly motivated by applications to higher structures of the Yoneda algebra.

Our results also allow us to give a combinatorial characterisation of the Gorenstein property for monomial algebras in terms of Anick chains of sufficiently large length. This builds on recent work of Chen, Shen, and Zhou [39] who introduced the notion of perfect paths for monomial algebras in their classification of indecomposable Gorenstein-projective modules. Their work also shows that the minimal resolution of $\Bbbk$ over a Gorenstein monomial algebra is eventually periodic. A consequence of our work is the following reflection of that periodicity property on the level of Hochschild cohomology as follows.

Theorem (Theorem 5.5.2)**.** *Let Λ be a Gorenstein monomial algebra, of Gorenstein dimension d. Then Hochschild cohomology is eventually periodic; more precisely, there exists an element $\chi \in$*

$\mathrm{HH}^*(\Lambda,\Lambda)$ *of even degree* p *such that taking cup product with* χ *gives an isomorphism*

$$\chi \smile -\colon \mathrm{HH}^n(\Lambda,\Lambda) \xrightarrow{\cong} \mathrm{HH}^{n+p}(\Lambda,\Lambda) \quad \text{for all } n \geq d+1.$$

For a Gorenstein algebra Λ, we let $\widehat{\mathrm{HH}}^*(\Lambda,\Lambda)$ denote its Tate–Hochschild cohomology. Our results imply the following elegant statement.

Theorem (Corollary 5.5.3)**.** *Let* Λ *be a Gorenstein monomial algebra. Then Tate-Hochschild cohomology is given by periodic Hochschild cohomology:* $\widehat{\mathrm{HH}}^*(\Lambda,\Lambda) \cong \mathrm{HH}^*(\Lambda,\Lambda)[\chi^{-1}]$.

4 Homotopical algebra and term rewriting theory

When studying algebras presented by generators and relations, the central general result is the statement known as the Diamond Lemma. Historically, one would say that it was already implicit in Shirshov's work [179] who proved an analogous but technically more involved result, the Composition Lemma, in the case of Lie algebras. About a decade after that, this result was proved in the case of associative algebras by Bokut in [27], and independently by Bergman [21], who also proposed the name "Diamond Lemma" in order to emphasize the analogy with the classical result of Newman [144]. Paraphrasing the opening phrase of [21], the main results of our paper [55] are doubly trivial: Bergman's Diamond Lemma is, in his own words, trivial, and our main goal is to explain a new trivial proof of this trivial statement, as well as some other trivial statements, from the point of view of homotopical algebra. However, we believe that readers of [55] can benefit from it in a number of ways. For a reader whose intuition comes from homotopical algebra, our proof will hopefully feel like a conceptual explanation of useful but seemingly technical criteria of "resolvable ambiguities" for uniqueness of normal forms. For a reader with background in Gröbner bases or term rewriting, our proof will offer intuition behind both the Diamond Lemma and its optimisation, known as the Chain Criterion in the commutative case [33, 57, 110] and as the Triangle Lemma in the case of noncommutative associative algebras [31, 117, 175], as well as precise guidance on how to generalise those for other algebraic structures. Specifically, our work means that computing models of algebras with monomial relations explicitly helps both to state the relevant Diamond Lemmas and to optimise them.

From Gröbner bases to deformation theory and back. The idea of using Gröbner bases for computing homological invariants of associative algebras is well known. The seemingly earliest instance appears in the work of Priddy [148] who constructed resolutions for the ground field, viewed as the trivial module, for algebras presented by quadratic–linear relations; this was later generalised by Anick [2] to arbitrary presentations. Existence of such resolutions is not at all surprising for the following reason. Defining relations of an associative algebra A form a Gröbner basis if the same monomials give a vector space basis for both algebras A and A_{mon}; the latter is the algebra with monomial relations given by the leading monomials of the relations of A with respect to a suitable

ordering. The trivial module for A_{mon} admits a combinatorially defined free resolution, implicit in the work [9] of Backelin, and explicitly determined by Green, Happel and Zacharia in [73]. Such resolution can be obtained as a contraction of the bar resolution; incorporating the lower terms of relations from a Gröbner basis can obtained from such a contraction by homological perturbation techniques [19, 81, 82, 116]. In fact, the one-sided module resolution of Anick can be generalised to a bimodule resolution, as established by Bardzell [11–13], leading to a computational method for determining the Hochschild (co)homology of an algebra, and thus for studying deformations of a given algebra. This construction works under a weaker assumption of a convergent rewriting system instead of a Gröbner basis, as shown by Chouhy and Solotar in [40].

Our main wish, motivated by interest in generalising these methods to algebraic structures other than associative algebras, is to go in the opposite direction and re-discover effective criteria for reduction systems with unique normal forms from the deformation theory viewpoint. Since in those situations the multiplication table of an algebra A is obtained from the simple combinatorial multiplication table of A_{mon} by incorporating appropriate lower terms, we naturally find ourselves in the framework of deformation theory. In fact, very recently the bimodule resolutions mentioned above have been used to study deformations of algebras with monomial relations by Barmeier and Wang in their work on deformation theory of quiver algebras [14] and by Redondo and Rossi Bertone in [155], using the following idea. A free bimodule resolution

$$A_{\text{mon}} \otimes C_\bullet \otimes A_{\text{mon}} \simeq A_{\text{mon}}$$

leads to the representative $\text{Hom}(C_\bullet, A_{\text{mon}})$ of the deformation complex, and one can use a strong deformation retract relating it to Hochschild complex to compute explicitly its L_∞-algebra structure using the homotopy transfer theorem. Studying deformations amounts to studying solutions to the Maurer–Cartan equation in that L_∞-algebra, and one may obtain various results this way. However, if one adopts this viewpoint, the Diamond Lemma criterion of "resolvable ambiguities" cannot be recovered instantly: it involves a calculation in the *free* associative algebra which cannot be reproduced directly since the target space of the deformation complex is the monomial algebra A_{mon}. The two deformation complexes are homotopy equivalent, so the necessary result can be proved in principle. However, recovering the classical Diamond Lemma will require some translation, in the spirit of Anick's slogan "the perturbative construction of the second differential of the resolution is precisely the procedure of resolving ambiguities", see [2, Sec. 2]. We also refer the reader to a discussion of this phenomenon in the survey of Ufnarovski [175, Sec. 3.8] where one also finds the Triangle Lemma as a way to optimise the algorithm.

Deformations via tangent complexes. It turns out that a natural way to remedy the situation is to move from homology to homotopy, and study deformations of algebras in terms of homotopical algebra. This means working with multiplicative free resolutions, that is resolutions which are free as algebras rather than as bimodules. In this case, the homotopy class of the deformation complex of an algebra can be represented by its tangent complex, that is, the differential graded

Lie algebra of derivations of its free resolution. Historically, this approach to deformations first emerged in the case of commutative associative algebras. In this context, it makes sense to note that some of the pioneering works both in the theory of Gröbner bases and in the deformation theory appeared in the algebro-geometric context: the former in the solution of the resolution of singularities problem in characteristic zero by Hironaka [92, 93], and the latter in Palamodov's work on deformation of analytic spaces [145], where the tangent complex approach is attributed to unpublished work of Tyurina (who tragically died in a kayaking accident at the age of thirty two). However, the standard bases of Hironaka were just one of many tools in a paper of more than two hundred pages, and the power of this method in algebraic geometry and commutative algebra became apparent only after work of Buchberger [34] who highlighted their algorithmic nature. The tangent complex approach to deformation theory of algebras became widely known from the famous manuscript [161] by Schlessinger and Stasheff. We would like to also remark that a seemingly completely independent path to multiplicative resolutions (which however stays away from deformation theory questions) emerges in higher-categorical rewriting theory. Original work of Squier [115, 165] on homological finiteness conditions for monoids admitting a convergent presentation first received a higher-categorical flavour in works of Lafont [113] and Citterio [46]; corresponding multiplicative resolutions appear in work of Lafont and Métayer [114] and Guetta [80] relying on theory of resolutions of categories by polygraphs developed by Métayer [140]. This formalism was recently extended to the $\Bbbk$-linear context by Guiraud, Hoffbeck, and Malbos [84], who in particular explained how to construct a polygraphic resolution of an associative algebra by techniques similar to the perturbative construction of A_∞-structures [83] and free resolutions of algebras [96].

Precursors: Evgeny Solomonovich Golod and Victor Nikolaevich Latyshev To conclude the introduction, we would like to mention two crucial sources of inspiration for our work. The first of them is the work of Golod who discovered a proof of the Diamond Lemma [68] using the non-commutative analogue of the Koszul complex, the Shafarevich complex associated to a system of elements in the free algebra originally introduced in [70]. That complex is a differential graded algebra whose homotopy type depends on the presentation of the original algebra. Our homotopy invariant free resolutions are obtained from the Shafarevich complex at the cost of adding extra generators of higher degrees; we believe that the benefit of the resulting clarity outweighs the cost. Our second inspiration comes from the work of Latyshev who used normal forms to resolve some particular cases [121, 122] of the celebrated Specht problem on identities of associative algebras [164], and advocated general importance of normal forms and standard bases [118–120]. Both Golod and Latyshev passed away relatively recently (in July 2018 and April 2020, respectively). The paper [55] Chapter 6 is based on is dedicated to their memory.

Chapter 2

Conventions and recollections

1 Conventions

Cross-referencing. The numbering in references is `maybeChapter.Section.Subsection`, where `maybeChapter` prints the chapter being referenced if it is not the current one. Thus, 'Section 2.1' refers to Section 1 in Chapter 2, while 'Section 1' refers to the same section in the current chapter. Similarly, 'Subsection 3.1.2' refers to Subsection 2 in Section 1 of Chapter 3, while 'Subsection 1.2' refers to Subsection 2 of Section 1 in the current chapter. The same remark applies to references to theorems, lemmas and corollaries.

1.1 Base field, suspensions and signs. We fix once and for all a field $\Bbbk$. All unadorned hom and $\otimes$, which denote the usual bifunctors on graded vector spaces, will be taken with respect to $\Bbbk$. We let $\Bbbk s^{-1}$ be the graded vector space concentrated in degree -1, where it is one dimensional, and write s^{-1} for its generator. If V is a graded vector space, we write $s^{-1}V$ for $\Bbbk s^{-1} \otimes V$, and denote $s^{-1} \otimes v$ by $s^{-1}v$, and write $V^\vee$ for the graded dual of V. Implicit signs follow the Koszul sign rule, and we make them as explicit as we can. By a *graded* vector space we mean a vector space V of the form

$$V = \bigoplus_{n \in \mathbb{Z}} V_n,$$

where we write $|v| = n$ for $v \in V_n$, and refer to n as the *homological degree* of an element v. The adjective 'homological' means that these degrees create "Koszul signs" arising from exchanging factors in tensor products. Recall that one defines the tensor product of two graded vector spaces by the formula

$$(V \otimes W)_n := \bigoplus_{i+j=k} V_i \otimes W_j$$

summarised in words by "degrees add under tensor products", and homological degrees enter the formula for the symmetry isomorphism

$$\tau_{V,W} : V \otimes W \to W \otimes V$$

given by $\tau(v \otimes w) = (-1)^{|v||w|} w \otimes v$. (This, for a trained eye, creates signs in a lot of places; for example, if one applies the tensor product of linear maps to a tensor product of two vectors, the formula $(\phi \otimes \psi)(v \otimes w) = (-1)^{|\psi||v|} \phi(v) \otimes \psi(w)$ has to be used.) Occasionally, our graded vector spaces will have extra gradings which do not create any extra signs in formulas; in such cases, we shall use the word "weight" for such degrees. In the last chapter, we will also use the word "grade" to refer to a refinement of the classical weight grading by $\mathbb{N}$ using free monoids instead.

1.2 Associative algebras and quivers. Elements of graded algebras and modules are always taken to be homogeneous. We will adhere to the convention that A or Λ will always denote a non-dg associative algebra concentrated in homological degree zero, while B will always denote a dg associative algebra concentrated in non-negative degrees. We use the language of quivers; a quiver is a directed graph $Q = (Q_0, Q_1, s, t)$ with the set of vertices Q_0 and the set of arrows Q_1, where for any arrow $\alpha \in Q_1$, we denote by $t(\alpha), s(\alpha) \in Q_0$ its target and source respectively. All quivers are assumed finite and connected. We use the notation Q_m for the set of paths in Q of length m, and $\mathscr{P}_Q = Q_0 \cup Q_1 \cup Q_2 \cup \cdots$ for the set of all paths in Q. We call a path in Q non-trivial if it has length at least one. We denote by $\mathfrak{m}$ the Jacobson radical $\operatorname{rad} \Bbbk Q$; it is spanned by all non-trivial paths.
Unless specified otherwise, we use the notation $\Lambda = \Bbbk Q/I$ for a finite dimensional quotient of the path algebra with $I \subseteq \mathfrak{m}^2$ admissible (so that $\mathfrak{m}^n \subseteq I$ for some n; this is automatic in the case of finite dimensional monomial algebras). We let $\underline{\Bbbk} := \Bbbk Q_0$ denote the corresponding semisimple $\Bbbk$-algebra, and use the notation $\mathfrak{r} = \operatorname{rad} \Lambda$ for the Jacobson radical of Λ. This allows us treat Λ as an augmented algebra over $\underline{\Bbbk}$ with augmentation $\varepsilon : \Lambda \twoheadrightarrow \Lambda/\mathfrak{r} = \underline{\Bbbk}$. All modules over finite dimensional algebras are taken to be right modules and finitely generated; we simply call them finite.

2 Recollections

2.1 Rewriting systems. In this section, we give a short introduction to term rewriting in the linear context. Mathematically, all of this goes back to [21]; terminologically, we choose to follow the recent literature on rewriting systems, see, for example, [84] and references therein.
Let us fix once and for all a finite set X; we shall denote by $\langle X \rangle$ the free monoid generated by X, so that the linear span $\Bbbk\langle X \rangle$ is the free associative $\Bbbk$-algebra generated by X. For an element $g \in \Bbbk\langle X \rangle$ we shall denote by $\operatorname{supp}(g)$ the set of all elements of $\langle X \rangle$ that appear in g with a non-zero coefficient.

Definition 1 (Rewriting system)**.** A *rewriting system* on $\Bbbk\langle X \rangle$ is a triple (X, W, f), where $W \subset \langle X \rangle$, and f is a function on W with values in $\Bbbk\langle X \rangle$.

The right way to think of a rewriting system is as of a collection of rules that allow to rewrite each monomial $w \in W$ into the element $f(w) \in \Bbbk\langle X\rangle$. In this way, each rewriting system Σ naturally gives rise to an associative algebra A_Σ with relations $R = \{w - f(w) \colon w \in W\}$:

$$A_\Sigma = \Bbbk\langle X|R\rangle.$$

Each rewriting rule of Σ becomes a way to replace elements of $\Bbbk\langle X\rangle$ by other representatives of the same coset in A_Σ, as follows.

Definition 2 (Reduction)**.** Suppose that $w \in W$, and $a, b \in \langle X\rangle$. A *basic reduction* associated to the triple (a, w, b) is the $\Bbbk$-linear endomorphism ρ of $\Bbbk\langle X\rangle$ such that for each $m \in \langle X\rangle$ we have

$$\rho(m) = \begin{cases} af(w)b, & \text{if } m = awb, \\ m & \text{otherwise.} \end{cases}$$

A finite sequence of basic reductions is called a *reduction*; it defines a $\Bbbk$-linear endomorphism of $\Bbbk\langle X\rangle$ by composing, in the given order, the basic reductions appearing in it.

Using reductions is intended to "simplify" representatives of cosets. This is formalised by the notion of a normal form.

Definition 3 (Normal form)**.** An element $g \in \Bbbk\langle X\rangle$ is called *irreducible* if $\rho(g) = g$ for every basic reduction ρ. A *normal form* of an element $g \in \Bbbk\langle X\rangle$ is an irreducible element $\bar{g}$ such that the cosets of g and $\bar{g}$ in A_Σ are equal.

In principle, it is possible that one can perform basic reductions infinitely many times. We shall only work with rewriting systems for which it does not happen. To formalise that, we give the following definition.

Definition 4 (Terminating rewriting system)**.** A *pseudo-reduction* is an infinite sequence

$$\rho = (\rho_1, \rho_2, \rho_3, \ldots)$$

of basic reductions. To every pseudo-reduction ρ and every $g \in \Bbbk\langle X\rangle$ we can associate the sequence

$$\rho(g) = (\rho_1(g), \rho_2\rho_1(g), \rho_3\rho_2\rho_1(g), \ldots)$$

of elements in $\Bbbk\langle X\rangle$. We say that an element $g \in \Bbbk\langle X\rangle$ is *reduction finite* if for every pseudo-reduction ρ, the associated sequence $\rho(g)$ is eventually constant. A rewriting system Σ is *terminating* if all elements $g \in \Bbbk\langle X\rangle$ are reduction finite.

Note that if an element g is reduction finite, there exists a reduction that sends this element to an irreducible one, for otherwise we could easily find a pseudo-reduction ρ for which $\rho(g)$ is not

eventually constant. Thus, for a terminating rewriting system, each element has at least one normal form. All rewriting systems considered in this chapter are assumed to be terminating. We shall now recall how to recast termination in terms of partial orderings.

Definition 5 (Compatible rewriting ordering)**.** For a given rewriting system Σ, we say that a partial order $\leqslant$ on $\langle X\rangle$ is a *rewriting ordering compatible with* Σ if the following conditions hold:

(1) *multiplicativity*: if $m < m'$ for two elements m, m', then $amb < am'b$ for any $a, b \in \langle X\rangle$;

(2) *well-ordering*: every non-empty set of monomials has a minimal element;

(3) *compatibility with* Σ: for all $w \in W$ and all $w' \in \mathrm{supp}(f(w))$, we have $w' < w$.

Among rewriting orderings compatible with Σ (if they exist at all), there exists the weakest possible one obtained as follows. We first define $<_\Sigma$ as the smallest transitive binary relation that is multiplicative and compatible with Σ. If $<_\Sigma$ is a well-ordering, the reflexive closure $\leqslant_\Sigma$ is a rewriting ordering compatible with Σ, and each other rewriting ordering compatible with Σ obviously refines the ordering $\leqslant_\Sigma$. According to [21, 40], the rewriting system Σ is terminating if and only if $<_\Sigma$ is a well-ordering.
For a terminating rewriting system, the associative algebra

$$A_{\Sigma_0} = \Bbbk\langle X|W\rangle$$

obtained from the rewriting system $\Sigma_0 = (W, 0)$ and called the *monomial algebra associated to* Σ is a relevant object of study. Indeed, an element $g \in \Bbbk\langle X\rangle$ is irreducible if and only if no monomial $m \in \mathrm{supp}(g)$ is divisible by elements of W, so irreducible monomials form a basis of the algebra A_{Σ_0}. However, since the same element may have several different normal forms, the cosets of irreducible monomials might not be linearly independent in the algebra A_Σ.

Definition 6 (Convergent rewriting system)**.** A (terminating) rewriting system is *convergent* if each element $g \in \Bbbk\langle X\rangle$ has exactly one normal form.

2.2 Gröbner bases. Among the rewriting systems, the best known class is given by *Gröbner bases*: those are rewriting systems where there is a total order $\leqslant$ on $\langle X\rangle$ satisfying the above three conditions. Let us recall an example [31, 84] showing that sometimes using a partial order is really advantageous. For that, we consider the algebra $A = \Bbbk\langle x, y, z|x^3 + y^3 + z^3 - xyz\rangle$. Note that for a total multiplicative order, we always have $xyz < \max(x^3, y^3, z^3)$, and one can check that for such an order the corresponding rewriting system is not convergent, so one needs to extend it by further rewriting rules. However, one can consider the rewriting system $\Sigma = (X, W, f)$ with $X = \{x, y, z\}$, $W = \{xyz\}$ and $f(xyz) = x^3 + y^3 + z^3$ which does not arise from a total multiplicative order. Let us explain why this rewriting system is terminating. For that, we define, for each monomial m, $\Phi(m) = 3A(m) + B(m)$, where $A(m)$ is the number of divisors of m equal to xyz and $B(m)$ is the number of divisors of m equal to y. Then one can check that any application of our rewriting rule

replaces a monomial by a linear combination of terms for which the parameter Φ is strictly smaller, which guarantees the termination. Later we shall see that this rewriting system is convergent, showing that in this case there is an advantage over the theory of Gröbner bases.

2.3 Diamond lemma. Bergman's Diamond Lemma [21] establishes a criterion of convergence in terms of the so called "resolvable ambiguities". Our goal is to explain how such criterion can be found, so that one can determine its analogues for other algebraic structures. It is easy to see that a rewriting system Σ is convergent if and only if the corresponding irreducible monomials are linearly independent. In other words, Σ is convergent if and only if the linear map $A_{\Sigma_0} \to A_\Sigma$ sending each element of the standard monomial basis of A_{Σ_0} to its coset in A_Σ is an isomorphism of vector spaces. This suggests that convergent rewriting systems may be studied from the viewpoint of deformation theory of associative algebras. In general, the notion of a *deformation* implies that we *deform* the structure by adding lower terms with respect to a suitable filtration. For example, for the classical deformation theory set-up [136] defined over an Artinian local ring containing $\Bbbk$, the filtration may be defined by powers of the maximal ideal $\mathfrak{m}$ of that ring. In our case, the filtration is defined out of the partial ordering $<_\Sigma$, or, better to say, out of any order that extends it to a total well-ordering. The multiplication table in the algebra A_{Σ_0} is very simple: the product of two basis elements is either a basis element or zero. For a convergent rewriting system, the product of two basis elements in the algebra A_Σ is either a basis element or the linear combination of smaller basis elements obtained by term rewriting. Thus, saying that Σ is convergent is equivalent to saying that the algebra A_Σ is a deformation of the algebra A_{Σ_0} with respect to the corresponding filtration. This suggests that using deformation theory of algebras might shed light on convergence of rewriting systems. The main part of chapter four uses this viewpoint extensively.

2.4 Monomial algebras. We recall that a monomial algebra over $\Bbbk$ is a quotient of a free algebra TV on a finite dimensional $\Bbbk$-module V by an ideal generated by finitely many monomials. We will follow the conventions and definitions from [129], and we refer the reader to it for the essentials on weight graded differential graded algebras. The grading of a graded algebra with a zero differential is homological: a weight grading is, for us, an *extra grading* coming from some additional structure so that, for example, our monomial algebras will be weight graded but concentrated in homological degree zero. All results in the third chapter can be proved for quiver algebras with monomial relations; for readability, we present all arguments in the case of associative algebras (that is, for one-vertex quivers) and then merely state the corresponding generalization.

2.5 Bar resolution and Tor. Let A be a weight graded $\Bbbk$-algebra. Observe that if $\Omega_\infty C \longrightarrow A$ is a minimal model of A, then the space of indecomposable elements C of $\Omega_\infty C$ can be identified with $\mathrm{Tor}_A(\Bbbk,\Bbbk)$ and is, in fact, the Quillen homology of $\Omega_\infty C$: it will become apparent in what follows that our choice of basis for $\mathrm{Tor}_A(\Bbbk,\Bbbk)$, that of Anick chains, will be central in describing our choice of minimal model of A.

Write ${}_A\mathsf{Mod}$ and Mod_A for the respective categories of left and right A-modules. The bifunctor $-\otimes_A - : \mathsf{Mod}_A \times {}_A\mathsf{Mod} \longrightarrow {}_{\Bbbk}\mathsf{Mod}$ gives us, upon derivation, the classical bifunctor $\mathrm{Tor}_A(-,-) : \mathsf{Mod}_A \times {}_A\mathsf{Mod} \longrightarrow {}_{\Bbbk}\mathsf{GMod}$ to the category of *graded* $\Bbbk$-modules, defined as follows. For $M \in \mathsf{Mod}_A$ and $N \in {}_A\mathsf{Mod}$, let us pick respective projective resolutions $P \to M$ and $Q \to N$ in Mod_A and ${}_A\mathsf{Mod}$. The diagram

$$P \otimes_A N \longleftarrow P \otimes_A Q \longrightarrow M \otimes_A Q$$

connects the above three complexes by natural quasi-isomorphisms, up to our choice of resolutions, and their homology is the graded $\Bbbk$-module $\mathrm{Tor}_A(M,N)$. Let us remark that $\mathrm{Tor}_A(M,N)$ is usually denoted by $\mathrm{Tor}^A(M,N)$ but that for typographical purposes we will instead write it $\mathrm{Tor}_A(M,N)$. When A is connected or, more generally, augmented, we will write Tor_A for $\mathrm{Tor}_A(\Bbbk,\Bbbk)$, where $\Bbbk$ is made into an A-module via the augmentation $A \longrightarrow \Bbbk$.

There is a particularly useful way we can construct such bifunctor following the definition above. Concretely, if $R \longrightarrow A$ is any projective resolution of the A-bimodule A, then the homology of the complex $M \otimes_A R \otimes_A N$ is $\mathrm{Tor}_A(M,N)$. The advantage of this is we need only choose one resolution, namely that of A as an A-bimodule, to obtain resolutions for every left or right A-module, and we now fix this choice. Define $B(A,A,A)$, the *double-sided bar resolution of* A, to be the chain complex such that for each $n \in \mathbb{N}_0$, we have $B_n(A,A,A) = A \otimes \overline{A}^{\otimes n} \otimes A$, the free A-bimodule with basis $\overline{A}^{\otimes n}$, where $\overline{A}$ is the kernel of the augmentation $A \longrightarrow \Bbbk$.

For each such integer, denote a generic bimodule basis element in degree n by $[a_1|\cdots|a_n]$. Its differential is then given by

$$-a_1[a_2|\cdots|a_n] + \sum_{i=1}^{n-1}(-1)^{i-1}[a_1|\cdots|a_ia_{i+1}|\cdots|a_n] + (-1)^{n-1}[a_1|\cdots|a_{n-1}]a_n$$

and is extended A-bilinearly. In particular, if $n = 0$ we have $B_0(A,A,A) = A \otimes A$ and there is an augmentation $B_0(A,A,A) \to A$ given by multiplication which renders the augmented complex $B(A,A,A) \to A$ contractible both as a complex of left and as a complex of right A-modules. From this it follows that if M is right A-module and N a left A-module, the complex $B(M,A,N) := M \otimes_A B(A,A,A) \otimes_A N$ computes $\mathrm{Tor}_A(M,N)$.

From now on we assume that A is connected, which makes it naturally augmented, and endows $\Bbbk$ with a trivial A-module structure on both sides, in which elements of positive degree act by zero. From the previous remarks it follows that the complex $B(\Bbbk,A,A)$ is a resolution of the right A-module $\Bbbk$ by free right A-modules, which we will denote by $B(A,A)$ and call the *bar resolution of* $\Bbbk$, so that $\mathrm{Tor}_A(\Bbbk,\Bbbk)$ may be computed as the homology of the complex $B(\Bbbk,A,\Bbbk)$, which we simply denote by BA and call the *bar construction of* A. Concretely, we have for each $n \in \mathbb{N}_0$ a natural isomorphism $(BA)_n \to \overline{A}^{\otimes n}$, which we consider an identification, with differential given on

basis elements $[a_1|\cdots|a_n]$ by

$$d[a_1|\cdots|a_n] = \sum_{i=1}^{n-1}(-1)^{i-1}[a_1|\cdots|a_ia_{i+1}|\cdots|a_n].$$

The complex BA admits a diagonal $\Delta_2' : BA \longrightarrow BA \otimes BA$ given by deconcatenation, that makes it into a non-unital dga coalgebra. Concretely, on basis elements $[a_1|\cdots|a_n]$ of degree $n \in \mathbb{N}_0$ we have that

$$\Delta_2'[a_1|\cdots|a_n] = \sum_{i=1}^{n-1}[a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

2.6 Anick's resolution. In his celebrated article [2], Anick constructs an A-free resolution of the trivial module for any augmented algebra A equipped with a Gröbner basis. This construction is generalized for quiver algebras in [3], where the authors use the notion of chains for such algebras from [73]. Since we will use the description of Tor_A by means of Anick's resolution, let us quickly recall his results.

Let us write S for a set of generators of A, the variables, and let $f : \Bbbk\langle S\rangle \longrightarrow A$ be the quotient map by the ideal of relations of A, which is a map of augmented $\Bbbk$-algebras. We weight grade A by the length of a monomial, and give S a total order. This induces on the monoid of monomials M_S on S a well ordering in such a way that $m < m'$ if $|m| < |m'|$ (here $|m|$ is the length of a monomial), or if m and m' are of the same length but $m < m'$ in the lexicographical order, also known as the dictionary order, induced from the total order of the letters in S: if $m = x_1 \cdots x_t$ and $m' = y_1 \cdots y_t$ are monomials of the same length, then $m < m'$ if $x_i < y_i$ for the first index $j \in [t]$ such that $x_j \neq y_j$. Given monomials $u, v \in M_S$, say *v is a divisor of u* if $u = u'vu''$ for monomials u', u'', to obtain a partial ordering $\subseteq$ on M_S. A subset I of M_S is an *order ideal of monomials* if it is a lower set for $\subseteq$. Note that the reverse ordering associated to $\subseteq$ is sometimes used in the literature: for us, $v \subseteq u$ means that v is a divisor of u. It is readily checked by induction that the set $N = \{x \in M_S : f(x) \notin \langle f(y) : y < x\rangle\}$ is an order ideal of monomials, and that $f(N)$ is a basis of A as a $\Bbbk$-module.

From N Anick extracts the basic building blocks for his resolution, the *obstructions*. Concretely, let V consist of those $x \in M_S$ that are not in N, but all $y \subsetneq x$ are in N. These are simply the maximal elements of the order ideal N, and thus form an anti-chain. Its elements are the obstructions. From the definitions it follows that an element is in $M_S \setminus N$ precisely when it contains as a divisor an obstruction. In case A is monomial, N consists of those monomials that contain no monomial relation as a divisor, and the obstructions are the minimal relations of A. Now set $V^{-1} = \Bbbk$, $V^0 = \Bbbk S$ and $V^1 = \Bbbk V$, to begin to construct a right A-free resolution

$$\cdots \longrightarrow V^2 \otimes A \xrightarrow{\delta_2} V^1 \otimes A \xrightarrow{\delta_1} V^0 \otimes A \xrightarrow{\delta_0} V^{-1} \otimes A \longrightarrow 0$$

of $\Bbbk$. For each $n \in \mathbb{N}$ we now obtain a vector space V^n with a basis of monomials, called the n-chains, in the following way. An *n-prechain* is a monomial $x_{i_1} \cdots x_{i_t}$ in B for which there exist

strictly increasing sequences of integers $(a_1,\dots,a_n)$ and $(b_1,\dots,b_n)$ with $a_1 = 1$ and $b_n = t$ such that the sequences are interlaced, meaning that $a_{i+1} \leqslant b_i$ for each $i \in \{1,\dots,n-1\}$, and such that for each $j \in \{1,\dots,n\}$, the monomial $x_{i_{a_j}} \cdots x_{i_{b_j}}$ is an obstruction.

In particular, the collection of 1-prechains, which coincides with that of 1-chains, is a basis for V^1. We say an n-prechain is an *n-chain* if the two previous sequences may be chosen so that $x_{i_1} \cdots x_{i_s}$ is not an m-prechain for any $s < b_m$ and $m \in \{1,\dots,n\}$. Plainly, a chain is a prechain that satisfies a minimality condition regarding the overlappings between the obstructions that constitute it. It is readily verified that in this case these two sequences are uniquely determined, there is a unique $s = b_{n-1} < t$ such that $x_{i_1} \cdots x_{i_s}$ is an $(n-1)$-chain and the tail $x_{i_s+1} \cdots x_{i_t}$ contains no divisor that is an obstruction. This is the key observation to construct a sequence of boundary maps $(\delta_n : V^n \otimes A \longrightarrow V^{n-1} \otimes A)_{n \geqslant 2}$ such that

$$\delta_n(x_{i_1} \cdots x_{i_t}) = x_{i_1} \cdots x_{i_s} \otimes x_{i_s+1} \cdots x_{i_t} + \text{lower terms}.$$

If A is monomial, there are no lower terms in the differential and this resolution is minimal, so that for each $n \in \mathbb{N}$, $\operatorname{Tor}_A^{n+1}$ is identified with the vector space V^n with basis consisting of the n-chains: this is the content of Lemma 3.3 in [2]. Let us make the important remark that, in what follows, we adhere to such identification strictly: our main result depends critically on using Anick chains to model Tor_A.

Since it will be useful to illustrate some of the rather technical constructions that will follow, let us define the algebra $J = T(x,y)/(x^2,[y^2,x]-xyx)$ and the monomial algebra $K = T(x,y,z)/(xy^2,y^2z)$. In the case of the algebra J with the order $y > x$, one can check that the specified generator for its ideal of relations constitute a Gröbner basis. In the case of K this is immediate for any order, since the relations are monomial and there are no redundant relations. For J, we obtain that for each $n \in \mathbb{N}$, the set of n-chains is $\{x^{n+1}, y^2x^n\}$, corresponding to n overlappings of the relation x^2 with itself, and of an overlapping of y^2x with $n-1$ copies of the relation x^n. In the case of K, we get finitely many Anick chains: the 0-chains are $\{x,y,z\}$, the 1-chains are the relations $\{xy^2, y^2z\}$, and the 2-chains are the overlappings $\{xy^2z, xy^3z\}$, and there are no other chains.

2.7 Algebraic discrete Morse theory. Let C be a non-negatively graded complex of $\Bbbk$-modules. Fix a basis $X = \{X_t\}_{t \geqslant 0}$ of homogeneous elements of C, so that for each $t \in \mathbb{N}_0$, the set X_t is a basis of C_t. Given $c \in X$ we introduce the notation

$$dc = \sum_{c' \in X} [c : c']c'$$

where $[c : c'] \in \Bbbk$. Let $G = G(C,X)$ be the directed weighted graph with vertices the set X and with an edge $c \to c'$ if c' appears in dc with non-zero coefficient $[c : c']$ which is, in that case, the weight of $c \to c'$. A finite subset M of edges of G is a *Morse matching* if it satisfies the following *Morse conditions*:

M1. Each vertex of G is in at most one edge of M.
M2. The weights of edges of M are invertible.
M3. The graph G_M obtained by inverting the edges of M in G has no directed cycles.

If $c' \to c$ is a edge in G_M with $c \to c' \in M$, we set its weight to be $-[c : c']^{-1}$. In our situation the coefficients $[c : c']$ will be either 1 or -1, which means **M2** is always satisfied. We write X^M for the collection of vertices not appearing in M, which we call *critical*. Write $P(c,c')$ for the set of paths in G_M from c to c', and assign a path the product of the weights of the edges it contains. Finally, write $\Gamma(c,c')$ for the sum of all the weights of paths from c to c' in G_M.
We define the *Morse complex of C with respect to M*, which we denote by C^M, as the complex with basis the critical vertices X^M and with differential given, on basis elements, by

$$dc = \sum_{c' \in X^M_{t-1}} \Gamma(c,c')c'$$

whenever $c \in X_t$. The result of main interest to us in [102] is the following theorem, which shows how to produce a homotopy retract datum from C to C^M given a Morse matching M on C relative to a basis X.

Theorem 2.1. *The complex C^M is homotopy equivalent to C. More precisely, there are maps $f : C \longrightarrow C^M$ and $g : C^M \longrightarrow C$ given on basis elements by*

$$f(c) = \sum_{c' \in X^M_t} \Gamma(c,c')c', \quad g(c) = \sum_{c' \in X_t} \Gamma(c,c')c'$$

for $c \in X_t$, respectively $c \in X^M_t$, which are inverse homotopy equivalences. In fact, $fg = 1$ and $gf - 1 = dh + hd$ where for a basis element $c \in X_t$,

$$h(c) = \sum_{c' \in X_{t+1}} \Gamma(c,c')c'.$$

Note that since for any two basis elements we have defined the coefficient $\Gamma(c,c')$ as a sum through paths, it is important that M is finite for the theorem above to hold. We can, however, consider matchings M of the complex C if C is the colimit of a finite sequence of finite subcomplexes $\{F^pC\}$ that is compatible with the matching, in the sense that $(F^pC)^M$ is a filtration by subcomplexes of C^M. This last condition means $\Gamma(c,c')$ is well defined and the last theorem extends in this situation. In particular, we will consider the situation of $\mathbb{N}$-multigraded complexes such that each homogeneous subcomplex is finite, and in this case the filtration by weight of tuples fulfills the condition above.
Let us note that in the homotopy h we can only have a path from an element in degree t to one in degree $t+1$ if it is given by a sequence of edges $e'_0 e_1 e'_1 \cdots e_j e'_j$ where e'_i is an inverted edge of the matching and e_i is a direct edge. Indeed, the first Morse condition forbids a concatenation of inverted edges, which means we also cannot have two consecutive non-inverted edges. Finally, let

us observe that if $c \in C^M$ is a cycle then $g(c) = c$, that the last observation means that $h^2 = 0$, and that $hg = 0$ and $fh = 0$. Thus (f,g,h) is a homotopy datum that satisfies the side conditions, as defined in Subsection 2.9.

2.8 Anick's resolution via Morse theory. Let A be a weight graded $\Bbbk$-algebra presented by generators $\{x_1,\dots,x_n\}$ and ideal of relations I, and assume that $\{f_1,\dots,f_m\}$ is a reduced Gröbner basis with respect to a fixed monomial order $<$. Following [102], we show how to obtain the Anick resolution of A as the Morse complex of an acyclic matching on the normalized bar resolution $B(A,A)$ of $\Bbbk$, which we now denote more simply by B.

Let $\mathrm{in}(I)$ denote the ideal of leading terms of elements of I, which is generated as an ideal by the leading terms of the elements in $\{f_1,\dots,f_m\}$. A monomial is *normal* if it is not divisible by a leading term of an element in $\{f_1,\dots,f_m\}$, and we write SM for the collection of such monomials. A monomial is *reducible* if it is not normal, and we say that $uv = 0$ *minimally* if for every prefix v' of v, the monomial uv' is normal. The set SM is a basis of A as a $\Bbbk$-module. In particular, given two normal monomials u and v we can write $uv = \sum_{w\in\mathsf{SM}} \lambda_w w$ where $|w| \leqslant |uv|$ for any w with $\lambda_w \neq 0$. Observe that if A is a monomial algebra, that is, if the relations of A are given by monomials, the normal monomials are those that do not contain as a subword any monomial relation, and reducible monomials are zero in A.

We now define a Morse matching on B by induction. Recall that we denote a generic basis element of the bar resolution by $[a_1|\cdots|a_n]$. Define M_1 to be the collection of edges of the form $[x_i|w_1|w_2|\cdots|w_t] \longrightarrow [x_iw_1|w_2|\cdots|w_t]$. The critical vertices $B^{(1)}$ with respect to M_1 are the variables $[x_1],\dots,[x_n]$ in degree 1, and those words $[x_1|w_1|\cdots|w_t]$ of normal monomials such that x_1w_1 can be reduced. We proceed inductively to define M_j for $j > 1$. Having defined M_{j-1}, let $B^{(j-1)}$ be the set of critical vertices with respect to $M_1 \cup \cdots \cup M_{j-1}$, and define E_j to be the set of edges that connect vertices of $B^{(j-1)}$. Suppose that

$$[x_{i_1}|w_2|\cdots|w_{j-1}|u_1|u_2|w_{j+1}|\cdots|w_t] \to [x_{i_1}|w_2|\cdots|w_{j-1}|w_j|w_{j+1}|\cdots|w_t]$$

is an edge in E_j. In particular, $w_j = u_1u_2$. We say e satisfies the *matching condition* if

B1. the monomial u_1 is a prefix of w_j,
B2. the source of e is in $B^{(j-1)}$ and,
B3. for each prefix v_1 of u_1 and each v_2 such that $v_1v_2 = w_j$, the vertex

$$[x_{i_1}|w_2|\cdots|w_{j-1}|v_1|v_2|w_{j+1}|\cdots|w_t]$$

is not in $B^{(j-1)}$.

We let M_j be the collection of edges in E_j that multiply monomials at the jth bar and satisfy the matching condition. Then the set $B^{(j)}$ is given by the variables $[x_i]$ in degree 1, the elements $[x_{i_1}|w]$ such that $x_{i_1}w$ is a minimal monomial generating the ideal of leading terms of I, and the elements of

the form $[x_i|w_2|w_3|\cdots|w_t]$ such that for each prefix u of w_j the vertex $[x_i|w_2|\cdots|w_{j-1}|u|w_{j+1}|\cdots|w_t]$ is not in $B^{(j-1)}$ and the term w_jw_{j+1} is reducible. We set $M=\bigcup_{j\geqslant 1}M_j$ to be the desired Morse matching, and let B^M be the collection of critical vertices with respect to M.

Lemma 2.2. *Assume that A is a weight graded monomial $\Bbbk$-algebra. Let $j\in\mathbb{N}$. The elements of M_j consist of those edges $[x_i|u_1|\cdots|u_{j-1}|u_j|\cdots]\to[x_i|u_1|\cdots|u_{j-1}u_j|\cdots]$ such that $x_iu_1=u_1u_2=\cdots=u_{j-2}u_{j-1}=0$ minimally and $u_{j-1}u_j\neq 0$. Moreover, the collection M is a Morse matching.*

Proof. This is a particular case of [102, Lemma 4.2]. □

We now describe the critical vertices B^M. Let $m_1,\ldots,m_l$ be minimal monomial generators of the ideal of leading monomials of I, such that for each $j\in\{1,\ldots,l\}$ we have $m_j=u_jv_ju_{j+1}$ where u_1 is a variable. We call the term $[u_1|v_1u_2|v_2u_3|\cdots|v_lu_{l+1}]$ *fully attached* if for all $j\in\{1,\ldots,l-1\}$ and each prefix u of $v_{j+1}u_{j+2}$ the monomial $v_ju_{j+1}u$ is normal. We denote by B_j the set of fully attached terms of degree $j\geqslant 2$ and let B_1 consist of the variables. We refer the reader to [102] for the proof of the following lemma, valid for any weight graded $\Bbbk$-algebra with a Gröbner basis, as in the beginning of this section.

Lemma 2.3. *The fully attached tuples are exactly the critical vertices, and the complex C^M is the Anick resolution of A. In case A is monomial, the critical vertices are the variables $[x_1],\ldots,[x_n]$ along with those terms $[x_i|u_1|\cdots|u_r]$ where if we set $x_i=u_0$, we have that $u_ju_{j+1}=0$ minimally for $j\in\{0,\ldots,r-1\}$.* □

Let us briefly explain how one can go back and forth from an Anick chain γ to a cycle in Tor_A. If γ is a 1-chain then it is a monomial relation $x_1\cdots x_t$, and the corresponding bar cycle (and critical vertex) is $[x_1|x_2\cdots x_t]$. Now suppose that γ is an n-chain. Then there exist a unique $(n-1)$-chain γ' and a unique monomial t such that $\gamma=\gamma' t$, and the bar term corresponding to γ is $[x_1|u_1|\cdots|u_l|t]$ where everything before t is the bar structure corresponding to γ'. In this way, for example, if γ is a 2-chain corresponding to an overlap of a relation $x_1\cdots x_t$ with another relation $x_j\cdots x_tx_{t+1}\cdots x_s$, then the bar structure of the corresponding critical vertex is $[x_1|x_2\cdots x_t|x_{t+1}\cdots x_s]$. To illustrate, in the case of J, for each $n\in\mathbb{N}$, the bar element corresponding to x^{n+1} is $[x|\cdots|x]$, where there are exactly n bars, and the bar element corresponding to y^2x^n is $[y|yx|x|\cdots|x]$ where, again, we have n bars. In the case of the algebra K, the Anick chains correspond to the bar elements

$$[x],\ [y],\ [z],\ [x|y^2],\ [y|yz],\ [x|y^2|z] \text{ and } [x|y^2|yz].$$

2.9 Homotopy transfer theorem and A_∞-coalgebras. Recall that an A_∞-coalgebra is a graded $\Bbbk$-module V along with sequence of locally finite maps $(\Delta_n : V\longrightarrow V^{\otimes n})_{n\in\mathbb{N}}$, where for each $n\in\mathbb{N}$ we have $|\Delta_n|=n-2$, that satisfy the following *Stasheff identities*

$$\mathrm{SI}(n):\quad \sum_{r+s+t=n}(-1)^{r+st}(1^r\otimes\Delta_s\otimes 1^t)\Delta_u=0.$$

That such sequence of maps be locally finite means that for each element $v \in V$ the set $\{\Delta_n(v) : n \in \mathbb{N}_0\}$ contains finitely many nonzero terms. We write (V,Δ) for an A_∞-coalgebra, and we call it *minimal* whenever Δ_1 vanishes. Observe that every graded vector space, every complex, and every dga coalgebra is, in an obvious way, an A_∞-coalgebra. Remark our A_∞-coalgebras are non-unital and positively graded. We warn the reader that conventions for the signs appearing in the Stasheff identities above vary in the literature. As explained in [137, Section 2], one can go from this convention to that of J. Stasheff by multiplying Δ_n by the sign $(-1)^{\binom{n}{2}}$.

We can associate to every A_∞-coalgebra (V,Δ) a dga algebra $(\Omega_\infty V, b)$, its *∞-cobar construction*, as follows. The underlying algebra to $\Omega_\infty V$ is the free associative algebra on the suspension $s^{-1}V$. Define the family of maps $(b_n : s^{-1}V \longrightarrow (s^{-1}V)^{\otimes n})_{n\in\mathbb{N}}$ by conjugation with the isomorphisms $s : s^{-1}V \longrightarrow V$ and $(s^{-1})^{\otimes n} : V^{\otimes n} \longrightarrow (s^{-1}V)^{\otimes n}$. This sequence gives a map $s^{-1}V \longrightarrow \Omega_\infty V$, and we then have a unique derivation $b : \Omega_\infty V \longrightarrow \Omega_\infty V$ that restricts to $s^{-1}V \longrightarrow \Omega_\infty V$ on $s^{-1}V$. A straightforward computation shows that $b^2 = 0$ is equivalent to the Stasheff identities, so we have a dga algebra. Observe that since V is positively graded, $\Omega_\infty V$ is non-unital and non-negatively graded. If V has a weight grading, as it happens for Tor_A whenever A is a weight graded algebra, $\Omega_\infty V$ inherits a weight-grading from V.

The ∞-cobar construction allows us to define the category of A_∞-coalgebras, which we denote by Cog_∞, quite painlessly: its objects are the A_∞-coalgebras and the hom-sets are given by

$$\hom_{\mathsf{Cog}_\infty}(C,C') = \hom_{\mathsf{Alg}}(\Omega_\infty C, \Omega_\infty C').$$

Plainly, Cog_∞ is the full subcategory of Alg consisting of dga algebras that are quasi-free, that is, those which are free as graded algebras if we forget about their differential. Since in the category Alg we have defined the notion of homotopy between maps and weak equivalences, the quasi-isomorphisms, these notions are available to us in Cog_∞ by creating them with the functor Ω_∞; see [123]. Observe, moreover, that if $F : V \rightsquigarrow W$ is a map between A_∞-coalgebras, it is determined uniquely by a sequence of maps $(f_n : V \longrightarrow W^{\otimes n})_{n\in\mathbb{N}}$ satisfying appropriate commutativity conditions with the coproducts of V and W. In view of this, we will identify such a map F with the sequence $f = (f_n)_{n\in\mathbb{N}}$, and write $\Omega_\infty(f)$ for F. Abusing notation a little, for a second sequence $(g_n)_{n\in\mathbb{N}}$, we write fg for the map corresponding to the composition $\Omega_\infty(f)\Omega_\infty(g)$.

Let C be a dga coalgebra, and assume that V is a complex of $\Bbbk$-modules which is a deformation retract of C, that is, there is a homotopy retract datum

$$V \underset{p}{\overset{i}{\rightleftarrows}} C \circlearrowright h\,, \qquad 1 - ip = dh + hd, \quad pi = 1,$$

which we denote by (i,p,h). We assume that such datum satisfies the side conditions, that is, all three maps h^2, hi and ph are zero, in which case we call it a *contraction*. The following result, which is a simplified form of Theorem 5 in [137], shows how to transfer on V an A_∞-coalgebra structure

from the dga coalgebra structure of C and, further, how to produce from a homotopy retract datum another homotopy datum of A_∞-coalgebras.

Theorem 2.4 (Homotopy Transfer Theorem)**.** *Let (C,Δ_2') be a dga coalgebra and consider a homotopy retract as above. There exists an A_∞-coalgebra structure on V and a homotopy retract datum*

$$\Omega_\infty V \underset{q}{\overset{j}{\rightleftarrows}} \Omega_\infty C \circlearrowright k\,, \quad 1-jq = bk+kb, \quad qj = 1.$$

The A_∞-coalgebra structure on V is given by $\Delta_1 = d_V$ and, for $n \in \mathbb{N}_{\geqslant 2}$, by $\Delta_n = p^{\otimes n}\Delta_n' i$, where for $n \in \mathbb{N}_{\geqslant 3}$ the arrows $\Delta_n' : C \longrightarrow C^{\otimes n}$ are defined by

$$\Delta_n' = \sum_{\substack{s+t=n \\ s,t>0}} (-1)^{s(t+1)} (\Delta_s' h \otimes \Delta_t' h)\Delta_2',$$

with the convention that $\Delta_1' h = 1$. □

There is a non-inductive definition of the maps $(\Delta_n)_{n\in\mathbb{N}}$ that will be useful to have in mind when we discuss A_∞-coalgebra structures on Tor_A, which can also be found in [137, Section 4]. Let T be a planar binary tree with n leaves, and let us assign to it a sign $\vartheta(T)$ as follows. For each vertex v of T, let r_1 be the number of paths from a leaf of T to the root that pass through the first (left) input of v, and let r_2 be the number of those that pass through the second (right). Set $\vartheta_T(v) = r_1(r_2+1)$ and $\vartheta(T) = \sum_{v\in T} \vartheta_T(v)$. It will be important later on to observe that if T is the right comb with n leaves then $\vartheta(T) = \binom{n+1}{2} - 1$. Let us write Δ_T for the cooperation of arity n obtained by decorating the leaves of T by p, the root of T by i, the inner vertices by Δ_2' and the inner edges by h; see Figure 1 in [137, Section 4]. We then have the following result.

Theorem 2.5. *Let $n \in \mathbb{N}$. Then Δ_n is given by the sum $\sum_T (-1)^{\vartheta(T)}\Delta_T$ as T ranges through all planar binary trees with n leaves.* □

2.10 Hochschild (co)homology. In this section we quickly recall the classical definitions of Hochschild homology and cohomology, and cyclic homology of algebras. For details, the reader is referred to [126] and [139].

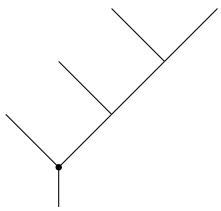


Figure 2.1: A right comb

Let us begin by recalling what the Hochschild (co)chain complex of a dga algebra (B,∂_B) looks like. For details, we refer the reader to the book [139, Section 7.1] and also to [1, Sections 1 and 2]. To this end form the first quadrant double complex $C_*(B)_*$ so that for each $p,q\in\mathbb{N}_0$ we have

$$C_p(B)_q = (B\otimes(s\overline{B})^{\otimes p})_{p+q},$$

the space of elements of $B\otimes(s\overline{B})^{\otimes p}$ of total homological degree $p+q$. Observe that since we have shifted the degree of $\overline{B}$, a generic element of bidegree (p,q) is of the form $b_0[b_1|\cdots|b_p]$ where $\sum_{i=0}^{p}|b_i| = q$.

The horizontal differential is the Hochschild differential that incorporates the Koszul signs of B, so that for a generic basis element $b_0[b_1|\cdots|b_p]$ of our double complex we have that

$$d_h(b_0[b_1|\cdots|b_p]) = -\overline{b_0}b_1[b_2|\cdots|b_p] + \sum_{i=1}^{p-1}\overline{b_0}[\overline{b_1}|\cdots|\overline{b_i}b_{i+1}|\cdots|b_p] + (-1)^{a|b_p|+|b_0|}b_pb_0[\overline{b_1}|\cdots|\overline{b_{p-1}}].$$

where we use the usual notation $\overline{b} = (-1)^{|b|+1}b$, and $a = |b_0|+|b_1|+\cdots+|b_{p-1}|+p-1$. The vertical differential is simply the induced differential on the tensor product of complexes $B\otimes(s\overline{B})^{\otimes p}$, which incorporates Koszul signs and suspension signs. Concretely, for a generic basis element $b_0[b_1|\cdots|b_p]$ we have that

$$d_v(b_0[b_1|\cdots|b_p]) = db_0[b_1|\cdots|b_p] - \sum_{i=1}^{p-1}\overline{b_0}[\overline{b_1}|\cdots|\overline{b_i}|db_{i+1}|\cdots|b_p]$$

Dually, we can form the double complex $C^*(B)^*$ so that for each $p,q\in\mathbb{N}_0$ we have

$$C^p(B)^q = \hom((s\overline{B})^{\otimes p},B)^{p+q},$$

where the outer index indicates the cohomological degree of a map. In this way, elements in $C^p(B)^q$ correspond to maps $\overline{B}^{\otimes p}\longrightarrow B$ that lower the homological degree by q. Observe this complex lives in the first two quadrants, since q can be negative. For a homogeneous map $f:(s\overline{B})^{\otimes p}\longrightarrow B$ and a basis element $[b_1|\cdots|b_p]$ of $(s\overline{B})^{\otimes p}$, we have

$$(d_hf)[b_1|\cdots|b_p] = (-1)^{|b_1||f|}b_1f[b_1|\cdots|b_p] + (-1)^{|f|}\sum_{i=1}^{p-1}f[\overline{b_1}|\cdots|\overline{b_i}b_{i+1}|\cdots|b_p] + (-1)^{|f|}f[\overline{b_1}|\cdots|\overline{b_{p-2}}]b_{p-1}.$$

2.11 Differential graded associative algebras. A sufficiently general recipe to define model categories is to consider simplicial algebras of the given type. However, in the case of associative algebras, it turns out to be possible to do homotopy theory in a slightly more hands-on way, using

differential graded algebras. These subsections give a recollection of the corresponding definitions to make Chapter 6 more readable to experts in Gröbner bases and rewriting.

Definition 7 (Differential graded associative algebra)**.** A *differential graded associative algebra*, is a pair (A,d) where A is a graded associative algebra, that is, a graded vector space equipped with the associative degree zero map $A\otimes A\to A$, and $d\colon A\to A$ is a degree -1 map satisfying the condition $d^2=0$ as well as the graded derivation property $d(a_1a_2)=d(a_1)a_2+(-1)^{|a_1|}a_1d(a_2)$.

We denote by DGA the category whose objects are differential graded associative algebras and whose morphisms from (A,d_A) to (B,d_B) are (degree zero) algebra homomorphisms $f\colon A\to B$ that are chain maps, that is satisfy $f\circ d_A=d_B\circ f$. Such a morphism f induces a morphism $f_\bullet\colon H_\bullet(A,d_A)\to H_\bullet(B,d_B)$ of graded associative algebras obtained by computing homology. We say that f is a *quasi-isomorphism* if the induced homology morphism $f_\bullet$ is an isomorphism.

2.12 Models of algebras. As indicated above, our general plan is to replace an associative algebra with a better behaved differential graded algebra of the same homotopy type. A mathematically precise meaning of the words "better behaved" is given by the notion of a cofibrant object in a closed model category. A motivated reader is invited to consult [94, 153] for foundations of the general theory of model categories. According to Hinich [91, Th. 4.1.1], the category DGA admits a closed model category structure for which weak equivalences are quasi-isomorphisms, and the fibrations are surjections. In fact, Hinich works in the general framework of algebras over operads, and in the only case relevant for this chapter, that of differential graded algebras concentrated in non-negative homological degrees, the same result is established by Jardine [101]. To describe cofibrant objects, we need one more definition.

Definition 8 (Triangulated quasi-free algebra)**.** A *quasi-free algebra* is a differential graded algebra (B,d_B) for which the underlying graded associative algebra is free, that is $B\cong T(V)$ for some graded vector space V. Such algebra is said to be *triangulated* if its space of generators admits an extra decomposition into a direct sum (of graded vector spaces)

$$V=V^{(1)}\oplus V^{(2)}\oplus V^{(3)}\oplus\cdots,$$

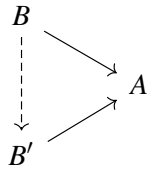
for which the differential of an element $v\in V^{(j+1)}$ belongs to the subalgebra generated by the subspace $\bigoplus_{1\leq k\leq j}V^{(k)}$ for all $j\geq 0$.

The cofibrant objects for Hinich's model category structure on DGA are exactly the retracts of triangulated quasi-free algebras. We remark that it is useful to think that triangulated algebras are an abstraction of the construction of "killing cycles" of Tate [103, 173]. The relevance of triangulated differential graded *commutative* algebras in homotopy theory became apparent after the seminal paper of Sullivan [168]; for this reason, they are frequently called Sullivan algebras.

Definition 9 (Model of an algebra)**.** A *model* of a differential graded algebra (A,d_A) is a triangulated quasi-free algebra $(T(V),d)$ equipped with a surjective quasi-isomorphism f to (A,d_A). Such

a model is said to be *minimal* if its differential is *decomposable*, that is for each $v \in V$, the element dv is a combination of products of length at least two in the tensor algebra.

While minimal models of algebras are unique up to isomorphism, their existence is a much more subtle matter; see, for example, the unpublished note of Keller [107]. However, the target algebras (A, d_A) of interest for us are in fact non-graded and non-differential, in other words, $A_n = 0$ for $n \neq 0$ and $d_A = 0$. Since we work over a field, minimal models for such algebras exist under very mild assumptions (see, for example, [15, 96]) which hold in all cases that are of interest to us.
There is a model structure on Alg whose weak equivalences are quasi-isomorphisms, the fibrations are the degree-wise surjections, and the cofibrant algebras are the retracts of triangulated quasi-free algebras; see [176]. In particular, minimal algebras are cofibrant, and may be used to cofibrantly resolve objects in Alg.
Minimal models of algebras, when they exist, are unique up to unique isomorphism, meaning that any solid diagram where the diagonal arrows are minimal models, can be uniquely completed to a homotopy commutative triangle where the vertical dashed map is an isomorphism in Alg. For brevity, we will use the term *model* to speak about triangulated quasi-free algebras with homology concentrated in degree zero.



2.13 Elements of operad theory. We now briefly recall here the elements from the theory of (colored) operads, their algebras, and Koszul duality theory which allows us to give explicit models of algebras 'up to homotopy' of certain well behaved operads. With this at hand, we can explain in Section 4.2.2 how to produce a homotopy coherent calculus on nc forms and nc fields obtained from any cofibrant replacement of an algebra. We also prove the seemingly ignored result that the operad Calc controlling Tamarkin–Tsygan calculi is inhomogeneous Koszul.
Let us write ${}_\Sigma\mathsf{Mod}$ for the category of Σ-modules, that is, collections of graded $\Bbbk$-modules $X = \{X(n)\}_{n\geqslant 0}$ where for each $n \in \mathbb{N}_0$ the space $X(n)$ is a graded S_n-module. There is a monoidal product on ${}_\Sigma\mathsf{Mod}$ that corresponds to the composition of formal series, so that for two Σ-modules X and Y with $Y(\varnothing) = 0$ we have, for each finite set I

$$(X \circ Y)(n) = \bigoplus_{k\geqslant 0} X(k) \otimes_{S_k} Y[k,n]$$

where for each $n \in \mathbb{N}$ and each $k \leqslant n$,

$$Y[k,n] := \bigoplus_{\lambda_1+\cdots+\lambda_k=n} {}^{S_n}_{S_\lambda}\!\uparrow Y(\lambda_1) \otimes \cdots \otimes Y(\lambda_k)$$

is the sum of induced representations from the subgroup $S_\lambda = S_{\lambda_1} \times \cdots \times S_{\lambda_k} \subseteq S_n$. Note it is a both an S_n-module and an S_k-module (by permutation of the k factors) in a compatible way.
We like to think of an element in $(X \circ Y)(n)$ as an n-corolla —a rooted tree with a single vertex of degree $n+1$— with its root decorated with an element of $X(n)$ and its leaves decorated with

elements of Y. This makes ${}_\Sigma\mathsf{Mod}$ into a monoidal category with unit the symmetric sequence $(0,\Bbbk,0,\ldots)$. Considering the particular case when $Y = Y(1) = V$ is a vector space, we obtain an endofunctor $X : \mathsf{Vect}_\Bbbk \longrightarrow \mathsf{Vect}_\Bbbk$ such that

$$X(V) = \bigoplus_{k\geqslant 0} X(k) \otimes_{S_k} V^{\otimes k}.$$

A symmetric operad is by definition an associative algebra in the monoidal category $({}_\Sigma\mathsf{Mod}, \circ, 1)$. It is customary to write $\gamma : P \circ P \longrightarrow P$ for the product of P which is instead called the composition map of P. In this way, to every element $(\mu;\nu_1,\ldots,\nu_t) \in P \circ P$ we assign an output $\gamma(\mu;\nu_1,\ldots,\nu_t)$ which we think of as grafting $\nu_1,\ldots,\nu_t$ at the leaves of μ to obtain a new operation in P. In particular, one can consider the case when all but one of the operations grafted at the leaves are the unit $1 \in P(1)$. This gives us for each partial composition operations

$$\circ_i : P(n) \otimes P(m) \longrightarrow P(m+n-1)$$

which are sufficient to describe γ: any element $\gamma(a;b_1,\ldots,b_t)$ can be obtained by iterated partial compositions. In this language, associativity of γ is equivalent to the parallel and sequential composition axioms: for operations a, b and $c \in P$, we have that,

$$(a \circ_j b) \circ_i c = \begin{cases} (-1)^{|a||b|}(a \circ_i c) \circ_{j+\mathrm{ar}(c)-1} b & \text{for } i-j<0 \\ a \circ_j (b \circ_{i-j+1} c) & \text{for } i-j \in [0,\mathrm{ar}(b)) \\ (-1)^{|a||b|}(a \circ_{i-j+1} c) \circ_j b & \text{for } i-j \geqslant m \end{cases}$$

We will often make use of the partial composition description of an operad to define the operads of interest to us.

To every operad P we can assign its category of algebras. These are vector spaces V endowed with a linear map $\gamma_V : P(V) \longrightarrow V$ that is compatible with the composition law of P in the sense that for $v_1 \otimes \cdots \otimes v_t \in V^{\otimes n_1} \otimes \cdots \otimes V^{\otimes n_t}$

$$\gamma_V(\gamma(a,b_1,\ldots,b_t),v_1,\ldots,v_t)) = \gamma_V(a,\gamma_V(b_1,v_1),\ldots,\gamma_V(b_t,v_t)),$$

and such that for each $\sigma \in S_n$ we have that

$$\gamma_V(a\cdot\sigma,v_1,\ldots,v_n) = \gamma_V(a,v_{\sigma 1},\ldots,v_{\sigma n}).$$

We like to think of $\gamma_V(a;v_1,\ldots,v_n)$ as the result of applying operation a to the string $v_1 \otimes \cdots \otimes v_n \in V^{\otimes n}$. We also require that the unit of P acts as the identity endomorphism of the space V.

Alternatively, one can consider the endomorphism operad End_V of a vector space V so that for each $n \in \mathbb{N}$ we have

$$\mathrm{End}_V(n) = \mathrm{End}(V^{\otimes n}, V)$$

and the operadic composition is given by

$$\gamma(f; g_1, \ldots, g_n) = f \circ (g_1 \otimes \cdots g_t)$$

while the symmetric group acts by permuting the variables of $V^{\otimes n}$. With this at hand, a P-algebra structure on V is given by a map of operads $P \longrightarrow \mathrm{End}_V$. In particular, we have that

$$\hom_{\Bbbk}(P(V), V) = \hom_{\mathsf{Op}}(P, \mathrm{End}_V).$$

A useful observation, that allows us to define operads through generators and relations, is that operads can be defined as algebras over a monad

$$\mathscr{T} : {}_{\Sigma}\mathsf{Mod} \longrightarrow {}_{\Sigma}\mathsf{Mod},$$

which we call the monad of trees. Concretely, for each Σ-module X, we define a new Σ-module $\mathscr{T}(X)$ so that for each $n \in \mathbb{N}$ we have that $\mathscr{T}(X)(n)$ is spanned by isomorphism classes of trees with n leaves with each vertex v decorated by en element in $X(k)$, where k is the number of inputs of v. The multiplication map

$$\mu : \mathscr{T} \circ \mathscr{T} \longrightarrow \mathscr{T}$$

is obtained by erasing the boundaries of vertices in a 'tree of trees'.

In this way, an operad is just an associative algebra over $\mathscr{T}$ and, in particular, $\mathscr{T}(X)$ is the free operad on X: $\mathscr{T}$ is the left adjoint to the forgetful functor from the category of operads to the category of ${}_{\Sigma}\mathsf{Mod}$. In this way, it makes sense to say that an operad P is generated by X: this means there is a surjective map of operads

$$\mathscr{T}(X) \longrightarrow P.$$

One then defines the notion of a double sided ideal, as usual, and hence it makes sense to say that an operad is generated by X subject to some Σ-module of relations R, as in the case of associative algebras.

Since it is useful for our purposes, let us consider the example where $X = X(2)$ consists of the trivial representation of the symmetric group S_2, with a generator m, so that $\mathscr{T}(X)(2)$ consists of a single binary tree which we identify with a commutative operation x_1x_2. The relation $m \circ_1 m = m \circ_2 m$, which we can write more familiarly as

$$x_1(x_2x_3) = (x_1x_2)x_3,$$

recovers the usual associativity of commutative algebras. We call $T(X)/(R)$ the operad of commutative algebras and write it Com. Note that for each $n \in \mathbb{N}$ the space $\mathsf{Com}(n)$ is the one dimensional trivial representation of S_n, and it is spanned by an operation which we can unambiguously write $x_1 \cdots x_n$. Similarly, we can consider the case $X(2)$ is the regular representation of S_2, in which case

following the same prescription we obtain the operad Ass governing associative algebras. In this case, for each $n \in \mathbb{N}$ the space $\mathsf{Ass}(n)$ is $n!$-dimensional and it is spanned by the operation $x_1 \cdots x_n$ along with all possible permutations of its variables.

We can similarly define the operad Lie governing Lie algebras: it is generated by a single binary operation b spanning the sign representation of S_2, subject to the Jacobi identity

$$(1+\tau+\tau^2)(b \circ_1 b) = 0$$

where $\tau = (123) \in S_3$. We can again write this more familiarly as

$$[[x_1,x_2],x_3]+[[x_3,x_1],x_2]+[[x_2,x_3],x_1] = 0.$$

For each $n \in \mathbb{N}$, the space $\mathsf{Lie}(n)$ has a basis consisting of $(n-1)!$ Lie words $[x_{\sigma 1}, x_{\sigma 2}, \ldots, x_{\sigma n}]$ where $\sigma \in S_n$ fixes 1, where we are using the convention that we bracket a Lie word to the left. Thus, for example, $[x_1,x_2,x_3] = [[x_1,x_2],x_3]$. We can use these two operads to define the operad Ger of Gerstenhaber algebras. It generated by an associative commutative product $m = x_1x_2$ in degree zero and a Lie bracket $b = [x_1,x_2]$ in degree one subject to the Leibniz rule $b \circ_1 m - m \circ_2 b - \sigma(m \circ_2 b) = 0$, where $\sigma = (12) \in S_3$, which we can again write in the familiar form $[x_1x_2,x_3] = x_1[x_2,x_3]+x_2[x_1,x_3]$.

We remark that both in the Jacobi relation and in the Leibniz rule are written without signs since they are relations in the operad Lie and Ger respectively and, since x_1 and x_2 are merely placeholders, they carry no degree. When these are evaluated at elements of a fixed algebra V, the Koszul sign rule will create the appropriate signs. For example, in our last equation, x_1 and x_2 are permuted in the last summand, so that when evaluating this on some $v_1 \otimes v_2 \otimes v_3$ in $V^{\otimes 3}$, we will create a sign in the step where we permute v_1 and v_2 and then apply $m \circ_2 b$:

$$v_1 \otimes v_2 \otimes v_3 \longmapsto (-1)^{|v_1||v_2|} v_2 \otimes v_1 \otimes v_3.$$

Given an algebra V over an operad P, let us consider the subspace $P(V,M) \subseteq P(V \oplus M)$ of elements of the form

$$(\mu; v_1, \ldots, v_{i-1}, m, v_{i+1}, \ldots, v_t), \quad \text{where } \mu \in P,\ v_j \in V \text{ and } m \in M.$$

It is useful to think of this as a corolla with its root decorated by an element of P, and its leaves decorated by elements of V except one, which is decorated by an element of M, which results on in an operation $\mu : V^{\otimes(i-1)} \otimes M \otimes V^{\otimes(t-i)} \longrightarrow M$. Note that $(P \circ P)(V,M)$ is naturally isomorphic to $P(P(V),P(V,M))$. A left V-module is a vector space M along with a structure map

$$\gamma_M : P(V,M) \longrightarrow M$$

compatible with the algebra structure of V, in the sense that, using the previous identification $\gamma_M \circ P(\gamma_V, \gamma_M) = \gamma_M \circ \gamma$.

To get acquainted with this notion of operadic module, it is useful to note that for V an associative algebra, a V-module is the same as a V-bimodule in the usual sense, the left and right actions given by the operations

$$mx = \gamma_M(\mu; m, x), \quad xm = \gamma_M(\mu; x, m).$$

In case V is a commutative algebra, a V-module is the same as a symmetric V-bimodule since in this case the symmetry of μ gives $xm = mx$, and for a Lie algebra V, we recover the usual definition of a representation of a Lie algebra.

Chapter 3

Models for monomial algebras

The chapter is organised as follows. In Section 1 we use the results of [102] to produce a homotopy retract datum from the bar construction of A to $\mathrm{Tor}_A(\Bbbk,\Bbbk)$ and therefore a minimal A_∞-coalgebra structure on $\mathrm{Tor}_A(\Bbbk,\Bbbk)$, which we describe explicitly in Section 2 in terms of decompositions of Anick chains into concatenations of smaller chains, and we note that our results generalize directly to the case of quiver algebras defined by monomial relations. In Section 3 we discuss various special features of the bar (co)homology of monomial algebras and the related higher structures. In particular, we establish some vanishing patterns for higher structures of Yoneda algebras that are at the heart of some of the main results of Chapter 5. We also present a new perspective on some classical results, including the bimodule resolution of Bardzell, and a previously unpublished example of a monomial algebra with highly nontrivial higher structure in its Yoneda algebra. We close the chapter with Section 4 containing some remarks concerning algebras with a chosen Gröbner basis and a conjecture regarding the construction of their minimal models.

1 The A_∞-coalgebra structure on Tor

1.1 The homotopy retract. Let A be an algebra with a reduced Gröbner as in the construction of Jöllenbeck–Welker. Using the contraction

$$\mathrm{Tor}_A \otimes A \underset{p}{\overset{i}{\rightleftarrows}} B(A,A) \circlearrowright h\,, \quad 1-ip = dh+hd, \quad pi = 1,$$

obtained from Subsection 2.2.7 and from the Morse matching for $B(A,A)$ described in Subsection 2.2.8 we obtain, upon tensoring to the right with $\Bbbk$, a contraction

$$\mathrm{Tor}_A \underset{p}{\overset{i}{\rightleftarrows}} BA \circlearrowright h\,, \quad 1-ip = dh+hd, \quad pi = 1,$$

from the dga coalgebra BA to its homology, Tor_A. This and the Homotopy Transfer Theorem 2.2.4 provide us with a minimal A_∞-coalgebra structure on Tor_A, and which we will describe in detail by means of the combinatorics of Anick chains. It is worthwhile to note that one may obtain this retract directly, by applying the methods of [102] to the bar construction BA.

We recall that by construction hi is zero, that is, the homotopy h vanishes on Tor_A. Suppose now that $\gamma = [x_{i_1}|u_1|\cdots|u_r]$ is a bar term representing an Anick chain in Tor_A. We then have that $\Delta'_2(\gamma) = \sum \gamma_{(i)} \otimes \gamma^{(i)}$ where each left term $\gamma_{(i)}$ is also a chain: $\gamma_{(i)}$ is the unique i-chain obtained from γ by removing a right divisor. Since Δ_n is obtained by projecting the map $\Delta'_n : BA \longrightarrow BA^{\otimes n}$, defined recursively in Theorem 2.2.4, we obtain the following.

Proposition 1.1. *For $n \geqslant 3$ we have that $\Delta'_n = (-1)^n(1\otimes\Delta'_{n-1}h)\Delta'_2$ on Tor_A.*

Proof. Let γ be a chain. Looking at the recursive definition of higher coproducts given by Theorem 2.2.4, any term which contains $\Delta'_s h$ on the left for some $s \geqslant 2$ will act by zero on $\Delta'_2(\gamma)$, since the all the terms to the left of the tensor appearing in this sum are also chains, and we already know $hi = 0$. □

1.2 Description of the homotopy. In this section, we focus on the case A is monomial. From the last proposition of the previous section, it follows in particular that $\Delta'_3 = -(1\otimes\Delta'_2 h)\Delta'_2$ on Tor_A, so the only tree that appears in Δ_3 is the right comb. Given $n \in \mathbb{N}_{\geqslant 3}$, we would like to show this is the case for the higher coproduct

$$\Delta_n : \mathrm{Tor}_A \longrightarrow \mathrm{Tor}_A^{\otimes n}.$$

defined by $p^{\otimes n}\Delta'_n i$. We will need an explicit description of the homotopy h. Because it will be useful later on, we also give a description of the projection p: this is the content of the following lemma. Let us say a bar term $[u_0|\cdots|u_r]$ is *attached* if for $i \in \{1,\ldots,r\}$, we have $u_{i-1}u_i = 0$.

Suppose that $\gamma = [u_0|\cdots|u_r]$ is attached but is not a chain. Then there is a largest i_1 such that $u_i = u'_i u''_i$ and such that $\eta^1 = [u_0|\cdots|u'_i]$ is a chain. Remark that by this we mean the *bar structure* is also the correct one; for example, $[t|t^2]$ and $[t^2|t]$ both have underlying monomial the chain t^3 in $\Bbbk[t]/(t^3)$ but only the first is a 1-chain. It may happen that $i = 0$, in which case u'_0 is simply the first variable in u_0, as it does for $[t^2|t]$. We define

$$\gamma^1 = (-1)^{i_1+1}[\eta^1|u''_{i_1}|u_{i_1+1}|\cdots|u_r], \quad \Gamma^1 = [\eta^1|u''_{i_1}u_{i_1+1}|\cdots|u_r].$$

If Γ^1 is a chain or zero, stop. Else, there is some largest $i_2 > i_1$ such that, keeping in with the notation above, $\eta^2 = [u_0|\cdots|u'_{i_1}|\cdots|u''_{i_2}]$ is a chain. In which case, set

$$\gamma^2 = (-1)^{i_2+1}[\eta^2|u''_{i_2}|u_{i_2+1}|\cdots|u_r], \quad \Gamma^2 = [\eta^2|u''_{i_2}u_{i_2+1}|\cdots|u_r].$$

Continuing in this way, we obtain terms $\gamma = \Gamma^0, \cdots, \Gamma^n$ and $\gamma^1, \cdots, \gamma^n$, where Γ^n is either zero or a chain. For convenience, we will agree that $\gamma^m = 0$ for $m > n$, and note that the sign accompanying γ^a is $(-1)^{i_a+1}$, where i_a is simply the length of the largest chain η^a contained in γ^a, starting from the left. If γ is a bar term in degree $r+1$ whose underlying monomial is an r-chain, we will write Γ for the r-chain obtained from γ at the end of the algorithm above, which we observe has no signs. Observe that by construction, the sequence $(i_a)_{a\geqslant 1}$ is strictly increasing, until it stabilizes.

Lemma 1.2. *With the notation above, we have that*

$$h(\gamma) = \sum_{i=1}^{n} \gamma^i, \quad p(\gamma) = \Gamma.$$

Proof. Suppose that $\gamma = [u_0|\cdots|u_r]$ is attached. The description of the Morse graph in Lemma 2.2.2 shows that there is a unique inverted edge from γ to the element γ^1 in the previous paragraph. The face maps of γ^1 are all zero or γ except possibly for Γ^1, up to sign. If Γ^1 is critical, there is no inverted edge leaving Γ^1, and so h is what we claim. Else, we can repeat the argument above. The claim for p follows, since there is a unique path from γ to Γ in the Morse graph, which is obtained by following the terms $\gamma^1, \Gamma^1, \gamma^2, \Gamma^2, \ldots$ that we described above. Finally, the signs can be read off from the definition of the Morse graph and the differential of the bar construction. Indeed, the signs in the bar construction are such that when a bar is removed and two terms are multiplied, there is a sign $(-1)^N$ where N is the number of terms preceding the first factor that was multiplied, and the Morse graph inverts and negates every sign, effectively changing every -1 to a 1, and vice-versa. A close look shows that this are precisely the signs we have incorporated in our description of the elements $\gamma^1, \gamma^2, \ldots$ and $\Gamma^1, \Gamma^2, \ldots$, which completes the proof the Lemma. □

It is useful to observe that uniqueness follows precisely because our algebra is monomial, and hence one can either "undo" a differential in the Morse graph in a unique way, or fail to do so. If we had more complex rewriting rules, it would be a priori possible to obtain a term in multiple ways from reduced monomials, causing our paths to branch and making the description of the action of h and p above much more complicated. In the language of the Morse graph of M, we have the following corollary.

Corollary 1.3. *Let c be a vertex in G^M of degree t that is not critical. There is a unique element c' of degree $t+1$ and a unique element c'' of degree t, which is either zero or critical, a unique path in G^M from c to c' and, if c'' is nonzero, a unique edge from c' to c''. Thus, the coefficients in the homotopy of Theorem 2.2.1 are all 1 or -1 and $p(c)$ coincides with c''.*

Proof. The proof of Lemma 1.2 shows there is a unique path to follow when computing the action of the homotopy h on a non-critical vertex, and is given by the successive terms $\gamma^1, \gamma^2, \ldots$ and $\Gamma^1, \Gamma^2, \ldots$ that we wrote down explicitly before its proof. The conclusion that the coefficients in the homotopy are all -1 or 1 follow also from the careful description of the terms above, since we gave the signs explicitly. The unique element c'' corresponds to Γ, while the unique element c' corresponds to the last non-zero term in the sequence $(\gamma^1, \gamma^2, \ldots)$. □

To illustrate, let us consider a third algebra $L = T(t)/(t^N)$ where $N > 2$. We then have

$$h[\overbrace{t^{N-1}|t}^{m}] = -\sum_{i=0}^{m} [\overbrace{t|t^{N-1}}^{i}|t|t^{N-2}|\overbrace{t|t^{N-1}}^{m-i-1}|t]$$

where the brackets mean the terms are repeated the indicated amount of times. Note that, since in every summand the homotopy extracted a chain of odd homological degree, all the signs are the same. Using the results of Section 2 the reader may recover the A_∞-coalgebra structure on Tor_A for p-Koszul monomial algebras, dual to the A_∞-algebra structure on Ext_A obtained in [86].

1.3 The exchange rule and the right comb. We now prove the desired result that when computing the higher coproducts in Tor_A obtained from the homotopy retraction datum of Subsection 2.2.8, the only contributing tree is the right comb. The following exchange rule for h and Δ'_2 will easily imply this result.

Lemma 1.4. *If γ is attached then $\Delta'_2(h(\gamma)) = (h \otimes 1)\Delta'_2(\gamma)$ modulo $\mathrm{Tor}_A \otimes BA$.*

Proof. This is a direct computation, albeit a bit cumbersome. We use the notation of Lemma 1.2. Let $\gamma = [u_0|\cdots|u_r]$ and write $\Delta'_2(\gamma) = \sum_{i=1}^{r} \gamma_{(i)} \otimes \gamma^{(i)}$. From the definitions it follows that if $j \in \{1, \ldots, r\}$ then:

(1) $(\gamma_{(j)})^a = 0$ if $j < i_a$.

(2) $(\gamma^a)_{(j)}$ is a chain for $j \leqslant i_a + 1$.

(3) $(\gamma^a)_{(j)} = (\gamma_{(j-1)})^a$ for $j \geqslant i_a + 1$.

(4) $(\gamma^a)^{(j)} = \gamma^{(j-1)}$ for $j > i_a + 1$.

That (1) holds follows, for if the chain we want to extract from γ appears after the jth bar, then $\gamma_{(j)}$ will be too short to contain it. It is clear that (2) holds, since γ^a has been "straightened" to chain structure up to the i_ath bar, and we already observed any initial bar term of a chain is again a chain. Note that (3) says that we can either straighten γ to a chain up to step a and then truncate far from where this chain ends, or we can truncate γ at worst at the boundary and then straighten it to a chain: the result is the same. Finally, (4) says that the tail of γ is not affected if cut beyond the

straightening bar. This means that we can write

$$\begin{aligned}\Delta_2'(h(\gamma)) &= \sum_{a\geqslant 1}\sum_{j\leqslant r+1}(\gamma^a)_{(j)}\otimes(\gamma^a)^{(j)}\\ &= \sum_{a\geqslant 1}\sum_{j\leqslant i_a+1}(\gamma^a)_{(j)}\otimes(\gamma^a)^{(j)} + \sum_{a\geqslant 1}\sum_{i_a<j-1\leqslant r}(\gamma^a)_{(j)}\otimes(\gamma^a)^{(j)}\\ &= \sum_{a\geqslant 1}\sum_{j\leqslant i_a+1}(\gamma^a)_{(j)}\otimes(\gamma^a)^{(j)} + \sum_{a\geqslant 1}\sum_{i_a<j-1\leqslant r}(\gamma_{(j-1)})^a\otimes\gamma^{(j-1)}\\ &= \sum_{a\geqslant 1}\sum_{j\leqslant i_a+1}(\gamma^a)_{(j)}\otimes(\gamma^a)^{(j)} + \sum_{a\geqslant 1}\sum_{i_a\leqslant j\leqslant r}(\gamma_{(j)})^a\otimes\gamma^{(j)}\end{aligned}$$

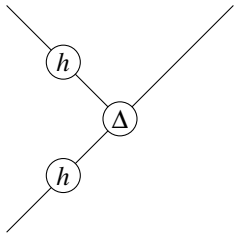
where the third equality uses (iii) and (iv), and from (ii) it follows the first summand is in $\mathrm{Tor}_A\otimes BA$. Finally, from (i) it follows that the second sum is, in fact, $(h\otimes 1)(\Delta_2'(\gamma))$, which completes the proof of the lemma. □

Corollary 1.5. *We have $(h\otimes 1)\Delta_2' h = 0$ on attached bar terms.*

Proof. This now follows from our exchange rule and the fact h has square zero and vanishes on Tor_A. □

Theorem 1.6. *Let $n\in\mathbb{N}_{\geqslant 3}$ and let $\gamma\in\mathrm{Tor}_A$ be an element represented by an Anick chain. The only tree that contributes to $\Delta_n'(\gamma)$, and hence to $\Delta_n(\gamma)$, is the right comb.*

Proof. The fact that h vanishes on Tor_A means that, at the root, the left edge must be a leaf. Knowing this, the exchange rule means that if T is planar and contains any subtree of the form



which corresponds to $(h\otimes 1)(\Delta_2' h)$, the operator Δ_T will vanish identically. This means that the only tree that may possibly give a nonzero contribution to Δ_n is the right comb. □

Let us also record here the following easy proposition, which means, plainly, that the computation of the A_∞-structure of Tor_A depends only on the local information on a given chain. Thus, there seems to be no upshot from looking at induced maps when relations are added.

Proposition 1.7. *Suppose A is a monomial algebra and B is obtained by adjoining to A a non-redundant monomial relation. Let $\varphi: A\to B$ be the quotient map. Then the map $\mathrm{Tor}_\varphi:\mathrm{Tor}_A\longrightarrow\mathrm{Tor}_B$ identifies Tor_A as a sub-A_∞-coalgebra of Tor_B in such a way that the coproducts of Tor_A are the restriction of those of Tor_B through Tor_φ.*

Proof. Since B is obtained from A by adjoining a non-redundant monomial relation, the collection of Anick chains for B can be computed from those of A by adding a (possibly infinite) new collection of chains, and the map Tor_φ is injective since it is induced by the inclusion map on Anick chains. To see that this a strict map of A_∞-coalgebras, meaning that it induces on Tor_A the correct higher coproducts, we note that we can arrange it so that the contraction on the bar complex of B onto Tor_A restricts to a contraction for the bar complex of A: according to the work of Jollenbeck–Welker, this datum can be produced exclusively from the Anick chains from B, and their procedure does not alter the underlying monomial of a chain in the monomial case, and hence restricts to the bar complex of A. Finally, the higher coproducts are built from the coproduct of the bar construction of B and the contraction, and for each Anick chain, this computation depends only on the underlying chain, and not on the inclusion of A into B. □

2 Description of the minimal model

We now aim to give a more refined description of the terms appearing in a higher coproduct of a fixed chain γ, as stated in the following theorem. It will follow immediately from Theorem 2.6 and its proof. Unless stated otherwise, we are working exclusively with monomial algebras in what follows.

Theorem 2.1. *Let γ be a chain and $n \in \mathbb{N}_{\geqslant 2}$. The terms that appear in $\Delta_n(\gamma)$ are exactly those of the form $\gamma_1 \otimes \cdots \otimes \gamma_n$ with $(\gamma_1, \ldots, \gamma_n)$ a decomposition of γ. Moreover, if γ_i is of length r_i for each $i \in \{1, \ldots, n\}$, the coefficient of $\gamma_1 \otimes \cdots \gamma_n$ is $(-1)^N$ where*

$$N = \binom{n+1}{2} + r_1 + \sum_{i=1}^{n-1} (n-i)(r_i+1).$$

2.1 Combinatorics of chains and tails. Suppose that $\gamma = x_{i_1} \cdots x_{i_s}$ is an Anick chain, with associated interlaced sequences $\{(a_j), (b_j)\}$. We will say a variable x_{i_s} is an overlapping variable if $s \in [a_{j+1}, b_j)$, and we will say that a bar is *inserted at* x_{i_s} if it is inserted immediately after it. A bar term obtained from γ is *regular* if it is obtained by inserting bars at non-overlapping variables, and it is *coregular* if it is obtained by inserting bars at overlapping variables. It may happen that $a_{j+1} = b_j$, in which case we agree that $x_{i_{a_{j+1}}}$ is both overlapping and non-overlapping. This always happens, for example, if A is quadratic. The following figure illustrates our definitions for the 4-chain $[t|t^3|t|t^3|t]$ in $\Bbbk\langle t|t^4\rangle$, where white circles represent overlapping variables, black ones represent non-overlapping variables, the cross represents the only variable that is both overlapping and non-overlapping, and bars mark the obstructions that constitute the chain.

Lemma 2.2. *Let γ be a monomial which is an r-chain. Any (co)regular bar term obtained by inserting*

(1) exactly r bars into γ is either attached and nonzero or is zero,



Figure 3.1: Schematics of the chain t^9 in $\Bbbk\langle t|t^4\rangle$.

(2) less than r bars into γ is zero, and
(3) more than r bars into γ is not attached and nonzero or is zero.

Proof. We prove this by induction on r. If $r = 1$, then γ is simply a monomial relation. Certainly inserting no bars gives a bar term of degree one which is zero and, since there are no overlapping variables to keep track of, inserting any bar gives a regular bar term, which is certainly nonzero, and inserting one more bar gives a non-attached term. Assume then $r \geqslant 1$ and that our claim holds for r-chains, and that we have an $(r+1)$-chain. We consider the three cases above separately:

(1) *We have inserted $r+1$ bars regularly:* if the bar term is zero, we are done. Else the bar term obtained in nonzero, and there must be at least one bar inserted in a non-overlapping variable of the last chain. Moreover, there must be exactly one, else, by removing the tail of the $r+1$ chain, we would obtain a regular bar term from an r-chain which is nonzero but has $r-1$ bars, which cannot happen. Having settled this, we now remove the tail and proceed by induction.
(2) *We have inserted less than $r+1$ bars regularly:* if no bar has been inserted on non-overlapping variables of the last monomial relation, we are done. Else, there is one variable inserted there. Removing the tail now gives a regular bar term obtained from an r-chain were less than r bars have been inserted, and induction does the rest.
(3) *We have inserted more than $r+1$ bars regularly:* if two or more bars have been inserted in non-overlapping variables of the last monomial relation, we get a zero term, since removing the tail gives a bar term where at most $r-1$ bars have been inserted regularly into an r-chain. If there is exactly one bar in the tail, we may remove it and proceed inductively.

Analogous considerations apply to coregular terms. □

We now note that the homotopy h, which introduces and shifts bars in bar terms, produces bar terms whose subchains, starting from the left, have bars introduced regularly.

Lemma 2.3. *If γ is an element of $\operatorname{Tor}_A^{r+1}$ corresponding to an r-chain, it has its r bars inserted regularly. In particular, if γ is an attached term, and if γ^a is a nonzero summand in $h(\gamma)$, following the notation of Lemma 1.2, then for $j \leqslant i_a$, the j-chain $(\gamma^a)_{(j+1)}$ has its j bars inserted regularly.*

Proof. The insertion of bars follows Anick's interlaced sequence associated to a chain in such a way that we inserts bars at variables $x_{i_1}, x_{i_{b_1}}, \ldots, x_{i_{b_{r-1}}}$ which are not overlapping, since the overlapping variables are precisely at the half-open intervals $[a_j, b_{j-1})$ for $j \in \{2, \ldots, r-1\}$. □

Let us now introduce the definitions that will be central to our proof of Theorems 2.1 and its equivalent formulation, Theorem 2.6, which we already stated the Introduction. Let γ be an r-chain and

$j \in \mathbb{N}$. We will say a bar term Γ is a *j-tail of γ* if there is a term of the form $\gamma_1 \otimes \cdots \otimes \gamma_j \otimes \Gamma$ in $\Delta'_{j+1}(\gamma)$ appearing with nonzero coefficient, where the first j tensors are chains, and, moreover, Γ is a concatenation of at least two chains $\gamma_{j+1}, \ldots, \gamma_n$, in this order. Moreover, if for $i \in [n]$ we have that γ_i is an r_i chain, we require that $r_1 + \cdots + r_n = r - 1$. The *length* of Γ is $n - j$. Let us call the n-tuple $(\gamma_1, \ldots, \gamma_n)$ a *decomposition* of γ. Remark that there is the notion of "tail" of a chain given in [2], but that this is not a special case of our definition, and that Γ may be a tail for several choices of the tuple $(\gamma_{j+1}, \ldots, \gamma_n)$.
We continue by observing that j-tails are obtained by cutting a chain in the form of a bar term either at a bar or at some place between bars.

Lemma 2.4. *Fix $j \in \mathbb{N}$ and suppose that $\gamma = [u_0|u_1|\cdots|u_r]$ is an r-chain, and that Γ is a j-tail of γ, with first chain γ_{j+1}. Then there exists $i \in \{1, \ldots, r\}$ and a decomposition $u_i = u'_i u''_i$ such that $u''_i \neq 1$, $u''_i u_{i+1} = 0$ minimally and $\Gamma = [u''_i|\cdots|u_r]$. Moreover:*

(1) This decomposition is nontrivial whenever $j > 1$
(2) The tail Γ contains exactly $r_{j+1} + \cdots + r_n$ bars.
(3) There is a unique $(j-1)$-tail Γ' and a unique term in $\Delta'_2 h(\Gamma')$ of the form $\gamma_j \otimes \Gamma$ that gives rise to Γ, and it appears with a sign as a coefficient.

Proof. The case that $j = 1$ and n is arbitrary is obvious, so let us assume $j > 1$, our claim true for $(j-1)$-tails, and analyse the claim for j.
Observe that by Theorem 1.6, if Γ is a j-tail of γ, it must come from a $(j-1)$-tail Γ' of γ by applying the operator $\Delta'_2 h$ on the last factor. We will prove that Γ has the desired form and, moreover, that there is a unique way to obtain Γ from Γ', so that if the term corresponding to Γ' appears with coefficient 1 or -1, then so does the term corresponding to Γ. The description of h from Lemma 1.2 and the inductive hypothesis applied to Γ' means that $\Gamma' = [u''_i|u_{i+1}|\cdots|u_r]$ with $u''_i u_{i+1} = 0$ minimally, or Γ' has no bars. In the latter case the chain γ_{j+2} is a variable and then Γ is obtained by removing this variable: we have that $h(\Gamma') = [\gamma_{j+1} \mid \cdots]$ and we obtain Γ uniquely from Γ'.
Let us then consider the case Γ' has bars, so that it contains $r - 1 - (r_1 + \cdots + r_j)$ bars by induction, and its first terms overlap minimally. We can certainly find some $k > i$ and a decomposition $u_k = u'_k u''_k$ in such a way that the underlying monomial of the bar term $[u''_i|u_{i+1}|\cdots|u_{k-1}|u'_k]$ is precisely γ_j. Observe that the bar structure of γ_j is coregular, for it interlaces with that of γ. By Lemma 2.2 there are exactly r_j bars in such term, so that $k = i + r_j$. We now analyse two cases.
Case 1: u''_i is a variable. In such case, it follows that $[u''_i|u_{i+1}|\cdots|u'_{i+r_j}]$ is a honest chain belonging to Tor_A. We claim that the decomposition $u_k = u'_k u''_k$ is non-trivial, that $u''_k u_{k+1} = 0$ minimally and that $h(\Gamma')$ is, up to signs, equal to the bar term

$$[u''_i|u_{i+1}|\cdots|u'_{i+r_j}|u''_k|u_{k+1}|\cdots|u_r]$$

which means, of course, that the description of Γ is the correct one. Indeed, note that if the decomposition were trivial, we would have a sequence of chains $\gamma_{j+1}\cdots\gamma_n$ underlying a $(j+1)$-tail with

less than $r_{j+1}+\cdots+r_n$ bars. As before, the starting chain γ_{j+1} appears with bars inserted coregularly, so we may remove it along with exactly r_{j+1} bars. Repeating this argument, we end up with a coregular bar term underlying an r_n-chain with less than r_n bars, which contradicts Lemma 2.2. To see that $u_k''u_{k+1}=0$, note that otherwise we again would have a bar term $[u_k''u_{k+1}|\cdots|u_r]$ whose underlying monomial has $r_{j+1}+\cdots+r_n-1$ bars. The fact that the overlap $u_k''u_{k+1}$ is minimal follows from the fact that overlap u_ku_{k+1} is minimal. The description of $h(\Gamma')$ shows that Γ is obtained uniquely from Γ', possibly with a sign.

Case 2: u_i'' is not a variable. Arguing as before, we see that the overlap $u_k=u_k'u_k''$ is not trivial, and that $u_k''u_{k+1}=0$ minimally. We can write $u_i''=xv$ were x is a variable and v a monomial, and we have that $h(\Gamma')$ has first term $[x|v|u_{i+1}|\cdots|u_r]$. Set j_* to be the last $k\geqslant j$ for which γ_k is a 0-chain. Our bar counting argument then shows that the concatenation $\gamma_j\cdots\gamma_{j_*}$ must be contained in the monomial u_i'', and then using our description of h it is clear we may extract the term $\gamma_j\otimes\cdots\otimes\gamma_{j_*}$ uniquely by iteration of $\Delta_2 h$. Let us assume then that γ_j is not a 0-chain. In such case, $vu_{i+1}\neq 0$, since γ_j begins with a minimal monomial relation which, by minimality, must in fact be xvu_{i+1}. Since $u_i''u_{i+1}$ is a minimal monomial relation of A, it follows that $[x|vu_{i+1}|\cdots|u_k']$ begins with the initial 1-chain from γ_j, so that if γ_j is a 1-chain, we are done: this term is of the form $[x|vu_k']$. Else, we can find the initial 2-chain of γ in the form $[x|vu_{i+1}|u_{i+2}']$: since $u_{i+1}u_{i+2}$ is a minimal monomial relation of A, the second monomial relation of γ_j must be contained in a monomial of the form $vu_{i+1}u_{i+1}'$ where u_{i+1}' is a proper initial divisor of u_{i+1}. Continuing this way, we see the Anick structure of γ_j is interlaced inside that of γ, and that the last term in $h(\Gamma')$ is $[x|vu_{i+1}|\cdots|u_k'|u_k''|u_{k+1}|\cdots|u_r]$, proving the description of Γ is the correct one.

We also observe that the summands of $h(\Gamma')$ different from this one cannot create a summand corresponding to Γ so that again Γ is obtained uniquely from Γ'. Indeed, the only way to produce a bar term in the left factor with the same underlying monomial as γ_j, we would have to use Δ_2' to break such a term of $h(\Gamma')$ precisely at the bar dividing u_k' and u_k'', presently only on the last term. If we do it at a bar before or after this one, the resulting term has either its left factor or its right factor non-attached, since it contains $[\cdots|u_k'|u_k''|\cdots]$. This same argument shows that the previous summands of $\Delta_2'h(\Gamma')$ cannot contribute to Δ_{j+1}: the only place where we may break them is at the last opened bar, say $[\cdots|u_t'|u_t''|\cdots]$, but the fact we can continue the algorithm of Subsection 1.2 means that u_t'' has nonzero product with u_{t+1}, and hence this term does not contribute to Δ_{j+1}.

The final claim regarding the number of bars in Γ is immediate from the above. □

The following proposition is the central result about tails and chains we were after.

Proposition 2.5. *Let γ be a chain, $n\in\mathbb{N}_{\geqslant 2}$ and let $(\gamma_1,\ldots,\gamma_n)$ be a decomposition of γ. For each $j\in[n-1]$ there is a unique j-tail Γ of γ with underlying monomial $\gamma_{j+1}\cdots\gamma_n$ and a unique term $\gamma_1\otimes\cdots\otimes\gamma_j\otimes\Gamma$ in $\Delta_{j+1}'(\gamma)$, and it appears with coefficient 1 or -1.*

Proof. Let $(\gamma_1,\ldots,\gamma_n)$ be a decomposition of γ and let Γ be a j-tail as in the statement of the Theorem. The claim is obvious for $j=1$. Moreover, Lemma 2.4 shows that once we know that a $(j-1)$-tail Γ' corresponding to this decomposition of appears in $\Delta_j'(\gamma)$, there is a unique summand

in $\Delta_2' h(\Gamma')$, with coefficient 1 or -1, that produces the term corresponding to Γ, which is what we wanted. □

Remark that the operators $(\Delta_j')_{j\geqslant 2}$ produce other terms than the ones described in the last proposition. However, the proof of Lemma 2.4 shows these terms have zero projection to tensor powers of Tor_A, since they contain factors that are not attached.

2.2 Main theorem. We now recall the promised description of the minimal model of a monomial algebra A. It follows immediately from Proposition 2.5 and Lemma 1.2, which in particular describes the signs appearing in the homotopy h.

Theorem 2.6. *For each monomial algebra A there is a minimal model $B \longrightarrow A$ where $B = \Omega_\infty \mathrm{Tor}_A$, and for a chain $\gamma \in \mathrm{Tor}_A$ the differential d acts by*

$$d\gamma = -\sum_{n\geqslant 2} (-1)^{\binom{n+1}{2}+|\gamma_1|} \gamma_1 \cdots \gamma_n,$$

where the sum ranges through all possible decompositions of γ.

Proof. We need only address the claim about signs and the differential b. We already know that whenever $\Delta_2 h$ extracts an r-chain, it produces a sign $(-1)^{r+1}$. Moreover, whenever h goes through an r-chain γ it produces a sign $(-1)^{r+1}$. Thus when creating the term $\gamma_1 \otimes \cdots \otimes \gamma_n$ by extracting γ_{n-1}, we have a sign $(-1)^L$ where $L = \sum_{i=1}^{n-1}(r_i+1)$. Inductively accounting for the signs created by $\Delta_3, \ldots, \Delta_{n-1}$, for the missing sign r_1+1 that is *not* created by Δ_2 and for the sign given by 2.2.5, we obtain a sign congruent to

$$\binom{n+1}{2} + r_1 + \sum_{i=1}^{n-1} (n-i)(r_i+1) \pmod 2,$$

which is the integer N in Theorem 2.1. To see the claim about the minimal model, we observe that $(s^{-1})^{\otimes n}(\gamma_1 \otimes \cdots \otimes \gamma_n) = (-1)^M s^{-1}\gamma_1 \otimes \cdots \otimes s^{-1}\gamma_n$ where M is $\sum_{i=1}^{n-1}(n-i)(r_i+1)$, giving the final result. □

The canonical identification of $\mathrm{Ext}_A := \mathrm{Ext}_A(\Bbbk, \Bbbk)$ as $\mathrm{Tor}_A^\vee$ gives us a result dual to Theorem 2.1 about the A_∞-algebra structure on Ext_A. Remark that it is quite crucial to have done all the work with A_∞-coalgebras and then dualizing to A_∞-algebras, and not otherwise, since not every A_∞-algebra is dualizable; see [32, §2.2]. It is important, however, to pay attention to the Koszul signs arising from the natural maps $D^n : \mathrm{Ext}_A^{\otimes n} \longrightarrow (\mathrm{Tor}_A^{\otimes n})^\vee$ for $n \in \mathbb{N}$: if $f_1 \otimes \cdots \otimes f_n$ is an element in the domain, and if we pick $c_1 \otimes \cdots \otimes c_n \in \mathrm{Tor}_A^{\otimes n}$, then

$$D^n(f_1 \otimes \cdots \otimes f_n)(c_1 \otimes \cdots \otimes c_n) = (-1)^N f_1(c_1) \otimes \cdots \otimes f_n(c_n)$$

where $N = \sum_{i=2}^n (|c_1| + \cdots + |c_{i-1}|)|f_i|$. Observe that if $f : V \longrightarrow W$ is a map between complexes, then $f^\vee(\varphi) = (-1)^{|f||\varphi|}\varphi f$, which explains the introduction of signs in the higher products of the

graded dual Ext_A of Tor_A. Concretely, for each $n \in \mathbb{N}_{\geqslant 2}$, define $\mu_n : \mathrm{Ext}_A^{\otimes n} \longrightarrow \mathrm{Ext}_A$ by $\mu_n(\varphi_1 \otimes \cdots \otimes \varphi_n) = (-1)^{n(|\varphi_1|+\cdots+|\varphi_n|)} D^n(\varphi_1 \otimes \cdots \otimes \varphi_n)\Delta_n$. Let us say an A_∞-algebra structure on Ext_A is *canonical* if it is A_∞-quasi-isomorphic to the dga algebra $BA^\vee$. We have the following result.

Theorem 2.7. *There is a canonical A_∞-algebra structure on Ext_A given as follows. If $n \in \mathbb{N}_{\geqslant 2}$ and if $\gamma_1^\vee, \ldots, \gamma_n^\vee$ are chains in Ext_A of lengths $r_1, \ldots, r_n$, respectively, then $\mu_n(\gamma_1^\vee \otimes \cdots \otimes \gamma_n^\vee) = (-1)^M \gamma^\vee$ if the concatenation $\gamma = \gamma_1 \cdots \gamma_n$ is a chain of length $r = r_1 + \cdots + r_n + 1$ where M is the integer $\binom{n+1}{2} - 1 + \sum_{i<j} r_i(r_j+1) + r_1 + r$. Otherwise, this higher product is zero.* □

Let us make the model explicit for the monomial algebra K, we will address the algebra J later (but note that its associated monomial algebra has the same Anick chains as J). The dg model for K is a free algebra $T(x,y,z,\alpha,\beta,\Gamma,\Lambda)$ where the first three generators are in homological degree 0, the next two in homological degree 1 and the last two in homological degree 2. The differentials are as follows:

$$d\alpha = xy^2, \quad d\beta = y^2x, \quad d\Gamma = x\beta - \alpha z, \quad d\Lambda = xy\beta - \alpha yz.$$

These can be read off the (unique) 3-decompositions of the relations into a concatenation of variables (0-chains), the the 2-decompositions of xy^2z into $x \cdot y^2z$ and $xy^2 \cdot z$, and the 3-decompositions of xy^3z into $x \cdot y \cdot y^2z$ and $xy^2 \cdot y \cdot z$.

2.3 The extension to monomial quiver algebras. We now observe that the results of these notes extend without any non-trivial modification to the more general class of monomial quiver algebras. Fix a quiver $Q = (Q_0, Q_1, s, t)$ and a set R of paths in Q of length at least two, none of which is a divisor of another. We call $A = \Bbbk Q/(R)$ a *monomial quiver algebra*. Let us write $\underline{\Bbbk}$ for the semi-simple $\Bbbk$-algebra $\Bbbk Q_0$, so that there is an augmentation $A \longrightarrow \underline{\Bbbk}$. We set $\mathrm{Tor}_A = \mathrm{Tor}_A(\underline{\Bbbk}, \underline{\Bbbk})$, and write BA for the bar construction of A, where unadorned $\otimes$ are now taken over $\underline{\Bbbk}$. Thus, a generic basis element of BA in degree $n \in \mathbb{N}$ is of the form $[a_1|\cdots|a_n]$ where $t(a_i) = s(a_{i+1})$ for each $i \in \{1, \ldots, n-1\}$. Since $\underline{\Bbbk}$ is semi-simple over $\Bbbk$, we can consider this alternative bar construction instead.

The methods of Subsection 2.3 go through to produce a homotopy retract datum from BA to Tor_A, and select a basis of Tor_A of critical vertices given by chains: Tor_A^1 has basis $\{[a] : a \in Q_1\}$, and for $n \in \mathbb{N}$ a basis of Tor_A^{n+1} is given by bar terms $[u_0|\cdots|u_n]$ where $t(u_i) = s(u_{i+1})$ and $u_iu_{i+1} = 0$ minimally for each $i \in \{0, \ldots, n-1\}$. The description of the action of the homotopy on fully attached terms is unchanged, as is the exchange rule.

The notion of decompositions of a chain carry through to this setting, as well as the technical work of Section 3. As an end result we obtain the following description of a minimal model for monomial quiver algebras: there is no change on the Morse matching, the action of the homotopy and the exchange rule, or any other detail: the fact that we can do things relative to $\underline{\Bbbk}$ makes any pathology that can arise in $\Bbbk Q$ due to non-concatenable arrows disappear, since elements of the bar construction just look like those in the usual one over $\Bbbk$, but with the extra condition of concatenability. This last condition ensures that spurious cycles, say of the form $[x|y]$, arising from

zero multiplication due x and y not being concatenable, disappear, so everything works like in the case Q is a bouquet. We refer the reader to [41, Lemma 2.1] where it is shown the relative double sided bar construction is a projective resolution of $A = \Bbbk Q/(R)$ as an A-bimodule (note the hypothesis that A be finite dimensional is not really needed there).
Let us remark that we also have, implicitly, obtained comparison maps between the bar resolution $B(A,A)$ of $\underline{\Bbbk}$ and the Green–Happel–Zacharia resolution $\mathrm{Tor}_A \otimes_\tau A$ of $\underline{\Bbbk}$ that are part of a homotopy retract datum; see [73] for details. Naturally, we have a dual result for the Yoneda algebra $\mathrm{Ext}_A(\underline{\Bbbk},\underline{\Bbbk})$ of A, which we also record.

Theorem 2.8. *For each quiver monomial algebra A there is a minimal model $B \longrightarrow A$ where $B = \Omega_\infty \mathrm{Tor}_A$, and for a chain $\gamma \in \mathrm{Tor}_A$ the differential d acts by*

$$d\gamma = -\sum_{n\geqslant 2} (-1)^{\binom{n+1}{2}+|\gamma_1|}\gamma_1\cdots\gamma_n,$$

where the sum ranges through all possible decompositions of γ. □

Theorem 2.9. *There is a canonical A_∞-algebra structure on Ext_A given as follows. If $n \in \mathbb{N}_{\geqslant 2}$ and if $\gamma_1^\vee,\ldots,\gamma_n^\vee$ are chains in Ext_A of respective lengths $(r_1,\ldots,r_n)$, then*

$$\mu_n(\gamma_1^\vee \otimes \cdots \otimes \gamma_n^\vee) = (-1)^M \gamma^\vee$$

if the concatenation $\gamma = \gamma_1\cdots\gamma_n$ is a chain of length $r = r_1+\cdots+r_n+1$ where M is the integer $\binom{n+1}{2} - 1 + \sum_{i<j} r_i(r_j+1) + r_1 + r$. Otherwise, this higher product is zero. □

Let us remark that the theorem above is a common generalisation of some of the results obtained in [72] and in [86], the latter in the case of monomial algebras. In the first the authors describe (among other results) a multiplicative basis of Ext_A for A a monomial quiver algebra given in terms of Anick chains, and show if γ_1 and γ_2 are chains, then $\gamma_1 \smile \gamma_2$ is zero unless the concatenation $\gamma_1\gamma_2$ is a chain, in which case $\gamma_1 \smile \gamma_2 = \gamma_1\gamma_2$. In the second, the authors describe (among other results) the higher products in Ext_A for algebras that are p-Koszul, and show that the chains involved in a product $\mu_p(\gamma_1 \otimes \cdots \otimes \gamma_p)$ are all of odd homological degree. A calculation shows that the only term that contributes to a sign in the integer M of Theorem 2.7 is the binomial coefficient $\binom{p+1}{2}$. Switching to the sign convention for the Stasheff identities used in [86] removes this sign, and then our result coincides with their result exactly in the case of monomial algebras: the higher product of $\gamma_1 \otimes \cdots \otimes \gamma_p \in \mathrm{Ext}_A^{\otimes n}$ is zero unless the chains $\gamma_1,\ldots,\gamma_p$ concatenate, in this order, to a chain γ of the correct homological degree, in which case $\mu_p(\gamma_1 \otimes \cdots \otimes \gamma_p) = \gamma$.

3 Non-vanishing products and vanishing patterns

3.1 Two formulas for Tor groups. In general, for an augmented algebra Λ, the Tor groups $\mathrm{Tor}_\bullet^\Lambda(\Bbbk,\Bbbk)$ of the trivial module can be computed as the homology of the bar construction $B_\bullet(\Lambda) =$

$\{(s\mathfrak{r})^{\otimes n}, d\}$. This formula is useful theoretically but in practice has a lot of limitations: the underlying space of bar construction is too big. In this section, we shall discuss two formulas that produce the Tor groups on the nose; one is valid for any augmented algebra Λ presented by generators and relations, and the other only makes sense for monomial algebras. A few of the results of this chapter arise from comparing those two formulas. To the best of our knowledge, this has not been done explicitly before; the only sources we know where the two formulas appear at the same time are the article [37] by Butler and King, and the survey of Ufnarovski [175].
The first formula for the Tor groups is as follows.

Theorem 3.1. *Let $\Lambda = kQ/I$ with $I \subseteq \mathfrak{m}^2$. For $p \in \mathbb{N}$ we have that*

$$\mathrm{Tor}^{\Lambda}_{2p}(\Bbbk,\Bbbk) \cong \frac{I^p \cap \mathfrak{m} I^{p-1}\mathfrak{m}}{I^p\mathfrak{m} + \mathfrak{m} I^p}, \tag{3.1}$$

$$\mathrm{Tor}^{\Lambda}_{2p+1}(\Bbbk,\Bbbk) \cong \frac{I^p\mathfrak{m} \cap \mathfrak{m} I^p}{I^{p+1} + \mathfrak{m} I^p\mathfrak{m}}. \tag{3.2}$$

Similar formulas were first discovered in the case of integral homology of groups presented by generators and relations by Gruenberg [79], and then established in the case of associative algebras over a field by Govorov [71]. The work of Govorov remained largely unnoticed, and it seems that the first prominent appearance of this result for quiver algebras is in a paper of Bongartz [30] who refers to a private communication from Butler inspired by work of Gruenberg. We shall refer to these formulas for Tor-groups as the Govorov–Gruenberg formulas.
The Govorov–Gruenberg formulas are way too general to be of computational relevance for arbitrary algebras. However, in the case of monomial algebras they can be made completely combinatorial, if we note that all the vector spaces in those formulas have compatible combinatorial bases, e.g. the vector space $\mathfrak{m} I^p$ has a basis of paths obtained as concatenations of a nontrivial path with a path containing at least p disjoint occurrences of the monomial relations. Once those compatible bases are identified, all intersections and quotients are computed on the level of combinatorial bases by taking the intersections and set-theoretic differences respectively.
The second formula for Tor-groups which is only valid for algebras with monomial relations utilises combinatorics of *Anick chains*. We shall use that terminology, even though it is inaccurate historically; in the case of algebras with monomial relations that combinatorics seems to have been first discovered by Green, Happel and Zacharia [73], but is better known from works of Anick [2] and Anick and Green [3] where the case of monomial algebras was upgraded to the case of augmented algebras with a Gröbner basis of relations.
Let us recall another definition of the set C_n of (right) Anick n-chains and of a tail of an Anick chain; done by induction on $n \geq 0$ instead of resorting to the notion of a 'pre-chain'. We let $C_0 = Q_1$ be the set of arrows, and any $a \in C_0$ is its own tail. Suppose that C_{n-1} is defined, as well as tails of $(n-1)$-chains. An n-chain is a path γ in $\mathscr{P}_Q$ such that

(i) we can write $\gamma = \gamma' t$ with $\gamma' \in C_{n-1}$,

(ii) if t' is the tail of γ', then $t't$ has a right divisor which is a monomial relation,

(iii) no proper left divisor of γ satisfies both (i) and (ii).

By definition, the tail of γ is the path $t_\gamma := t$. Note that C_1 is the set of the defining relations of Λ, with the tail of each monomial relation given by the path obtained after removing the leftmost arrow in it. It is shown in [2, 3] that a path γ admits at most one structure of a chain: if γ is an n-chain with tail t_γ, then n and t_γ are uniquely determined. The following result is essentially due to Green, Happel, and Zacharia [73]; although we mentioned it in passing before, we state it here for convenience.

Theorem 3.2. *For each $n \in \mathbb{N}$ we have isomorphisms* $\operatorname{Tor}_n^\Lambda(\Bbbk,\Bbbk) \cong kC_{n-1}$.

To prove this result, one uses the combinatorial definition above to prove a slightly stronger statement: the minimal free right module resolution of $\Bbbk$ over Λ can be proved to have the form

$$\cdots \to kC_n \otimes_\Bbbk \Lambda \xrightarrow{d} kC_{n-1} \otimes_\Bbbk \Lambda \xrightarrow{d} \cdots \xrightarrow{d} kC_1 \otimes_\Bbbk \Lambda \xrightarrow{d} kC_0 \otimes_\Bbbk \Lambda \to \Lambda \to 0 \tag{3.3}$$

with $d(\gamma \otimes 1) = \gamma' \otimes t_\gamma$. (Of course applying the functor $- \otimes_\Lambda \Bbbk$ to this resolution, we immediately see that $\operatorname{Tor}_n^\Lambda(\Bbbk,\Bbbk) \cong kC_{n-1}$, as required.)

Similarly, one can define the set of left Anick chains C'_n by considering *heads* instead of tails, and thus obtain the minimal free left module resolution of $\Bbbk$ over Λ. Bardzell [12] proved that $C'_n = C_n$ for all $n \geq 0$, and so for any n-chain γ, we can write $\gamma = \gamma' t_\gamma = h_\gamma \gamma''$ for unique $(n-1)$-chains γ', γ'', with tail t_γ and head h_γ. Another explanation of this symmetry from the point of view of the Govorov–Gruenberg formulas was given by Butler and King, see [37, Th. 8.2, Remark 8.3].

At this point, it is appropriate to make a remark on how the extra grading mentioned earlier can be utilised. Since the path algebra kQ, as well as its ideals generated by monomials, is graded by the category $\mathfrak{C}(Q)$ generated by Q, that grading descends to all monomial quotients of kQ, and to bar constructions of those quotients, where the boundary map manifestly respects the grading. Thus, every group $\operatorname{Tor}_n^\Lambda(\Bbbk,\Bbbk)$ is graded by $\mathfrak{C}(Q)$. From the description of Tor via Anick chains, it is clear that for every path $p \in \mathscr{P}_Q$, the p-graded component of Tor is of dimension at most one (and when it is of dimension one, the corresponding one-dimensional space has the Anick chain p as its basis). Since the Tor groups (together with their $\mathfrak{C}(Q)$-gradings) are well defined, all the descriptions we outlined (via the bar construction, of Govorov and Gruenberg, and of Green–Happel–Zacharia) must give the same result. The first immediate consequence of that is the result of Bardzell on the coincidence of left and right Anick chains mentioned in the previous paragraph: both of those index nonzero graded components in the homology of the bar construction. Another interesting observation is that the above property of Tor stating that for each path $p \in \mathscr{P}_Q$, the p-graded component of Tor is of dimension at most one is precisely the key result of the paper of Iyudu [99] who suggests that it was conjectured by Kontsevich in 2015.

3.2 Higher structures of Tor and Ext. Recall that an A_∞-algebra over $\Bbbk$ is a graded $\Bbbk$-bimodule E equipped with multilinear operations $m_n\colon E^{\otimes n}\to E$ of degree $2-n$ satisfying the Stasheff identity

$$\sum_{r+s+t=n}(-1)^{rs+t}m_{r+1+t}\circ(\mathrm{id}^{\otimes r}\otimes m_s\otimes\mathrm{id}^{\otimes t})=0 \tag{3.4}$$

for each $n\geq 1$. These identities imply that m_1 is a differential for E and that m_2 is associative up to coboundary given by m_3, and we call $(E,\{m_n\})$ minimal when $m_1=0$. In this case, (E,m_2) is an honest graded augmented $\Bbbk$-algebra with some extra data given by the higher products $\{m_n\}_{n\geq 3}$. We refer the reader to [130, Chapter 9] for further information on A_∞-algebras.

The A_∞-algebra E is said to be strictly unital if it contains an element 1 for which

$$m_2(1,x)=x=m_2(x,1)\text{ and }m_n(\dots,1,\dots)=0\text{ for }n\geq 3.$$

The A_∞-algebra E is augmented over $\Bbbk$ if it is equipped with degree zero morphisms $\varepsilon\colon E\leftrightarrows\Bbbk\colon\eta$ satisfying $\varepsilon\eta=\mathrm{id}$, so that it is strictly unital for $1:=\eta(1)$, and the products preserve the augmentation ideal $\overline{E}:=\ker(\varepsilon)$. We always take E to be augmented.

In particular, the Yoneda algebra $E=\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ of any augmented dg $\Bbbk$-algebra admits a minimal augmented A_∞-algebra structure that is A_∞-equivalent to the dg algebra $\mathrm{RHom}_\Lambda(\Bbbk,\Bbbk)$. Such a structure is unique up to A_∞-isomorphism; it recovers Λ up to an A_∞-quasi-isomorphism. For the rest of this article $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ will always be considered equipped with such an A_∞-structure, which we shall call canonical.

For a monomial algebra Λ, the canonical A_∞-structure on Ext was determined in [171]; this result generalises the classical formula of Green and Zacharia [77] for the Yoneda product. By Theorem 3.2, if we denote by $(-)^\vee$ the $\Bbbk$-dual, we have identifications

$$\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)=(\Bbbk\oplus kC_{*-1})^\vee=\Bbbk\oplus kC^\vee_{*-1},$$

where we set $C_{-1}=\emptyset$ and moreover abuse notation and let $C_r^\vee\subseteq\mathscr{P}_{Q^{op}}$ consist of the paths in C_r with opposite orientation. More generally we identify the sets of paths $\mathscr{P}_Q$ and $\mathscr{P}_{Q^{op}}$ via the notation $p\leftrightarrow p^\vee$ which reverses orientation, so that $(pq)^\vee=q^\vee p^\vee$.

Consider the Yoneda algebra $E=\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)=\Bbbk\oplus kC^\vee_{*-1}$. For each $n\geq 2$, let us define morphisms $m_n\colon E^{\otimes n}\to E$ on (duals of) Anick chains $\gamma_i\in C^\vee_{r_i}$ by

$$m_n(\gamma_1,\dots,\gamma_n)=\begin{cases}(-1)^N\gamma_1\dots\gamma_n & \text{if }\gamma_1\dots\gamma_n\in C^\vee_{r_1+\dots+r_n+1},\\ 0 & \text{else,}\end{cases} \tag{3.5}$$

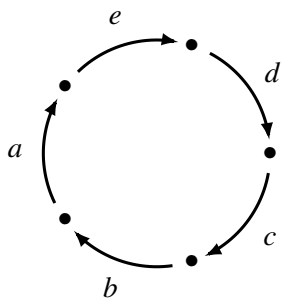
where $N=\sum_{i<j}r_i(r_j+1)+r_1+\sum_i r_i$.

Theorem 3.3. *These multilinear maps model the canonical A_∞-algebra structure on* $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$.

Proof. This is analogous to [171, Theorem 4.2] but uses left Anick chains instead of right ones; one just has to remark that in this case the only non-vanishing tree appearing in the homotopy transfer formula for the higher products is the left comb, and that no extra signs arise, as opposed to what happens for the right comb. □

3.3 Example of non-vanishing higher products. At this point we already have enough information to present an interesting example due to Chuang and King of a very simple monomial quiver algebra which however has highly nontrivial higher structure on its Yoneda algebra. The only available source in the literature where an example of a similar nature appears is a paper of Conner and Goetz [47]; however, in that case the authors show that *a certain* canonical A_∞-structure is highly nontrivial, and do not establish that it holds for *every* canonical A_∞-structure. Hence we believe that this example should be of interest to experts.

Let us consider the cyclic quiver C_5 with the arrows a,b,c,d,e in the cyclic order. The algebra CK_5 is defined by generators and relations as $kC_5/(abcd,bcde,deab)$, which is a finite-dimensional algebra. We will prove that a canonical A_∞-structure on its Yoneda Ext-algebra cannot be "too simple".



Theorem 3.4 (Chuang–King, unpublished)**.** *The Yoneda algebra* $B = \operatorname{Ext}^*_{CK_5}(\Bbbk,\Bbbk)$ *has infinitely many non-vanishing higher products, for any canonical* A_∞*-structure.*

Proof. The Anick 1-chains of the algebra CK_5 are the relations $abcd$, $bcde$, and $deab$, and the Anick 2-chains are $abcde$, $bcdeab$, $deabcd$. Furthermore, it turns out that for each $n \geq 3$ there are just two Anick chains: for $n = 2s-1$, the monomials

$$bcde(abcde)^{s-2}abcd \quad \text{and} \quad de(abcde)^{s-1}ab,$$

and for $n = 2s$, the monomials

$$bcde(abcde)^{s-1}ab \quad \text{and} \quad de(abcde)^{s-1}abcd.$$

This can be easily proved by induction on n. We identify these chains with their duals in the Yoneda algebra B.

We prove the statement of the theorem in a very concrete way. Namely, we demonstrate that for the element

$$\omega_n = d \otimes e \otimes (abcde)^{\otimes(n-4)} \otimes a \otimes b \in B^{\otimes n},$$

and for any canonical A_∞-structure $\{\nu_n\}$ on B, we have $\nu_n(\omega_n) \neq 0$.

First, we note that the total degree of ω_n is $1+1+3(n-4)+1+1 = 3n-8$, and the concatenation $de(abcde)^{n-4}ab$ is a chain of length $2(n-3)-1$, so an element of homological degree $2(n-3) =$

$2n-6$. Since $2n-6 = 3n-8+2-n$, the homological degrees match, so for the canonical structure $\{\mu_n\}$ obtain in this chapter, we have $\mu_n(\omega_n) \neq 0$.
Suppose that ν_n is another canonical structure, and $\{f_n\}$ is an ∞-equivalence between them; without loss of generality, $f_1 = \mathrm{id}$. Let us denote by Φ_n the set of all n-fold products of chains $\gamma_1 \otimes \cdots \otimes \gamma_n$ in $B^{\otimes n}$ with the following properties:

(i) all degrees $|\gamma_i|$ are odd,
(ii) if $|\gamma_i| = 1$, then $i \in \{1, 2, n-1, n\}$,
(iii) if $|\gamma_1| = |\gamma_2| = 1$, then $\gamma_1 = d$, $\gamma_2 = e$,
(iv) if $|\gamma_{n-1}| = |\gamma_n| = 1$, then $\gamma_{n-1} = a$, $\gamma_n = b$,
(v) $|\gamma_i| > 1$ for some i.

By direct inspection, every connected r-fold subtensor ψ of $\phi \in \Phi_n$ is in $\Phi_r \setminus \{\omega_r\}$. Also, for the canonical structure of Chapter 1, if $\mu_n(\phi) \neq 0$ for some $\phi \in \Phi_n$, then $\phi = \omega_n$, which easily follows from the observation that concatenations of pairs of elements of degree greater than 2 cannot be found as divisors of any Anick chain.
Using these observations, we prove by induction on n that for any minimal canonical A_∞-structure $\{\nu_m\}$ and for any $\phi \in \Phi_n$ we have $\nu_n(\phi) = \mu_n(\phi)$. The morphism identity states that

$$\sum_{i+j+k=n} \pm f_{i+1+k}(1^{\otimes i} \otimes \mu_j \otimes 1^{\otimes k})(\phi) = \sum_{i_1+\ldots+i_r=n} \pm \nu_r(f_{i_1} \otimes \cdots \otimes f_{i_r})(\phi).$$

By induction and above observations, the only element on the left that does not vanish is $f_1(\mu_n)$, and the only element on the right that does not vanish is $\nu_n(f_1^{\otimes n})$ (since for $r < n$, in order for ν_r to not vanish, $(f_{i_1} \otimes \cdots \otimes f_{i_r})(\phi)$ must create ω_n with a nonzero coefficient, which is impossible for degree reasons), so we are done. □

3.4 Vanishing patterns for higher products. Let us now demonstrate how one can combine the two formulas for Tor groups to prove some nontrivial results.

Proposition 3.5. *Consider the canonical A_∞-algebra structure on $\mathrm{Ext}^*_\Lambda(\Bbbk, \Bbbk)$ from Theorem 3.3. Suppose $\gamma_1, \ldots, \gamma_n$ are elements of the* Ext *algebra. Then $m_n(\gamma_1, \ldots, \gamma_n)$ vanishes:*

(i) if at least three of them are of even degree, or
(ii) if exactly two of them are of even degree, unless those elements are γ_1 and γ_n, or
(iii) if exactly one element is of even degree, unless that element is γ_1 or γ_n.

Proof. By linearity, it is enough to prove this result in the case of all those elements being (duals of) Anick chains. Suppose that among them, we have $k$elements of even degrees $2d_1, \ldots, 2d_k$, and ℓ elements of odd degrees $2e_1+1, \ldots, 2e_\ell+1$; of course, $n = k+\ell$. We once again use the fact that Tor groups and their combinatorial bases are well defined, and identify those elements

with the elements of the Govorov–Gruenberg spaces; from Formulas (3.1) and (3.2) it follows in particular that those elements are represented by paths in $I^{d_1}, \dots, I^{d_k}, I^{e_1}\mathfrak{m} \cap \mathfrak{m}I^{e_1}, \dots, I^{e_\ell}\mathfrak{m} \cap \mathfrak{m}I^{e_\ell}$ respectively. By Formula (3.5), the value of the operation μ_n on these elements is proportional to the concatenation of the corresponding paths.
Note that the concatenation of paths from $I^{d_1}, \dots, I^{d_k}, I^{e_1}\mathfrak{m} \cap \mathfrak{m}I^{e_1}, \dots, I^{e_\ell}\mathfrak{m} \cap \mathfrak{m}I^{e_\ell}$ (in any order) is in $I^{d_1+\cdots+d_k+e_1+\cdots+e_\ell}$. At the same time, the result of applying μ_n to such words has homological degree

$$2(d_1+\cdots+d_k+e_1+\cdots+e_\ell)+\ell+2-n = 2(d_1+\cdots+d_k+e_1+\cdots+e_\ell)+2-k.$$

In particular, for $k>2$, the degree of $\mu_n(\gamma_1,\dots,\gamma_n)$ is less than

$$d = 2(d_1+\cdots+d_k+e_1+\cdots+e_\ell).$$

Applying Formulas (3.1) and (3.2) again, we conclude that the concatenation of all our elements is in the zero coset of the corresponding Govorov–Gruenberg space.
Similarly, for $k=2$, the degree of $\mu_n(\gamma_1,\dots,\gamma_n)$ is the even number

$$2(d_1+\cdots+d_k+e_1+\cdots+e_\ell).$$

If we assume that at least one of the two elements γ_1, γ_n is of odd degree, we observe that the concatenation of all our elements is in $\mathfrak{m}I^d + I^d\mathfrak{m}$, which is in the zero coset of the corresponding Govorov–Gruenberg space.
Finally, for $k=1$, the degree of $\mu_n(\gamma_1,\dots,\gamma_n)$ is the odd number

$$2(d_1+\cdots+d_k+e_1+\cdots+e_\ell)+1.$$

If we assume that both elements γ_1, γ_n are of odd degree, we observe that the concatenation of all our elements is in $\mathfrak{m}I^d\mathfrak{m}$, which is in the zero coset of the corresponding Govorov–Gruenberg space. □

As a first consequence, one may simplify the signs arising in the higher products, at least to some extent.

Corollary 3.6. *Formula (3.5) can be simplified to*

$$m_n(\gamma_1,\dots,\gamma_n) = \begin{cases} (-1)^{nr_1+r_1r_n+r_1+r_n}\gamma_1\dots\gamma_n & \text{if } \gamma_1\dots\gamma_n \in C^\vee_{r_1+\cdots+r_n+1}, \\ 0 & \text{else.} \end{cases} \tag{3.6}$$

3.5 Application: Bardzell's resolution. In [12], Bardzell constructed explicitly the minimal bimodule resolution of the diagonal bimodule over any algebra Λ with monomial relations. By

an intricate study of combinatorics of Anick chains, he established that the shape of the differential of such resolution alternates, depending on the homological degree. Let us explain how this phenomenon is an immediate consequence of Proposition 3.5.

Repeating the argument of [89, Th. 4.2] *mutatis mutandis* for the case of quiver algebras, we see that the canonical A_∞-algebra structure on $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ gives rise to a free bimodule resolution of the diagonal bimodule. The corresponding twisting cochain $\tau\colon \mathrm{Tor}^\Lambda_*(\Bbbk,\Bbbk) \to \Lambda$ is extremely easy to describe for a monomial quiver algebra $\Lambda = kQ/I$: it annihilates elements of homological degree different from one, and on elements of degree one it is given by the identity map under the identification $\mathrm{Tor}^\Lambda_1(\Bbbk,\Bbbk) = kQ_1 \subset \Lambda$.

Now, the only thing that remains is to examine carefully the dual formulas of 3.5. We shall use the variant of that formula which describes the boundary map in the minimal model of Λ, and can be chosen in the form similar to [171, Th. 4.1], but with a different choice of signs arising from considering left Anick chains:

$$b(s^{-1}\gamma) = \sum_{n\geq 2} (-1)^{|s^{-1}\gamma_1|} s^{-1}\gamma_1 \otimes \cdots \otimes s^{-1}\gamma_n, \tag{3.7}$$

where the sum ranges through all decompositions of an Anick chain γ. This shape of the formula is particularly convenient for computing the differential of the bimodule resolution of the diagonal module, since the twisting cochain τ is precisely the operator $\gamma \mapsto s^{-1}\gamma$ on γ of degree 1. Thus, to obtain the answer, we need to classify all possible decompositions

$$\gamma = \gamma_1 \cdots \gamma_n$$

with $|s^{-1}\gamma| = |s^{-1}\gamma_1| + \cdots + |s^{-1}\gamma_n| + 1$ and where in addition all $s^{-1}\gamma_i$ but one are of homological degree 0. By Proposition 3.5, this depends on the parity of $|\gamma|$. If $|\gamma|$ is odd, then for the only element $s^{-1}\gamma_i$ of positive homological degree, $|\gamma_i|$ is even, and so $i = 1$ or $i = n$. On the other hand, if $|\gamma|$ is even, then for the only element $s^{-1}\gamma_i$ of positive homological degree, $|\gamma_i|$ is odd, so there are no constraints on the position of this element. Thus, Formula 3.7 suggests that the image of the generator $e_{t(\gamma)} \otimes \gamma \otimes e_{s(\gamma)}$ of the bimodule resolution $\Lambda \otimes kC_* \otimes \Lambda$ under the differential of that resolution is given by the formula

$$\begin{cases} h_\gamma \otimes \gamma'' \otimes e_{s(\gamma)} - e_{t(\gamma)} \otimes \gamma' \otimes t_\gamma, & |\gamma| \equiv 1 \pmod 2, \\ \sum \alpha \otimes \tilde{\gamma} \otimes \beta, & |\gamma| \equiv 0 \pmod 2, \end{cases}$$

where $\gamma = h_\gamma \gamma'' = \gamma' t_\gamma$ are the decompositions of c as the left chain and the right chain that factor out the head and the tail, and the sum is over all decompositions $c = \alpha\tilde{\gamma}\beta$ with $\tilde{\gamma}$ a chain of homological degree one less than γ. This is precisely the result of [12, Th. 4.1], the main theorem of that paper.

4 Final remarks and a conjecture

4.1 Streamlining homotopy transfer methods. It is fair to observe that the construction of our minimal model requires the construction of a homotopy retract datum from BA to Tor_A, and thus of comparison morphisms, which are usually difficult to produce. However, the construction of this retraction is streamlined by the machinery of algebraic discrete Morse theory and, in fact, one may attempt to apply the methods outlined in [102] to any algebra admitting a Gröbner basis to produce a model of it. Let us also remark that one need not recourse to comparison maps to produce models of algebras. In the article [52], for example, the authors produce models for monomial operads, in particular for monomial algebras, without doing this. As explained in that article, one may use this model to understand not necessarily monomial algebras admitting a Gröbner basis by the method of homological perturbation theory. Remark, too, that in [156] the authors produce chain comparison maps between the Bardzell resolution of a monomial quiver algebra and its usual bar resolution, and succeed in using them to compute the Gerstenhaber bracket on Hochschild cohomology of some examples. It may be the case that the maps of [156] are a part of a homotopy retract datum provided by algebraic discrete Morse theory [102, 162].

4.2 The case of algebras with a Gröbner basis. Let us put ourselves in the situation where A is a finitely generated algebra with generators V and ideal of relations (R). Pick a Gröbner basis with respect to a monomial order on TV, and let us write A' for the monomial algebra associated to A and $B' = (TW, d')$ for the minimal model of Theorem 2.6. Note that since W consists of monomials of TV, this graded space is partially ordered by looking at the support of a chain, and this order extends to monomials lexicographically.

We claim that there exists a model $B = (TW, d)$ of A such that for any $w \in W$, the terms appearing in $(d - d')(w)$ are smaller than w, and such that the associated graded morphism to $B \longrightarrow A$ is the model $B' \longrightarrow A'$ in the main theorem of these notes. As before, let (C, d) denote the complex obtained from the Anick resolution of A that computes Tor_A. Note that Proposition 1.1 is still valid if we replace Tor_A with C, since at no point we used A is monomial to prove it. We also observe that the differential on B' preserves the support of a chain.

Naturally, to prove our claim, it suffices we do it for each higher coproduct, including the possibly non-zero differential Δ'_1 on C. The work of Anick shows this differential decreases the order of a chain, and the claim is obvious for Δ'_2, so we may only worry about Δ'_n for $n \in \mathbb{N}_{\geqslant 3}$. In this case, the recursive formula of Proposition 1.1 means it suffices we do this for the homotopy h. But this follows from the fact it is built from the differential of BA, which, after rewriting possible non-zero products that appear, decreases the order of the underlying monomial of any bar term, independent of them being a cycle or not. From this we obtain the desired result, which in fact follows from the more detailed results in Chapter 6:

Theorem 4.1. *Let A be a finitely generated algebra with a finite Gröbner basis, and let A' be its associated monomial algebra. There exists a (possibly non-minimal) model $(B, d) \longrightarrow A$ such that*

the associated graded morphism $(B,d') \longrightarrow A'$ is the model of Theorem 2.6. More precisely, we can arrange it so that $d-d'$ decreases the order of the underlying monomial of a chain in B.

Proof. We have given some details in the discussion preceding the statement of the theorem to obtain a proof following the strategy used to prove our main theorem. Alternative, one can use a homological perturbation argument completely analogous to [52, Theorem 4.1], where instead of starting with the (usually non-minimal) model of the authors, one starts with the minimal model of our main theorem with the internal grading given by the underlying monomial of an Anick chain. □

We remark that this theorem is not too surprising, since it is the non-linear analog of the work of S. Chouhy and A. Solotar [40]. The lack of an explicit formula for the perturbed differential makes this theorem uninteresting for computations: in concrete examples, what we usually do is produce a perturbed differential which squares to zero, since it is usually possible to come up with a candidate of model and, through a filtration argument, show it is indeed acyclic. However, we would like to state the following

Conjecture. *Let A be as before, and let w be a chain in the generators of the model $(B,d) \longrightarrow A$. Then the basis elements appearing in dw are obtained as follows:*

- **C1.** *Compute all possible decompositions of the chain w.*
- **C2.** *Starting from the left, rewrite the chain w once, and obtain all possible decompositions into chains of the terms that appear after this.*
- **C3.** *Repeat this procedure until all terms that appear are in normal form.*

As an example, let us consider the algebra J with two generators x and y subject to the relations $x^2=0$ and $y^2x=xy^2+xyx$, and lexicographical order with respect to $y>x$. The associated monomial algebra J' has relations $x^2=0$ and $y^2x=0$, and its model has generators $x_0,y_0,x_1,y_1\ldots$ with differential

$$dy_{n+1}=y^2x_n+\sum_{\substack{s+t=n\\ s\geqslant 1}}(-1)^s y_s x_t, \quad dx_{n+1}=\sum_{s+t=n}(-1)^s x_s x_t.$$

Here, for $n\in\mathbb{N}$, the generator y_n has underlying ambiguity y^2x^n while x_n has underlying ambiguity x^{n+1}, which our differential preserves. The differential then codifies all possible 2-decompositions of y^2x^n into $y^2x^s\cdot x^t$ for $s+t=n$, and the unique 3-decomposition $y\cdot y\cdot x^n$. Similarly, x^{n+2} only admits 2-decompositions of the form $x^{s+1}\cdot x^{t+1}$ where $s+t=n$. The model corresponding to the original algebra J incorporates lower order terms as follows:

$$dy_{n+1}=[y^2,x_n]-\sum_{s+t=n}x_s y x_t-\sum_{\substack{s+t=n\\ t\geqslant 1}}(x_s y_t-(-1)^t y_t x_s), \quad dx_{n+1}=\sum_{s+t=n}(-1)^s x_s x_t.$$

It is routine to check this perturbed differential squares to zero, so that we have obtained a model of J. To illustrate our conjecture, let us consider the term $y_2 = y^2x^2$. This can be decomposed into the chains $y_0^2x_1$ and y_1x_0 and no others. Rewriting, we obtain two terms, xy^2x and xyx^2. The first can be decomposed into x_0y_1 only, and the second into xyx_1. We can only rewrite the first monomial, and we obtain x^2y^2 and x^2yx which rewrite to zero. We can decompose these into x_1y^2 and $x_1y_0x_0$, and no other terms. Summing up, the basis elements that appear are the following: y^2x_1, x_1y^2, y_1x_0, x_0y_1, x_1yx_0, x_1yx_0. These are precisely those appearing in the formula for dy_2 above.

Chapter 4

Non-commutative calculus

The chapter is organised as follows. In Section 1 we recall the basics of (colored) operads, their algebras, and the relevant Koszul duality theory for them, which we use to give a inhomogeneous Koszul model of Calc. In Section 2, we introduce the main algebraic structure of this chapter, Tamarkin–Tsygan calculi. We then give two complexes to compute Hochschild (co)homology in terms of models, and give the promised formulas for the Tamarkin-Tsygan calculus of an algebra in terms of these. Finally, in Section 3, following the notion of Koszul A_∞-algebras of Berglund–Börjeson [20], we show that the Tamarkin–Tsygan calculus of an A_∞-Koszul algebra is dual to that of its Koszul dual algebra, extending a result of E. Herscovich [87]. In Section 3 we give some examples of computations to illustrate our methods.

1 Colored operads and Koszul duality

Operads allow us to describe algebras of certain kind. We can embellish them to describe pairs of algebras of a certain kind along with a module over this algebra, as follows.
Let P be an operad, which we think of as having color $\circ$ (white). We define a 2-colored operad P^* with colors $\circ$ and $\bullet$ (black) as follows, where we use the notation $P^*(\varepsilon;a,b)$ for the spaces of operations with a inputs of color $\circ$, b inputs of color $\bullet$, and the single output of color $\varepsilon \in \{\bullet,\circ\}$.

$$(1)\ \ P^*(\circ;a,b) = \begin{cases} 0 & \text{if } a \geqslant 1, \\ P(b) & \text{if } a = 0. \end{cases}$$

$$(2)\ \ P^*(\bullet;a,b) = \begin{cases} 0 & \text{if } a = 0 \text{ or } a \geqslant 2, \\ (\partial P)(b+1) & \text{if } a = 1. \end{cases}$$

Here ∂ is the pointing operator on symmetric sequences: $\partial X(n)$ is a copy of $X(n+1)$ where $n+1$ is 'marked' and S_n acts on the remaining ones. Alternatively, we are taking the restriction of $X(n+1)$ for the inclusion $S_n \longrightarrow S_{n+1}$ as the stabilizer of $n+1 \in [n+1]$. We then color the marked point

black, as well as the output. In terms of finite sets, we have that $(\partial X)(I) = X(I \sqcup \{I\})$ with the obvious action of the symmetric group S_I.
The composition law in P^* is dictated by the simple rule we may only graft a tree with root colored ε into a leaf of the same color. Hence, we only have compositions coming from those of P, and two new compositions that either graft an operation of P into a white leaf of a marked operation of ∂P, or a new composition that grafts a black root into a black leaf of ∂P, both resulting in an operation in $P^*(\bullet; 1, a)$.
If $V = (V_\circ, V_\bullet)$ is a pair of vector spaces, which we think of put in color $(\circ, \bullet)$, then we can consider the colored operad End_V, where for each $s,t \in \mathbb{N}$ and $\varepsilon \in \{\circ, \bullet\}$,

$$\mathrm{End}_V(\varepsilon; s, t) = \hom(V_\circ^{\otimes s} \otimes V_\bullet^{\otimes t}, V_\varepsilon).$$

In this way, a P^*-algebra structure on V is equivalent to a morphism of colored operads

$$P^* \longrightarrow \mathrm{End}_V .$$

With this at hand, we have the following elementary result. We point the reader to [109] for details.

Proposition 1.1. *A P^*-algebra structure on $V = (V_\circ, V_\bullet)$ is equivalent to the datum of a P-algebra structure on $V_\circ$ and a $V_\circ$-module structure on $V_\bullet$.* □

Let us record the following result, which we will use later.

Proposition 1.2. *The operad P^* is Koszul if and only if P is Koszul.*

Proof. This follows from the fact the homology of the free P^*-algebra on $(V_\circ, V_\bullet)$ is nothing but the Koszul homology of the free P-algebra over $V_\circ$ in the first color and the Koszul homology of the free P-algebra over $V_\circ$ with coefficients in the free $V_\circ$-module on $V_\bullet$, and both these complexes are acyclic since P is Koszul. Since an operad is Koszul if and only if its free algebras have trivial Koszul (co)homology in non-negative degrees, this proves the theorem. □

Observe that this implies both operads Com^* and $\mathscr{S}^{-1}\mathsf{Lie}^*$ are Koszul. From the operad Ger we form the 2-colored operad $\mathsf{PreCalc} = (\mathsf{Ger}, \mathsf{M})$ by adding two operations $[v,x]$ and $v \cdot x$ in $\mathsf{PreCalc}(\bullet; 1, 1)$, of degrees 0 and -1 respectively, and relations as follows:

(1) the product $x_1 x_2$ and the right action $v \cdot x_1$ satisfy the same relations defining Com^*,

(2) the bracket $[x_1, x_2]$ and the odd action $[v, x_1]$ satisfy the same relations defining $\mathscr{S}^{-1}\mathsf{Lie}^*$,

(3) the Lie action and the commutative action satisfy the following mixed Leibniz relations:

$$v \cdot [x_1, x_2] = [v, x_1] \cdot x_2 - [v \cdot x_2, x_1] \tag{L1}$$

$$[v, x_1 x_2] = [v \cdot x_1, x_2] + [v, x_1] \cdot x_2. \tag{L2}$$

Theorem 1.3. *The operad* PreCalc *is Koszul.*

Proof. We use the distributive law criterion, adapted to 2-colored operads. Namely, the result that if P and P' are Koszul operads and if P'' is a third operad obtained from a distributive rule $P' \circ P \longrightarrow P \circ P'$, then P'' is itself Koszul [129, Section 8.6]. We will apply this criterion in the case $P = \mathsf{Com}^*$ and $P' = \mathscr{S}^{-1}\mathsf{Lie}^*$.

To do this orient the Leibniz rule (L1) to display it in the form

$$[v \cdot x_2, x_1] = [v, x_1] \cdot x_2 - v \cdot [x_1, x_2]$$

and rewrite the second Leibniz rule (L2) incorporating (L1) as follows:

$$[v, x_1 x_2] = [v, x_2] \cdot x_1 - v \cdot [x_1, x_2] + [v, x_1] \cdot x_2.$$

In this way, along with the usual Leibniz rule, we have defined a map $V \circ_{(1)} V' \longrightarrow V' \circ_{(1)} V$ on the generators V of Com^* and V' of $\mathscr{S}^{-1}\mathsf{Lie}^*$ which, in turn, induces a surjective map

$$\mathsf{Com}^* \circ \mathscr{S}^{-1}\mathsf{Lie}^* \longrightarrow \mathsf{PreCalc}.$$

It is well known that, on the sub-operad where all outputs and inputs are of the same (white) color, this is indeed an isomorphism, so it suffices we check that this on the components of mixed color. It is straightforward to check that $(\mathsf{Com}^* \circ \mathscr{S}^{-1}\mathsf{Lie}^*)(n)$ is of dimension $2 \cdot n!$, so it suffices we check that $\mathsf{PreCalc}(n)$ is of the same dimension. To do this, we recall that in [49] the authors provide us with a geometrical model of PreCalc. Indeed, their results show there exists a topological operad PreCyl along with homotopy equivalences

$$\mathsf{PreCyl}(\circ; n, 0) \simeq \mathrm{Conf}_n(\mathbb{R}^2) \quad \mathsf{PreCyl}(\bullet; n-1, 1) \simeq \mathrm{Conf}_{n-1}(\mathbb{R}^2 \smallsetminus 0),$$

and a surjection $\mathsf{PreCalc} \longrightarrow H_*(\mathsf{PreCyl})$, and the Poincaré series of these spaces are known [8]. This gives us the requisite lower bound on the dimensions of PreCalc, and shows that PreCalc is obtained from a distributive law from two Koszul operads, and hence is Koszul. □

Remark 1. Observe that this dimension count along with the methods developed in [108] and the quantum order of [54], allows us to exhibit a quadratic Gröbner basis of PreCalc where the leading terms of the mixed Leibniz rules (L1) and (L2) are $[v \cdot x_1, x_2]$ and $[v, x_1 x_2]$, as it was done for the Poisson operad in [50], for the pre-Poisson operad in [54], and the for Lie–Rinehart operad in [108].

Let us now add a new operator δ of degree 1 to PreCalc along with the non-quadratic relation encoding the magic formula of Cartan

$$\delta(v \cdot x) - \delta v \cdot x = [v, x].$$

We write this operad Calc; it is the operad governing Tamarkin–Tsygan calculi. If we consider the increasing filtration on Calc by the number of uses of δ, we obtain an associated quadratic operad qCalc governing PreCalc-algebras with a degree -1 square zero operation δ such that

$$[\delta v, x] = \delta[v, x], \qquad \delta(v \cdot x) = \delta v \cdot x.$$

Let us consider the (colored) operad of dual numbers $\mathsf{Diff} = T(\delta)/(\delta^2)$ put in color $\bullet$. Then we can present qCalc through a distributive rule λ between PreCalc and Diff, and we claim that this is a distributive law. To do this, we need to check that

$$\rho : \mathsf{Diff} \circ \mathsf{PreCalc} \longrightarrow \mathsf{PreCalc} \vee_\lambda \mathsf{Diff} = q\mathsf{Calc}$$

is an isomorphism.

Theorem 1.4. *The operad* qCalc *is Koszul and* Calc *is inhomogeneous Koszul. Moreover, the map* $q\mathsf{Calc} \longrightarrow \mathrm{gr}\,\mathsf{Calc}$ *is an isomorphism of symmetric sequences, so that the underlying collections of* Calc *and* qCalc *are isomorphic to* $\mathsf{Diff} \circ \mathsf{PreCalc}$.

Proof. Since Diff and PreCalc are both Koszul, we deduce that qCalc is also Koszul by a another use of the distributive law argument of [129, Section 8.6]. To apply it, we need to check that ρ is an isomorphism, and do this, we only need to check that ρ is injective, which is immediate, since it is surjective and by [49] the dimensions of the domain and the codomain match. This implies that Calc is inhomogeneous Koszul, and hence the Poincaré–Birkhoff–Witt theorem for Koszul operads shows that $q\mathsf{Calc} \longrightarrow \mathrm{gr}\,\mathsf{Calc}$ is an isomorphism. □

From this, we obtain a model Calc_∞ of Calc with generators

$$\mathsf{Calc}^! = \mathsf{PreCalc}^! \circ \mathsf{Diff}^!$$

through the methods of [129, Chapter 7]. Note that $\mathsf{Diff}^! = T[u]$ is a polynomial algebra generated by $u = \delta^\vee$, and when we write $\mathsf{Calc}^!$ we are doing it in the sense of inhomogeneous Koszul duality theory.

Remark 2. In [170] the authors state that Calc is Koszul and give a description of what they refer to as Calc_∞-algebras. However, it is not clear that Calc admits a quadratic presentation, although it does admit a quadratic-cubic presentation (where the quadratic-cubic equations only appear in the operations of mixed color).

2 Main results

Throughout, we fix an associative algebra A concentrated in homological degree 0 and a model B of A, and show how to compute the Tamarkin–Tsygan calculus of A using only the dga algebra B and

two small complexes $\mathscr{X}^*(B)$ and $\Theta_*(B)$ built out of B, which we like to think of as the complexes of non-commutative (nc) polyvector fields and non-commutative differential forms on B. We begin by recalling the essential details of this algebraic structure we want to compute.

2.1 Tamarkin–Tsygan calculi. We have already introduced the operad Calc. It turns out the pair

$$(\mathrm{HH}^*(A), \mathrm{HH}_*(A))$$

admits a structure of a Calc-algebra consisting of the cup product, the cap product, Gerstenhaber's bracket and Connes' operator. The Lie action on $\mathrm{HH}_*(A)$ is defined so that for each $\omega \in \mathrm{HH}_*(A)$ and each $X \in \mathrm{HH}^*(A)$,

$$L_X(\omega) = di_X(\omega) - (-1)^{|X|} i_X(d\omega),$$

where we are using the notation i_X for the cup product action of X on $\mathrm{HH}_*(A)$; this will be convenient later.

That these operations define a Calc-algebra structure was originally proved in [48]. See also [169]. We call this algebra–module pair $(\mathrm{HH}^*(A), \mathrm{HH}_*(A))$ along with the operator d the *Tamarkin–Tsygan calculus* of A. We will write Calc_A for this Calc-algebra, hoping it does not give rise to any confusion, in that we are not evaluating A on the endofunctor Calc, for example.

Proposition 2.1. *Let A be an associative algebra. Then $(\mathrm{HH}^*(A), \smile, [-,-])$ is a Gerstenhaber algebra, and $(\mathrm{HH}_*(A), i, L)$ is a module over $(\mathrm{HH}^*(A), \smile, [-,-])$, so that in particular, if for each $X \in \mathrm{HH}^*(A)$ we write i_X for the action of X, for $Y \in \mathrm{HH}^*(A)$,*

$$[L_X, i_Y] = i_{[X,Y]} \quad \text{and} \quad L_{X\cdot Y} = L_X i_Y + (-1)^{|X|} i_X L_Y.$$

We can describe all the operators of Proposition 2.1 on the cochain level using the pair of classical complexes $(C^*(A), C_*(A))$ as follows. For cochains $f, g \in C^*(A)$ homogeneous of degrees p and q and a chain

$$z = a[a_1|\cdots|a_{p+q}] \in C_*(A)$$

of degree $n = p+q$, the cup product, the cap product, the Lie bracket and Connes' differential are defined by the following formulas, where $\circ$ is Gerstenhaber's pre-Lie bracket

$$f \circ g = \sum_{i=1}^{n} f \circ_i g$$

on the endomorphism operad $C^*(A) = \mathrm{End}_A$ (see the Section 2.2.13 for details):

- Cup product: $f \smile g = \mu\,(f \otimes g)\,\Delta$,
- Cap product: $i_f(z) = af[a_1|\cdots|a_p][a_{p+1}|\cdots|a_{p+q}]$,
- Lie bracket: $[f,g] = f \circ g - (-1)^{(p-1)(q-1)} g \circ f$,
- Connes' differential: $dz = \sum_{j=0}^{n} (-1)^{jn} [a_{j+1}|\cdots|a_n|a|a_1|\cdots|a_j]$

- Lie action: $L_f(\omega) = [d, i_f](\omega)$

In other words, the homology of the pair of complexes $(C^*(A), C_*(A))$ along with the operators $(\smile, [-,-], i, L, d)$ recovers the Tamarkin–Tsygan calculus of the associative algebra A.

The work of B. Keller [105, 106], and later of M. Armenta and B. Keller [6, 7] shows that the Tamarkin–Tsygan calculus of an algebra is derived invariant. Although our main result does imply it is homotopy invariant, in the sense it induces a well-defined functor $\mathrm{Ho}(\mathsf{Alg})^{\mathrm{iso}} \longrightarrow \mathrm{Ho}(\mathsf{Calc})^{\mathrm{iso}}$ (as in Theorem 8.5.3 in [91]), invariance is not a novel result, since homotopy equivalent algebras are derived equivalent. Rather, our main result is more concerned with the explicit computation of this calculus using a choice of resolution in Alg.
To be precise, the result of Armenta–Keller implies that one may *attempt* to compute the calculus of an associative algebra A by choosing some model B of it and, in some way or another, obtain a description of the spaces

$$(\mathrm{HH}^*(A), \mathrm{HH}_*(A))$$

and formulas for the Calc-algebra structure here. Knowing this, our goal here is, having chosen B, to carry out the last step as explicitly as possible, and to give a datum depending on B that makes such computation feasible.

2.2 Fields and forms. Let us consider the B-bimodule resolution of B given by

$$S_*(B) : 0 \longrightarrow B \otimes V \otimes B \longrightarrow B \otimes B \longrightarrow 0,$$

where the only non-trivial map is given by $b \otimes v \otimes b' \longmapsto bv \otimes b' - b \otimes vb'$. In case B has no differential, this is the usual small resolution of the free algebra TV. In our case, there is a caveat: this is a semi-free resolution, in the sense that $B \otimes V \otimes B$ can be filtered by sub-bimodules (using the triangulation of V) in such a way that the successive quotients are free B-bimodules on a basis of cycles. Indeed, once we have considered the filtration on $B \otimes V \otimes B$ using a triangulation of V, the differential that is nontrivial on V will vanish. Since this is precisely the kind of resolutions needed to compute Hochschild (co)homology in the dg setting, we may use this resolution to proceed with our computations.
To understand where the differential of $B \otimes V \otimes B$ of comes from, it is useful to note it identifies with the bimodule of *Kahler differentials* Ω^1_B of B. This is the free B-bimodule on B, say spanned by basis elements $b \otimes b' \otimes b''$ subject to the relations

$$b' \otimes b_1 b_2 \otimes b'' = b' b_1 \otimes b_2 \otimes b'' + b' \otimes b_1 \otimes b_2 b''.$$

Note that since B is generated by V, any class in Ω^1_B, which we write $bdb'b''$, can be written in the form $b_1 dv b_2$, and the identification is such that $g : b \otimes v \otimes b' \longmapsto bdvb'$. The following lemma makes this precise.

Lemma 2.2. *The map $g : B \otimes V \otimes B \longrightarrow \Omega^1_B$ is an isomorphism of B-bimodules.*

Proof. The B-bimodule Ω^1_B satisfies the universal property that the restriction map induced from $B \longrightarrow \Omega^1_B$ sending $b \longmapsto db$ induces a bijection

$$\hom_{B^e}(\Omega^1_B, -) \longrightarrow \mathrm{Der}(B, -).$$

Since B is free on V, this last functor is naturally isomorphic to the functor $\hom(V, -)$. It is immediate that the inclusion of V into $B \otimes V \otimes B$ satisfies the analogous universal property, which gives the result. □

Since $S_*(B)$ gives us a resolution of B in B-bimodules the complexes

$$\mathscr{X}^*(B) = \hom_{B^e}(S_*(B), B), \quad \Theta_*(B) = B \otimes_{B^e} S_*(B)$$

compute Hochschild (co)homology of B and hence, by invariance, that of A. We call $\mathscr{X}^*(B)$ the *space of nc poly vector fields on B*, and $\Theta_*(B)$ the *space of nc differential forms on B*. Let us now observe that from Lemma 2.2 it follows these complexes take a very simple form, which also explains the origin of their names.

Corollary 2.3. *We have natural identifications of complexes*

$$\mathscr{X}^*(B) = \mathrm{cone}(\mathrm{Ad} : B \longrightarrow \mathrm{Der}(B)), \quad \Theta_*(B) = \mathrm{cone}(\mathrm{Co} : B \otimes V \longrightarrow B).$$

Here, the map Ad is the adjoint map of B and the map Co is the commutator map of B.
It is worthwhile to observe that since $H_*(B) = H_0(B) = A$, there is a four term exact sequence coming for the long exact sequence of the cone

$$0 \longrightarrow \mathrm{HH}_1(A) \longrightarrow H_0(B \otimes V) \longrightarrow H_0(B) \longrightarrow \mathrm{HH}_0(A) \longrightarrow 0$$

and it is straightforward to check that the image of the middle arrow is

$$[A, A] \subseteq A = H_0(B),$$

which recovers the usual description of $\mathrm{HH}_0(A)$. Moreover, we see that $\mathrm{HH}_1(A)$ is the kernel of this map, and that for $n \in \mathbb{N}_{\geqslant 2}$, we an identification $\mathrm{HH}_n(A) = H_{n-1}(B \otimes V)$. Dually, for $n \in \mathbb{N}_{\geqslant 2}$ we have an identification $\mathrm{HH}^n(A) = H^{n+1}(\mathrm{Der}(B))$, and a four term exact sequence

$$0 \longrightarrow \mathrm{HH}^0(A) \longrightarrow H^0(B) \xrightarrow{j^*} H^0(\mathrm{Der}(B)) \longrightarrow \mathrm{HH}^1(A) \longrightarrow 0,$$

that shows that $\mathrm{HH}^1(A) = H^0(\mathrm{Der}(B))/\mathrm{im}(j^*)$. A far-reaching generalization of this, explaining the relation of operadic cohomology and Hochschild cohomology of operads under reasonable homotopical hypotheses is present in [157, Theorem 1.3.8].

We now show that the resolution $S_*(B)$ is retraction of the double sided bar resolution of B, a fact known since the inception of Hochschild cohomology. Once we have developed the homotopy theory of calculi, this will give us a way of transporting the homotopy coherence calculus of chains on B onto the pair $(\mathscr{X}^*(B), \Theta_*(B))$. For convenience, we recall that the data of a homotopy deformation retract is a triple of maps of complexes (i, π, h)

$$i : C' \longrightarrow C, \quad \pi : C \longrightarrow C'$$

such that $\pi i = 1$ and $i\pi - 1 = dh + hd$.

Lemma 2.4. *There is a homotopy deformation retract (i, π, h) between $S_*(B)$ and the double sided bar resolution $\mathsf{Bar}_*(B)$ of B, where we may take*

$$\pi : \mathsf{Bar}_*(B) \longrightarrow S_*(B)$$

to be either zero or the identity except on $B \otimes B \otimes B$ where for $b = v_1 \cdots v_n$ we have

$$\pi(1 \otimes b \otimes 1) = \sum_{i=1}^{n} v_1 \cdots v_{i-1} \otimes v_i \otimes v_{i+1} \cdots v_n.$$

In particular, the chain complex $C_*(B)$ is homotopy equivalent to $\Theta_*(B)$ and the cochain complex $C^*(B)$ is homotopy equivalent to $\mathscr{X}^*(B)$.

Proof. Let us begin by noting that the non-trivial component of π identifies with the quotient map

$$q : B \otimes B \otimes B \longrightarrow \Omega^1_B$$

under the isomorphism of Lemma 2.2, so it is immediate it is a map of complexes. To construct i, we observe that if B has zero differential, we may choose it to be the inclusion of $B \otimes V \otimes B$ into $B \otimes B \otimes B$.
In case B has a differential, we can filter the total complexes above using its columns, and then the extra differential coming from B lowers this filtration degree. In this way, the homological perturbation lemma guarantees that we can obtain again a homotopy retract. □

It would be desirable to have an explicit formula of the maps i and h.

2.3 The homotopy calculus. Having already set up things to obtain a retract from the Hochschild pair of B onto the pair $(\mathscr{X}^*(B), \Theta_*(B))$, we recall that Hochschild (co)chains on B carry a homotopy coherent calculus structure (which we can understand at the 'strict' level) and use this to obtain a homotopy coherent calculus structure on $(\mathscr{X}^*(B), \Theta_*(B))$ whose homotopy type recovers Calc_A.
Let us recall from the Section 2.2.13 that there is a 2-colored operad $\mathsf{PreCalc}$ so that a $\mathsf{PreCalc}$ algebra is the same as a pair (V, M) where V is a Gerstenhaber algebra and M is a V-module. As

explained there, extending the operad PreCalc by a square zero unary operation d of the second color and imposing the Cartan formula gives us the operad Calc.
We proved that the operad PreCalc is quadratic Koszul, while the operad Calc is inhomogeneous Koszul. In particular, Calc admits a model where the only non-quadratic differential comes from the Cartan formula and, in particular, there is Ger_∞ placed in the first color. We will write Calc_∞ for this model, although it is not the same model that the authors consider in [49]. With this at hand, we recall from [49, Section 4.3] the following technical results:

Theorem 2.5. *There is a dg colored operad* KS*, the Kontsevich–Soibelman operad, and a topological colored operad* Cyl *such that*

(1) The operad $C_*(\mathsf{Cyl})$ is formal and its homology is Calc*.*
(2) There is a quasi-isomorphism $\mathsf{KS} \longrightarrow C_*(\mathsf{Cyl})$ of dg-operads.
(3) KS acts on the pair $(C^*(-), C_*(-))$ in such a way that
(4) On homology we obtain the usual Calc*-algebra structure on $(\mathrm{HH}^*(-), \mathrm{HH}_*(-))$.*

In particular, for every cofibrant replacement Q of Calc*, the pair $(C^*(A), C_*(A))$ is a Q-algebra which on homology gives Calc_A.*

Proof. Claims (1)-(4) are in the cited references, and the last claim is standard, but let us explain it, since we will apply it for our choice of cofibrant replacement of Calc. That Q is a cofibrant replacement of Calc means that there is a quasi-isomorphism $Q \longrightarrow \mathsf{Calc}$ and Q is a cofibrant operad in the model category of dg operads. Since we have a surjective quasi-isomorphism $\mathsf{KS} \longrightarrow \mathsf{Calc}$ or, what is the same, a trivial fibration. But Q is cofibrant, so we can obtain a lift $Q \longrightarrow \mathsf{KS}$. Finally, since KS acts on the pair $(C^*(A), C_*(A))$, so does Q, and on homology we obtain Calc_A. □

In particular, from the theorem it follows that $(C^*(B), C_*(B))$ admits a Calc_∞-algebra structure which on homology gives the Calc-algebra $\mathsf{Calc}_B = \mathsf{Calc}_A$, and we will write $\mathsf{Calc}_{\infty,B}$ for this structure. We point out that the explicit construction of the operads KS and Cyl are not necessary for us here.
We remind the reader that Calc_∞ is our choice of cofibrant replacement obtained from the inhomogeneous Koszul duality theory. Our main result is that the pair $(\mathscr{X}^*(B), \Theta_*(B))$ admits Calc_∞-algebra structure which on homology gives the usual Calc-algebra structure; for some formulas see Theorem 2.7.

Theorem 2.6. *For any quasi-free model B of A, the pair $(\mathscr{X}^*(B), \Theta_*(B))$ admits a Calc_∞-algebra structure which is ∞-quasi-isomorphic to the Calc_∞-algebra $\mathsf{Calc}_{\infty,A}$. In particular, this structure recovers Calc_A by taking homology.*

Proof. We have already proved that the operad Calc is inhomogeneous Koszul. The methods developed in [63] imply that Calc-algebras then have the same rich homotopy theory that algebras over Koszul operads do. In particular by [63, Theorem 33], any Calc_∞-algebra structure may be

transported through a homotopy equivalence of chain complexes. By Lemma 2.4, there is a homotopy equivalence

$$(C^*(B), C_*(B)) \longrightarrow (\mathscr{X}^*(B), \Theta_*(B)),$$

which implies the Calc_∞-algebra structure on the pair $(C^*(B), C_*(B))$ may be transported to one on the pair $(\mathscr{X}^*(B), \Theta_*(B))$. The fact this structure is ∞-quasi-isomorphic to that one on $\mathsf{Calc}_{\infty,A}$ follows by another use of the transfer theorem, this time using $\mathsf{Calc}_{\infty,A}$ and $\mathsf{Calc}_{\infty,B}$. □

The space $\mathscr{X}^*(B)$ of nc poly vector fields on B is a dg Lie algebra under the usual bracket between derivations, with differential $[\partial_B, -]$. We can now state the next theorem, which in particular tells us that to compute Gerstenhaber brackets in A we may do so by choosing any model B of A and computing the usual Lie bracket in $\mathscr{X}^*(B)$. This last statement is a special case of [91, Theorem 8.5.3].

Theorem 2.7. *Let $B = (TV, d)$ be a quasi-free model of A.*

(1) the cup product can be computed through the symmetrization of dual cobrace operation $x_1 x_2 = \{x_1, x_2; d\}$ of Theorem 3.1.
(2) the Gerstenhaber bracket is the Lie bracket on the semi-direct product of a shifted copy of B with its Lie algebra of derivations,
(3) the Connes boundary can be computed by

$$d\omega = \sum_{i=1}^{n} (-1)^{\varepsilon} v_{i+1} \cdots v_n v_1 \cdots v_{i-1} dv_i$$

for $\omega = b + b' dv$ be a differential form in $\Theta_*(B)$ and $b = v_1 \cdots v_n$.

Proof. In Theorem 3.1 we will see that the cup product above is part of a brace algebra structure on $\mathscr{X}(B)$ which has the same quasi-isomorphism type as the brace algebra $C^*(B)$, which implies that the symmetrization of the operation in that formula gives the desired cup product in cohomology. Similarly, the Lie bracket corresponds to the antisymmetrization of the brace operation $\{x_1; x_2\}$ which is just the composition. Finally, the claim for the Connes boundary can actually be checked directly, using, for example, the definition of the boundary map of the usual LES of Hochschild and cyclic cohomology. □

3 Other results and computations

3.1 Brace operations. Let V be a graded vector space. A B_∞-algebra structure on V is the datum of a structure of dg bialgebra on $T(sV)$ where the comultiplication is given by deconcatenation [67]. It follows that the data required to define such structure amounts to a differential on $T(sV)$, which gives V the structure of an A_∞-algebra, along with a multiplication on $T(sV)$. The fact this is a map

of coalgebras means it is completely determined by a map $T(sV)\otimes T(sV)\longrightarrow sV$. This defines, for each $(p,q)\in\mathbb{N}\times\mathbb{N}$, a map of degree $1-p-q$

$$\mu_{p,q}: V^{\otimes p}\otimes V^{\otimes q}\longrightarrow V.$$

Let B be a quasi-free dga algebra. We proceed to show that its space of nc-vector fields $\mathscr{X}^*(B)$ admits a B_∞-algebra structure. We have already noted it admits an A_∞-structure, so it suffices we define the family of maps corresponding to the multiplication. As it happens for Hochschild cochains, we can arrange it so that for each $(p,q)\in\mathbb{N}\times\mathbb{N}$ we have $\mu_{p,q}=0$ whenever $q>1$, in which case what we have is an algebra over the operad of braces Br.

For linear maps $f_1,\dots,f_n,g\in\hom(V,B)$ corresponding to a derivation in $\mathscr{X}^*(B)$, we define an operation $\{f_1,\dots,f_n;g\}$ as follows. Let $\mathrm{sh}(f_1,\dots,f_n)$ be the unique derivation on B that acts by zero on monomials of length less than n, and acts, for $k\in\mathbb{N}$ on monomials of length $n+k$ by the sum

$$\sum_{\sigma}\sigma(f_1\otimes\cdots\otimes f_n\otimes 1^{\otimes k})$$

as σ runs through (n,k)-shuffles in S_{n+k}. In other words, each σ keeps $f_1,\dots,f_n$ in order and puts the identities in the middle slots. If G is the derivation that corresponds to g, we set $\{f_1,\dots,f_n;g\}$ to be the linear map corresponding to the derivation $\mathrm{sh}(f_1,\dots,f_n)\circ G$. We call $\{f_1,\dots,f_n;g\}$ a *dual brace operation*.

Recall from [66] that if A is an associative algebra, there are *brace operations* defined on $C^*(A)$ that make it, along with its usual structure of a dga algebra, into a brace algebra. Concretely, for each $n\in\mathbb{N}$ and for $f,g_1,\dots,g_n\in C^*(A)$ we have that

$$\{f;g_1,\dots,g_n\}=f\circ\mathrm{sh}(g_1,\cdots,g_n).$$

In other words, we are inserting $g_1,\dots,g_n$ into f in all possible ways preserving their order. For example, $\{f;g\}$ is the circle product of Gerstenhaber whose antisymmetrization gives the Gerstenhaber bracket. Note our notation is not ambiguous here: $\{f;g\}$ is both a brace and a dual brace operation, in both cases it is induced by composition of (co)derivations.

Theorem 3.1. *Let $B=(TV,d)$ be a quasi-free dga algebra. The dual braces give the space of nc fields $\mathscr{X}^*(B)$ the structure of a* Br*-algebra. In particular, for every quasi-free algebra B, the space of nc fields on B is a* Ger_∞*-algebra. Moreover, $\mathscr{X}^*(B)$ is* Br*-quasi-isomorphic to the brace algebra $C^*(A)$.*

Proof. The first part of the first claim is proved, mutatis mutandis, in the same way one shows brace operations on $C^*(A)$ satisfy the defining equations of a brace algebra: the definitions are dual to each other. The second claim follows from a deep result of D. Tamarkin [90], which shows that there is a quasi-isomorphism of operads $\mathsf{Ger}_\infty\longrightarrow\mathsf{Br}$. In this way, one can obtain from the brace algebra structure of $\mathscr{X}^*(B)$ to a Ger_∞-structure.

For the last claim, we note that $C^*(B)$ is Br-quasi-isomorphic to $C^*(A)$. Our theorem implies that $\mathscr{X}^*(B)$ is, in particular, Ger_∞-quasi-isomorphic to $C^*(B)$ and, by Tamarkin's result, we deduce that these two are also Br-quasi-isomorphic. □

A word of caution. The cup product we have defined using the differential d of B is not commutative on the nose, and in fact the cup product that one obtains using the result of Tamarkin is given by the symmetrization of this product. Since the cup product is commutative when passing to homology, there is no harm in using the non-symmetrized version for computations.

3.2 Duality. In [87], E. Herscovich fixes an augmented weight graded connected dg algebra A and proceeds to show that, writing E_A for the dual of the dg coalgebra BA, the Tamarkin–Tsygan calculi of A and E_A are dual to each other.
In the same context, let us pick a quasi-free model $(TV,d) = B \longrightarrow A$ of A, where V is a weight graded A_∞-coalgebra. It then makes sense to consider the Tamarkin–Tsygan calculus of the A_∞-algebra $E_A = (sV)^\#$. From our main theorem, one obtains the, now tautological, extension of Herscovich's result.
We also point out that, following A. Berglund and K. Börjeson [20], the algebra A is A_∞-Koszul with Koszul dual E_A. Their result also implies that $\mathscr{X}^*(B)$ is A_∞-quasi-isomorphic to the Hochschild dg-algebra $C^*(A)$ with its canonical product and differential.

Theorem 3.2. *The Tamarkin–Tsygan calculi of A and E_A are dual to each other in the sense that there is an isomorphism of Gerstenhaber algebras and an isomorphism of Gerstenhaber modules,*

$$f : \mathrm{HH}^*(E_A) \longrightarrow \mathrm{HH}^*(A) \quad g : \mathrm{HH}_*(A)^\# \longrightarrow \mathrm{HH}_*(E_A)$$

respectively, where $\mathrm{HH}_*(A)$ is a module over $\mathrm{HH}^*(E_A)$ through f. Moreover, we have that $\delta_E g = -g\delta_A^\#$ where δ_E is the Connes boundary of $\mathrm{HH}_*(E_A)$ and δ_A is that of $\mathrm{HH}_*(A)$.

Proof. We already know that to compute the Tamarkin–Tsygan calculus of A we may use nc-fields and nc-forms obtained from B. To compute the Tamarkin–Tsygan calculus of E_A, we may use a quasi-free dg model of the dg coalgebra BE_A —the ∞-bar construction on E_A— and nc-cofields, that is, coderivations, and nc-coforms on it: this is just the classical definition. These two constructions are dual to each other: the space $\hom(BE_A, E_A)$ is equal to $\hom(V, TV)$ while $BE_A \otimes E_A$ is dual to $V \otimes TV$, and as explained in [87], these isomorphisms are compatible with the cup and cap products, the Lie bracket, and Connes' boundary. □

3.3 Examples of computation. We now give two examples where, in the spirit of [62], we compute Hochschild cohomology of an algebra A using a (minimal) model of it. We will also compute, in some cases, Hochschild homology and cyclic homology, and the action of Hochschild cohomology on Hochschild homology using the results of the previous sections. We remark that the computations carried out here are many and for convenience we have omitted them. However, they

are all simple and straightforward computations with no intricacies whatsoever, so nothing should be lost to this omission.

A crown quiver algebra. Let us consider the quiver as in the next figure with the single relation $\alpha_1\alpha_2\cdots\alpha_r\alpha_1$, and its associated algebra A. In [156], the authors compute its Hochschild cohomology, including the Gerstenhaber bracket and the cup product. We will recover their results using the minimal model of A.

We begin by noting that for $n \in \mathbb{N}$ we have that $\mathrm{Tor}_A^{n+1}(\Bbbk,\Bbbk)$ is one dimensional spanned by the class of the chain

$$\varepsilon_n = [\alpha_1|\alpha_2\cdots\alpha_r\alpha_1|\cdots|\alpha_2\cdots\alpha_r\alpha_1]$$

while, as usual, Tor_A^1 is spanned by the arrows. One way to see this is to argue that the Anick resolution of the trivial A-module $\Bbbk$, which has the overlappings

$$\tau_n = \alpha_1(\alpha_2\cdots\alpha_r\alpha_1)^n$$

corresponding to ε_n as A-module generators, is minimal, and hence these generate the corresponding Tor-groups. The minimal model B has then r generators in degree 0, and we write ε_0 the one corresponding to $[\alpha_1]$, and for each $n \in \mathbb{N}$ a generator ε_n in degree n whose differential is, by the main result of [171], as follows:

$$\partial(\varepsilon_n) = \sum_{s+t=n-1} (-1)^s \varepsilon_s\alpha_2\cdots\alpha_r\varepsilon_t.$$

We are intentionally suppressing the sign given by the binomial coefficient since in this case there is exactly one non-vanishing higher coproduct, Δ_{r+1}.

To find $\mathrm{HH}^*(A)$, observe that for each natural number $n \in \mathbb{N}_0$ there is an obvious cycle f_n of degree $-n$ in $\mathrm{Der}(B,A)$ such that $f_n(\varepsilon_n) = \varepsilon_0$, and one can check, as it is done in [156], that it provides a generator for $\mathrm{HH}^{n+1}(A)$, which is therefore one dimensional. We will now find a derivation of

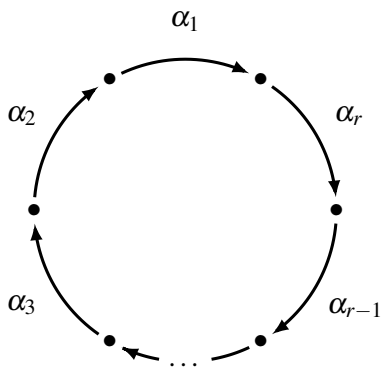


Figure 4.1: The wheel

B that covers f_n under $\alpha : B \longrightarrow A$, and then compute the Gerstenhaber bracket with these cycles: since the arrow α_* is a quasi-isomorphism, we deduce that these cycles represent generators for the cohomology groups of $\mathrm{Der}(B)$.

To do this, let us fix $n \in \mathbb{N}_0$ and let F be a derivation of degree $-n$ such that $F(\varepsilon_n) = \varepsilon_0$. Recursively solving for the values of F on generators using the equation

$$\partial F = (-1)^n F \partial,$$

shows that the following choice of lift works:

$$F_{2n}(\varepsilon_t) = (t-2n+1)\varepsilon_{t-2n}, \quad F_{2n+1}(\varepsilon_t) = \begin{cases} \varepsilon_{t-2n-1}, & \text{for } t \text{ odd} \\ 0, & \text{for } t \text{ even.} \end{cases}$$

With this choice of generators of $\mathrm{HH}^*(A)$, we compute that

$$[F_m, F_n] = \begin{cases} (m-n)F_{m+n}, & \text{for } m,n \text{ even,} \\ 0, & \text{for } m,n \text{ odd,} \\ mF_{m+n}, & \text{for } n \text{ even, } m \text{ odd.} \end{cases}$$

This coincides with the formulas obtained in [156]. However, observe that since we are using the natural grading in $\mathrm{Der}(B)$, the derivations in odd degree represent elements of even degree in $\mathrm{HH}^*(A)$, and those of even degree represent elements of odd degree in $\mathrm{HH}^*(A)$, which explains the shift in our formulas.

Note that since B has no quadratic part in its differential, the cup product structure in $\mathrm{HH}^*(A)$ is trivial. This means, in particular, that the Gerstenhaber algebra structure on $\mathrm{HH}^*(A)$ is independent of the parameter r, but one can check that the higher products can be used to distinguish them: the A_∞-structures on Hochschild cohomology are not quasi-isomorphic for distinct parameters, which shows in particular that these algebras are not derived equivalent; that is, their derived categories are not equivalent.

We now observe that $\mathrm{HC}_*(A)$ is easily computable: it is nothing but the homology of the abelianization of B, and this has a simple description. Indeed, since B is quasi-free, the space $B/[B,B]$ is spanned by equivalence classes of cyclic words in B with respect to cyclic shifts. Moreover, the differential of B in 3.3 lands in $[B,B]$: this follows from the fact that the non-cyclic arrow ε_n lies in the commutator subspace, and the differential preserves it, hence we deduce that $\mathrm{HC}_*(A) = B/[B,B]$.

A non-3-Koszul algebra. Let us consider the following quiver Q with relations $R = \{xy^2, y^2z\}$. We will compute its minimal model and with it its Hochschild cohomology, including the bracket. We will also compute the cup product; since the coproduct on $\mathrm{Tor}_A(\Bbbk, \Bbbk)$ is non-vanishing only on the generator which we call Γ, this computation is straightforward.

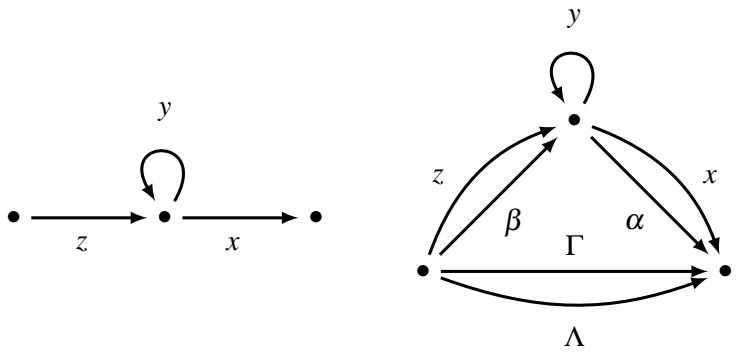


Figure 4.2: A quiver and its dg resolution with respect to the relations R

By the main result in [171], the algebra $A = \Bbbk Q/(xy^2, y^2z)$ has minimal model B given by the free algebra over the semisimple $\Bbbk$-algebra of vertices $\underline{\Bbbk}$ with set of homogeneous generators $\{x, y, z, \alpha, \beta, \Gamma, \Lambda\}$ such that

$$\partial x = \partial y = \partial z = 0, \quad \partial\alpha = xy^2, \quad \partial\beta = y^2z, \quad \partial\Gamma = \alpha z - x\beta, \quad \partial\Lambda = xy\beta - \alpha yz.$$

Here Γ corresponds to the overlap xy^2z while Λ corresponds to the overlap xy^3z, and α and β correspond to the relations they cover under the differential ∂ of the model, so that x, y and z are in degree 0, α and β in degree 1, and Γ and Λ in degree 2. Thus B is the path algebra of the dg quiver in Figure 2.
The elements in B are the following, where r, t are elements of $\{0, 1\}$ and $s \in \mathbb{N}_0$:

- *degree zero:* $x^r y^s z^t$,
- *degree one:* $\alpha y^s z^t, x^r y^t \beta$,
- *degree two:* $\Gamma, \Lambda, \alpha y^s \beta$.

Since we know that $\mathrm{Tor}_A^{\geqslant 4}(\Bbbk, \Bbbk)$ is zero, it follows that $\mathrm{HH}^{\geqslant 4}(A)$ is zero. Moreover, it is straightforward to see that $\mathrm{HH}^0(A) = Z(A)$ has basis

$$\{xyz, xz, y^2, y^3, \ldots\},$$

so we may focus our attention on derivations of degree 0, -1 and -2 to obtain bases for the remaining groups $\mathrm{HH}^1(A)$, $\mathrm{HH}^2(A)$ and $\mathrm{HH}^3(A)$.
The following is a basis of derivations for the $\underline{\Bbbk}$-bilinear 0-cycles, where $s \in \mathbb{N}_0$, and we adopt the convention that $\alpha y^{-1}\beta = \Lambda$ and $\alpha y^{-2}\beta = \Gamma$:

$$\begin{aligned}
&E_s(x) = 0, && E_s(y) = y^{s+1}, && E_s(z) = 0,\\
&E_s(\alpha) = 2\alpha y^s, && E_s(\beta) = 2y^s\beta && E_s(\Lambda) = 3\alpha y^{s-1}\beta,\\
&E_s(\Gamma) = -2\alpha y^{s-2}\beta,
\end{aligned}$$

$$
\begin{aligned}
&F_s(x) = xy^s, && F_s(y) = 0, && F_s(z) = 0,\\
&F_s(\alpha) = \alpha y^s, && F_s(\beta) = 0 && F_s(\Lambda) = \alpha y^{s-1}\beta,\\
&F_s(\Gamma) = -\alpha y^{s-2}\beta,\\
&G_s(x) = 0, && G_s(y) = 0, && G_s(z) = y^s z,\\
&G_s(\alpha) = 0, && G_s(\beta) = y^s\beta, && G_s(\Lambda) = \alpha y^{s-1}\beta,\\
&G_s(\Gamma) = -\alpha y^{s-2}\beta.
\end{aligned}
$$

We now compute the $\underline{\Bbbk}$-bilinear 0-boundaries. A basis for them is given by the following family of derivations, where $s \in \mathbb{N}_0$:

$$
\begin{aligned}
&T_s(x) = xy^{s+2}, && T_s(y) = 0, && T_s(z) = 0, && T_s(\alpha) = \alpha y^{s+2},\\
&T_s(\beta) = 0, && T_s(\Lambda) = \alpha y^{s+1}\beta, && T_s(\Gamma) = -\alpha y^s\beta,\\
&R_s(x) = 0, && R_s(y) = 0, && R_s(z) = y^{s+2}z, && R_s(\alpha) = 0,\\
&R_s(\beta) = y^{s+2}\beta, && R_s(\Lambda) = -\alpha y^{s+1}\beta, && R_s(\Gamma) = \alpha y^s\beta,,
\end{aligned}
$$

By direct inspection, we have that $F_{s+2} = T_s, G_{s+2} = R_s$ for $s \in \mathbb{N}_0$, and no other relations, so that $H^0(\mathrm{Der}(B))$ has infinite dimension and is spanned by the classes in the set $\{F_0, F_1, G_0, G_1, E_s : s \in \mathbb{N}_0\}$. Moreover, a basis for $H^0(B)$ is of course given by the monomials belonging to A. We may only worry about the adjoint action on B of y and e_2, since they are the only ones that are $\underline{\Bbbk}$-bilinear. We have that

$$
\begin{aligned}
&\mathrm{Ad}_y(x) = -xy, && \mathrm{Ad}_y(y) = 0, && \mathrm{Ad}_y(z) = yz,\\
&\mathrm{Ad}_y(\alpha) = -\alpha y, && \mathrm{Ad}_y(\beta) = y\beta,\\
&\mathrm{Ad}_y(\Lambda) = 0, && \mathrm{Ad}_y(\Gamma) = 0.
\end{aligned}
$$

so that $F_1 + \mathrm{Ad}_y = G_1$. Similarly, $F_0 + \mathrm{Ad}_{e_2} = G_0$, so that $\mathrm{HH}^1(A)$ is infinite dimensional with basis the classes in $\{F_0, G_0, E_s : n \in \mathbb{N}_0\}$. Moreover, for each $s,t \in \mathbb{N}$,

$$[E_s, E_t] = (s-t)E_{s+t},$$

and that $[F_0, -]$ and $[G_0, -]$ are identically zero. This determines $\mathrm{HH}^1(A)$ as a Lie algebra: it consists of abelian algebra $\Bbbk^2$ acting trivially on the Witt algebra. The following derivations form a basis of the 1-cycles in $\mathrm{Der}(B)$, (unspecified values are zero and we agree that $y^{-1} = y^{-2} = 0$):

$$
\begin{aligned}
&\Phi_s(\alpha) = xy^s, && \Phi_s(\beta) = y^s z, && \Phi_s(\Lambda) = \alpha y^{s-1}z, && \Phi_s(\Gamma) = -\alpha y^{s-2}z,\\
&\Phi'_s(\alpha) = 0, && \Phi'_s(\beta) = y^{s+2}z, && \Phi'_s(\Lambda) = -\alpha y^{s+1}z, && \Phi'_s(\Gamma) = \alpha y^s z,
\end{aligned}
$$

$$\begin{array}{llll}
\Pi_s(\alpha)=0, & \Pi_s(\beta)=y^{s+2}, & \Pi_s(\Lambda)=\alpha y^{s+1}, & \Pi_s(\Gamma)=\alpha y^s,\\
\Pi'_s(\alpha)=xy^sz, & \Pi'_s(\beta)=0, & \Pi'_s(\Lambda)=0, & \Pi'_s(\Gamma)=0,\\
\Psi_s(\alpha)=0, & \Psi_s(\beta)=y^{s+2}z, & \Psi_s(\Lambda)=-\alpha y^{s+1}z, & \Psi_s(\Gamma)=xy^s\beta,\\
\Theta_s(\alpha)=0, & \Theta_s(\beta)=0, & \Theta_s(\Lambda)=\alpha y^sz-xy^s\beta, & \Theta_s(\Gamma)=0,\\
\Xi_s(\alpha)=0, & \Xi_s(\beta)=0, & \Xi_s(\Lambda)=0, & \Xi_s(\Gamma)=\Theta_s(\Lambda).
\end{array}$$

Let us now compute the 1-boundaries of $\mathrm{Der}(B)$. We observe that for every $s\in\mathbb{N}_0$, the elements $\Xi_s,\Phi'_s,\Psi_s,\Theta_s$ have zero projection onto A under $\alpha:B\longrightarrow A$, so these are boundaries. It is easy to check that the following set of derivations completes the list of 1-boundaries, where $s\in\mathbb{N}_0$:

$$\begin{array}{llll}
X_s(\alpha)=xy^{s+2}, & X_s(\beta)=0 & X_s(\Lambda)=xy^{s+1}\beta, & X_s(\Gamma)=-xy^s\beta\\
Y_s(\alpha)=2xy^{s+1}, & Y_s(\beta)=2y^{s+1}z, & Y_s(\Lambda)=xy^s\beta-\alpha y^sz, & Y_s(\Gamma)=0,
\end{array}$$

Moreover, we have that $\alpha(\Phi_{s+2}+\Phi'_s)=0$, so Φ_{s+2} is a boundary for $s\in\mathbb{N}_0$. It follows that a basis of $H^1(\mathrm{Der}(B))$ is given by the classes of the derivations Φ_0,Φ_1 so that $\mathrm{HH}^2(A)$ is two dimensional.

Every derivation of degree -2 is a cycle and vanishes on every generator except, possibly, Λ and Γ, so that a basis for the 2-cycles is given by the following family of derivations, where $s\in\mathbb{N}_0$ and $t\in\{0,1\}$:

$$\Omega^t_s(\Lambda)=0, \qquad \Omega^t_s(\Gamma)=xy^sz^t, \qquad \Upsilon^t_s(\Lambda)=xy^sz^t, \qquad \Upsilon^t_s(\Gamma)=0.$$

It is straightforward to check that all of these are boundaries except for Υ^1_0 and Υ^0_0. More precisely, the following is a complete list of the 2-boundaries, where $s\in\mathbb{N}_0$ and $t\in\{0,1\}$:

$$\Omega^t_s(\Lambda)=0, \qquad \Omega^t_s(\Gamma)=xy^sz^t, \qquad \Upsilon^t_{s+1}(\Lambda)=xy^{s+1}z^t, \qquad \Upsilon^t_{s+1}(\Gamma)=0.$$

From this it follows that $\mathrm{HH}^3(A)$ is also two dimensional. Finally, we compute the Gerstenhaber algebra structure. We already determined the bracket in $\mathrm{HH}^1(A)$, while the bracket in $\mathrm{HH}^2(A)$ is trivial, since both generators vanish on Λ and Γ. The action of $\mathrm{HH}^1(A)$ on $\mathrm{HH}^2(A)$ and $\mathrm{HH}^3(A)$ is as follows, where $s,t\in\mathbb{N}_0$ and $r\in\{0,1\}$.

$$\begin{array}{ll}
[E_{s+2},\Phi_t]=3\Xi_{s+t+1}-2\Theta_{s+t}, & [F_{s+2},\Phi_t]=\Theta_{t+s+1}-\Xi_{t+s},\\
[G_{s+2},\Phi_t]=\Theta_{t+s+2}-\Xi_{t+s+2}, & [F_1,\Phi_t]=[G_1,\Phi_t]=\Theta_t,\\
[E_s,\Upsilon^r_t]=(t-3\delta_{s,0})\Upsilon^r_{s+t}. & [E_s,\Omega^r_t]=(t+2\delta_{s,0})\Omega^r_{s+t},\\
[F_0,-]=[G_0,-]=2 & \text{on } \langle\Omega^t_s,\Omega'^t_s : s\in\mathbb{N}_0\rangle,\\
[F_0,-]=[G_0,-]=[E_0,-]=0 & \text{on } \langle\Upsilon^t_s,\Upsilon'^t_s,\Phi_s : s\in\mathbb{N}_0\rangle,\\
[E_1,\Phi_t]=3\Xi_t. &
\end{array}$$

To compute cyclic and Hochschild homology of A, we begin by noting that for $i \in \mathbb{N}$, we have that $[B,B]_i = B_i$, and that $(B/[B,B])_0$ is generated by the classes of y^j for $j \in \mathbb{N}_0$. This means that $\mathrm{HC}_*(A)$ is concentrated in degree zero where it coincides with $A/[A,A] = \Bbbk[\bar{y}]$. The long exact sequence shows that $\mathrm{HH}_*(A)$ is trivial in degrees larger than 1, and that $d : \mathrm{HH}_0(A) \to \mathrm{HH}_1(A)$ is an isomorphism.

The case of dg algebras that are not 'aspherical'. In the above, we focused our attention on cofibrant algebras B for which $H_*(B)$ is concentrated in degree zero. It is important to point out the work done here works equally well if B is only assumed to be cofibrant and non-negatively graded. Indeed, the only crucial property we used is that B is free as an associative algebra, so that the complexes of fields and forms compute what they are supposed to.

Chapter 5

A_∞-algebras and support variety theory

In Section 1, we discuss the combinatorics of perfect cycles for monomial algebras due to Chen, Shen, and Zhou, and present an elegant combinatorial description of Anick chains for Gorenstein monomial algebras. In Section 2, we recall the **FG** conditions of Snashall–Solberg, and reinterpret them in terms of higher commutators and A_∞-centres of $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$. We use the previously established vanishing patterns of higher structures to obtain a combinatorial criterion for A_∞-centrality in the monomial case. In Section 3, we attach "periodicity operators" in $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ to any stable relation cycle, prove that periodicity operators are A_∞-central, and present the ring of periodicity operators explicitly. In Section 4, we prove the main result of this chapter, namely that the monomial algebras satisfying the **FG** conditions are exactly the Gorenstein ones. In Section 5, we offer a combinatorial characterisation of the **FG** conditions, and record some interesting results on the structure of Hochschild cohomology.

1 Perfect paths and Gorenstein algebras

1.1 Gorenstein-projective modules and related combinatorics. Let us recall that a module $M \in \operatorname{mod}\Lambda$ is called Gorenstein-projective if it satisfies the following conditions:

i) There exists an acyclic complex of finite projectives

$$C_* : \cdots \to C_3 \xrightarrow{d} C_2 \xrightarrow{d} C_1 \xrightarrow{d} C_0 \xrightarrow{d} C_{-1} \xrightarrow{d} C_{-2} \to \cdots \tag{5.1}$$

such that $\operatorname{coker}(C_1 \xrightarrow{d} C_0) \cong M$.

ii) The dual complex $\operatorname{Hom}_\Lambda(C_*,\Lambda)$ is also acyclic.

In this case the non-negative truncation of C_* resolves M, and we call C_* a complete resolution of M. Equivalently [35, Chp. 4], M is Gorenstein-projective if and only if

$$\mathrm{Ext}^i_\Lambda(M,\Lambda) = 0 \text{ and } \mathrm{Ext}^i_{\Lambda^{op}}(M^*,\Lambda^{op}) = 0 \text{ for all } i > 0 \tag{5.2}$$

and $M \cong M^{**}$ is reflexive, where $M^* = \mathrm{Hom}_\Lambda(M,\Lambda)$ is the dual module.
In [39], Chen–Shen–Zhou gave a full classification of the indecomposable Gorenstein-projectives over monomial algebras. Let us recall the combinatorial notions that underpin that classification. Note that for a monomial algebra Λ, the coset of a path $p \in \mathscr{P}_Q$ is zero in Λ if and only if it is divisible by a relation. We let

$$\mathscr{P}_\Lambda = \{p \in \mathscr{P}_Q \mid p \text{ is not divisible by any relation}\} \tag{5.3}$$

be the subset of nonzero paths in Λ, and the canonical surjection $kQ \twoheadrightarrow \Lambda$ identifies $\mathscr{P}_\Lambda$ with a basis for Λ.
Given any path $p \in \mathscr{P}_\Lambda$, we define minimal left zero cofactors of p to be those non-trivial paths $q \in \mathscr{P}_\Lambda$ with $s(q) = t(p)$, $qp = 0$, and for which no proper end segment q' has the same property: if $q = q''q'$, and $q'p = 0$, then $q = q'$. Similarly one defines minimal right zero cofactors. We let

$$L(p) = \{q \in \mathscr{P}_\Lambda \mid q \text{ is a minimal left zero cofactors of } p\} \tag{5.4}$$
$$R(p) = \{q \in \mathscr{P}_\Lambda \mid q \text{ is a minimal right zero cofactors of } p\}. \tag{5.5}$$

Definition 10. A pair of non-trivial paths (p,q) in Λ with $s(p) = t(q)$ is a perfect pair if $R(p) = \{q\}$ and $L(q) = \{p\}$.

Definition 11. Let $\mathbf{p} = (p_0, p_1, \ldots, p_{r-1})$ be a sequence of non-trivial paths in Λ such that $s(p_i) = t(p_{i+1})$ for all $i \in \mathbb{Z}/r\mathbb{Z}$. We say that $\mathbf{p}$ is a perfect cycle if all pairs (p_i, p_{i+1}) are perfect; in such case we call the paths p_i perfect as well. If r is the minimal integer with this property, we call r the period of $\mathbf{p}$.

Given a perfect cycle $\mathbf{p} = (p_0, p_1, \ldots, p_{r-1})$, one may extend it by periodicity to $\mathbf{p} = (p_i)_{i\in\mathbb{Z}}$ by setting $p_{i+r} = p_i$. We then attach an unbounded, periodic complex of finite right projectives $Q^{(\mathbf{p})}_*$, with terms $Q^{(\mathbf{p})}_i = e_{s(p_i)}\Lambda$ and differentials given by left multiplication by the p_i:

$$Q^{(\mathbf{p})}_* : \cdots \to e_{s(p_{i+3})}\Lambda \xrightarrow{l_{p_{i+3}}} e_{s(p_{i+2})}\Lambda \xrightarrow{l_{p_{i+2}}} e_{s(p_{i+1})}\Lambda \xrightarrow{l_{p_{i+1}}} e_{s(p_i)}\Lambda \to \cdots \tag{5.6}$$

That this is a complex follows from the zero cofactor relation $p_ip_{i+1} = 0$ in Λ. Moreover, $Q^{(\mathbf{p})}_*$ is acyclic as each pair (p_i, p_{i+1}) is perfect [39, Prop. 4.4].
Let $\mathscr{GP}(\Lambda)$ be the set of isomorphism classes of non-projective, indecomposable Gorenstein-projective right modules over Λ.

Theorem 1.1 ([39, Th. 4.1]). *Let Λ be a monomial algebra. Then the map $p \mapsto p\Lambda$ is a bijection*

$$\{\text{perfect paths in } \mathscr{P}_\Lambda\} \leftrightarrow \mathscr{GP}(\Lambda).$$

Moreover, given a perfect path $p = p_0$ in a perfect cycle $\mathbf{p} = (p_0, p_1, \ldots, p_{r-1})$, the complex $Q_*^{(\mathbf{p})}$ is a complete resolution of $p_0\Lambda$, and in particular its non-negative truncation

$$\cdots \to e_{s(p_3)}\Lambda \xrightarrow{l_{p_3}} e_{s(p_2)}\Lambda \xrightarrow{l_{p_2}} e_{s(p_1)}\Lambda \xrightarrow{l_{p_1}} e_{s(p_0)}\Lambda \xrightarrow{l_{p_0}} p_0\Lambda \to 0 \tag{5.7}$$

is a minimal projective resolution of $p\Lambda = p_0\Lambda$.

A monomial algebra Λ may contain no perfect path and so may admit no non-trivial Gorenstein-projective module at all. At the opposite end of the spectrum, Gorenstein monomial algebras (of infinite global dimension) have an abundance of non-trivial Gorenstein- projective modules, and so have a large supply of perfect paths in view of the classification result of Chen–Shen–Zhou.

Recall that Λ is Gorenstein if the two-sided injective dimensions $\operatorname{injdim}{}_\Lambda\Lambda < \infty$ and $\operatorname{injdim}\Lambda_\Lambda < \infty$ are finite, in which case they are equal by a theorem of Zaks [182]. In this case, we denote this common number by $\dim\Lambda$ and call it the Gorenstein dimension of Λ. When Λ is Gorenstein, [35, 4.2] shows that $\operatorname{Ext}^i_\Lambda(M,\Lambda) = 0$ for all $i > 0$ is necessary and sufficient for M to be Gorenstein-projective and so the n-th syzygy $\Omega^n N$ of any module $N \in \operatorname{mod}\Lambda$ is Gorenstein-projective for $n \geq \dim\Lambda$.

For finite dimensional algebras, by work of Bergh–Jorgensen–Oppermann [17] one can in fact characterise the Gorenstein property for Λ in terms of the syzygies of $\underline{\Bbbk}$. We will use this in Section 3 to characterise Gorenstein monomial algebras in terms of Anick chains. The following easy consequence of [17, Th. 4.1] will be the relevant result for us.

Proposition 1.2. *Let $\Lambda = \Bbbk Q/I$ be a finite-dimensional path algebra. Then Λ is Gorenstein if and only if $\Omega^n\underline{\Bbbk}$ is Gorenstein-projective for some $n \geq 0$. In this case, the minimal such n equals the Gorenstein dimension of Λ.*

Proof. If Λ is Gorenstein then $\Omega^n\underline{\Bbbk}$ is Gorenstein-projective whenever $n \geq \dim\Lambda$. Conversely, if $\Omega^n\underline{\Bbbk}$ is Gorenstein-projective for some $n \geq 0$ then it is easy to see that the *Gorenstein defect category* (that is, the singularity category of Λ modulo Gorenstein-projectives, see [17]) vanishes by Jordan-Hölder filtration arguments, and this characterises the Gorenstein property by [17, Th. 4.1]. For the last statement, let Λ be Gorenstein and let n be the minimal such integer. It is immediate that $n \leq \dim\Lambda$. Since $\Omega^n\underline{\Bbbk}$ is Gorenstein-projective, we have

$$\operatorname{Ext}^{n+i}_\Lambda(\underline{\Bbbk},\Lambda) = \operatorname{Ext}^i_\Lambda(\Omega^n\underline{\Bbbk},\Lambda) = 0 \text{ for all } i > 0. \tag{5.8}$$

From Jordan-Hölder filtrations one obtains that $\operatorname{Ext}^{n+i}_\Lambda(N,\Lambda) = 0$ for all $i > 0$ and $N \in \operatorname{mod}\Lambda$, and so $\dim\Lambda \leq n$. □

1.2 Anick chains over Gorenstein algebras. We now know how to test the Gorenstein property for a monomial algebra Λ in terms of the syzygy modules $\Omega^n\underline{\Bbbk}$ via Proposition 1.2. In this section, we translate this into the combinatorial language of Anick chains, and deduce normal forms for chains of sufficiently large degree over Gorenstein monomial algebras.

Up to now we have taken modules without further qualifiers to mean right modules. However, certain results below will need to be stated for both left and right modules, and similarly results on the structure of Anick chains will need to be stated for both right and left Anick chains, which correspond to the right and left minimal projective resolution of $\underline{\Bbbk} = \Lambda/\mathfrak{r}$, respectively. In what follows we will continue working with right modules and right Anick chains, and simply note that all dual statements for left modules and chains can be obtained formally by passing to the opposite algebra $\Lambda \mapsto \Lambda^{\mathrm{op}}$.

We start by relating the syzygy modules to Anick chains.

Proposition 1.3. *Let Λ be a monomial algebra and $n \geq 1$. We have isomorphisms of right and left modules*

$$\Omega^n(\underline{\Bbbk}_\Lambda) \cong \bigoplus_{\gamma \in C_{n-1}} t_\gamma \Lambda, \qquad \Omega^n({}_\Lambda\underline{\Bbbk}) \cong \bigoplus_{\gamma \in C_{n-1}} \Lambda h_\gamma. \tag{5.9}$$

Proof. Recall that the minimal resolution of $(P_*, \partial) \xrightarrow{\sim} \underline{\Bbbk}$ is given by the Anick resolution

$$\cdots \to \underline{\Bbbk} C_{n-1} \otimes_{\underline{\Bbbk}} \Lambda \xrightarrow{\partial_{n-1}} kC_{n-2} \otimes_{\underline{\Bbbk}} \Lambda \xrightarrow{\partial_{n-2}} \cdots \xrightarrow{\partial_1} kC_0 \otimes_{\underline{\Bbbk}} \Lambda \xrightarrow{\partial_0} \Lambda \to 0$$

with n-th term $P_n = kC_{n-1} \otimes_{\underline{\Bbbk}} \Lambda$ and differential $\partial_{n-1}(\gamma) = \gamma' \otimes t_\gamma$, and we can identify $\Omega^n\underline{\Bbbk} = \mathrm{Im}(\partial_{n-1})$ explicitly. Consider the identification of projective modules

$$kC_{n-1} \otimes_{\underline{\Bbbk}} \Lambda = \bigoplus_{\gamma \in C_{n-1}} e_{s(\gamma)} \Lambda \tag{5.10}$$

given by sending $\gamma \otimes 1$ to $e_{s(\gamma)}$. The image of ∂_{n-1} then takes the form

$$\mathrm{Im}(\partial_{n-1}) = \sum_{\gamma \in C_{n-1}} t_\gamma \Lambda \subseteq \bigoplus_{\gamma' \in C_{n-2}} e_{s(\gamma')} \Lambda \tag{5.11}$$

where by abuse of notation we let $\gamma' \in C_{n-2}$ run over all $(n-2)$-chains, and we write a $(n-1)$-chain as $\gamma = \gamma' t_\gamma$ so that $t_\gamma \Lambda \subseteq e_{s(\gamma')}\Lambda$ sits as a submodule in the corresponding copy of $e_{s(\gamma')}\Lambda$. Moreover, here we must assume that $n \geq 2$, but the case $n = 1$ is treated similarly by replacing the righthand side by Λ.

Now note that this sum is in fact direct. Assume that $\gamma_1 = \gamma_1' t_{\gamma_1}$ and $\gamma_2 = \gamma_2' t_{\gamma_2}$ are $(n-1)$-chains and $t_{\gamma_1}\Lambda$, $t_{\gamma_2}\Lambda$ are sent into the same summand $e_{s(\gamma')}\Lambda$, meaning that $\gamma_1' = \gamma_2' = \gamma'$. If $t_{\gamma_1}\Lambda \cap t_{\gamma_2}\Lambda \neq 0$, then there is a path left divisible by both tails t_{γ_1} and t_{γ_2}, and so one tail must divide the other. As

$\gamma_1' = \gamma_2'$, one of γ_1, γ_2 then divides the other, contradicting minimality. Hence we obtain:

$$\mathrm{Im}(\partial_{n-1}) = \bigoplus_{\gamma \in C_{n-1}} t_\gamma \Lambda \subseteq \bigoplus_{\gamma' \in C_{n-2}} e_{s(\gamma')} \Lambda, \tag{5.12}$$

as required. The second claim is dual. □

Our next step is to characterise Gorenstein monomial algebras in terms of Anick chains. We first need two auxiliary statements. The first of them collects several results of [39].

Lemma 1.4 (See Lem. 3.1, Prop. 4.6 in [39])**.** *Let Λ be a monomial algebra and p a non-trivial path in Λ. Then:*

(1) $p\Lambda$ is Gorenstein-projective and non-projective if and only if $R(p) = \{q\}$ for q a perfect path.
(2) Λq is Gorenstein-projective and non-projective if and only if $L(q) = \{p\}$ for p a perfect path.
(3) $p\Lambda$ is projective if and only if $R(p)$ is empty.
(4) Λq is projective if and only if $L(q)$ is empty.

Remark 3. This lemma is "best possible" in that Chen-Shen-Zhou construct [39, Ex. 4.5] an example of a non-perfect path p with Λp Gorenstein-projective, along with an isomorphism $\Lambda p \cong \Lambda p'$ for some perfect path p'.

The second result we need is as follows.

Lemma 1.5. *Let Λ be a Gorenstein algebra, and let $(P_*, \partial) \xrightarrow{\sim} \underline{\Bbbk}$ be the minimal projective resolution. If $\Omega^n \underline{\Bbbk} = \mathrm{Im}(\partial : P_n \to P_{n-1})$ has a non-trivial projective summand, then $n \leq \dim \Lambda$.*

Proof. Decompose $\Omega^n \underline{\Bbbk} = N \oplus P$ with P a non-trivial projective summand. Consider the short exact sequence

$$0 \to \Omega^n \underline{\Bbbk} \xrightarrow{\iota} P_{n-1} \xrightarrow{\partial} \Omega^{n-1} \underline{\Bbbk} \to 0. \tag{5.13}$$

If $n > \dim \Lambda$, we show that the embedding $\iota_{|P} : P \hookrightarrow P_{n-1}$ splits, and this will contradict minimality of the resolution. Consider the commutative diagram of short exact sequences of Figure 5.1. A diagram chase shows that we have an induced exact sequence

$$0 \to N \to M \to \Omega^{n-1} \underline{\Bbbk} \to 0. \tag{5.14}$$

Now, Gorenstein-projectives are closed under summands and extensions, and must be projective the moment they have finite projective dimension. If $n > \dim \Lambda$, then $\Omega^{n-1} \underline{\Bbbk}$ and $\Omega^n \underline{\Bbbk}$, and therefore also N, are Gorenstein-projective, and thus so is the extension M. But the diagonal short exact sequence shows that $\mathrm{projdim}\, M < \infty$ and so M must be projective. Hence the map $\iota_{|P} : P \hookrightarrow P_{n-1}$ is split as claimed, contradicting minimality of the resolution. □

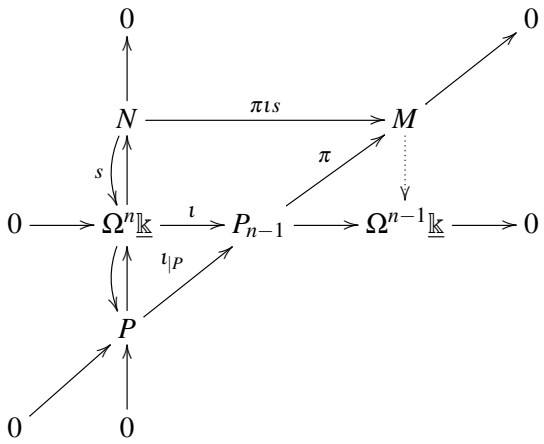


Figure 5.1: The diagram of Lemma 1.5

Results of Lu–Zhu [133, Rem. 4.12] give a simple description of the self-injective monomial algebras: these are precisely the algebras $\Lambda = \Lambda_{m,n}$ whose quiver is a simple oriented cycle on m vertices with relations consisting of all paths of length n. We now show how to characterise the Gorenstein algebras of dimension at most $d+1$ in terms of the structure of d-chains; note that this forces $0 \leq d < \infty$, and so our results complement the case of Gorenstein dimension zero of Lu and Zhu. Their paper also characterises monomial algebras of Gorenstein dimension at most 1; we leave it to the reader to compare [133, Th. 5.4] with the corresponding particular case of the result below.

Theorem 1.6. *Let Λ be a monomial algebra. The following are equivalent, for $d \geq 0$:*

i) Λ is Gorenstein of dimension at most $d+1$.

ii) For every d-chain γ, either $R(t_\gamma)$ is empty or equal to $\{p\}$ with p perfect.

iii) For every d-chain γ, either $L(h_\gamma)$ is empty or equal to $\{q\}$ with q perfect.

Moreover if $\dim\Lambda < d+1$, then t_γ and h_γ themselves are perfect, and in particular the sets above are never empty.

Proof. Prop. 1.2 and Prop. 1.3 show that Λ is Gorenstein of dimension $\dim\Lambda \leq d+1$ if and only if $\Omega^{d+1}\underline{\Bbbk} = \bigoplus_{\gamma\in C_d} t_\gamma\Lambda$ is Gorenstein-projective, and by Lemma 1.4 this occurs if and only if $R(t_\gamma)$ is empty or consists of a single perfect path. This shows the equivalence i)-ii), with i)-iii) formally dual.

For the last claim assume that actually $\dim\Lambda \leq d$. First assume that $d \geq 1$. Let $\gamma \in C_d$, and write it as $\gamma = \gamma' t_\gamma$ for $\gamma' \in C_{d-1}$. By the above we have $R(t_{\gamma'}) = \emptyset$ or $R(t_{\gamma'}) = \{p\}$ for p perfect. But $t_\gamma \in R(t_{\gamma'}) \neq \emptyset$, which shows that $t_\gamma = p$ is perfect. Dually h_γ is perfect.

For the case $d = 0$ we have $\gamma = t_\gamma$, which must then be an arrow. Lemma 1.5 shows that $t_\gamma\Lambda$ cannot be projective and so $R(t_\gamma) \neq \emptyset$, which forces $R(t_\gamma) = \{p\}$ for p perfect. Letting $L(p) = \{q\}$, we see

that $t_\gamma p = 0$ forces q to be a right divisor of t_γ. But t_γ is an arrow and so $t_\gamma = q$ is perfect, as claimed. The case of h_γ is dual as before. □

Ignoring the precise dimension, we obtain a cleaner characterisation of Gorenstein monomial algebras.

Corollary 1.7. *Let Λ be a monomial algebra. The following are equivalent:*

i) The algebra Λ is Gorenstein.

ii) There is $n_0 \in \mathbb{N}$ such that every n-chain γ for $n \geq n_0$ has tail $t_\gamma = p$ given by a perfect path.

iii) There is $n_0 \in \mathbb{N}$ such that every n-chain γ for $n \geq n_0$ has head $h_\gamma = q$ given by a perfect path.

As an example of how our results can be applied, let us give a characterisation of local Gorenstein monomial algebras. By a local monomial algebra, we mean algebras Λ of the form $\Lambda = \Bbbk\langle S\rangle/I$ with S a non-empty finite set and $I \subseteq (S)^2$ generated by monomial relations. (We still assume Λ finite dimensional.)

Proposition 1.8. *Let $\Lambda = \Bbbk\langle S\rangle/I$ be a local monomial algebra. Then Λ is Gorenstein if and only if $\Lambda \cong \Bbbk[t]/(t^n)$ for some $n \geq 2$.*

Proof. We show that $|S| \geq 2$ implies that Λ is not Gorenstein. Let $s \neq t \in S$ be distinct elements, and note that $s^m, t^n \in R$ for some minimal $m,n \geq 2$ by finite dimensionality of Λ. The elements $t, t^n, t^{n+1}, t^{2n}, \ldots, t^{kn}, t^{kn+1}, \ldots$ are all Anick chains and in particular t^{kn+1} has tail t. If Λ is Gorenstein then t^{kn+1} has perfect tail for $k \gg 0$ and so t must be a perfect path. Similarly s is perfect.
We have $(ts)^l \in I$ for $l \gg 0$ and so the set of relations of Λ must contain a string of the form $tstst\ldots$ or $ststs\ldots$, say the first one without loss of generality. The perfect pair (t, t^{n-1}) then shows that t^{n-1} left divides $stst\ldots$, a contradiction. □

We were not able to locate a precise reference for Proposition 1.8 in the literature, although we have no doubts that it is known to experts. In particular, it admits the following short proof not relying on combinatorics of Anick chains. A local Gorenstein algebra A must be self-injective, as follows from the Auslander–Buchsbaum Formula for noncommutative local rings of Wu-Zhang (see [181, Th. 0.3], using $\mathrm{Hom}_k(A,k)$ for the pre-balanced dualizing complex). Applying [133, Rem. 4.12], we see that A is a local Gorenstein monomial algebra if and only if $A \cong \Bbbk[t]/(t^n)$ for some n.

1.3 Example of a Gorenstein monomial algebra. Let $d \geq 2$ and $\Lambda_d = \Bbbk Q_d/I_d$, with the quiver Q_d as shown in the picture and relations $I_d = (R_d)$ given by $R_d = \{\beta_1\beta_2,\ \beta_2\beta_1,\ \alpha_i\alpha_{i+1} \mid 1 \leqslant i \leqslant d-1\}$. Then Λ_d is Gorenstein of dimension d.
Indeed all Anick $(d-1)$-chains are of the form $\beta_1\beta_2\beta_1\ldots$ or $\beta_2\beta_1\beta_2\ldots$ with tails β_i perfect, or by the path $\alpha_1\alpha_2\ldots\alpha_d$ whose tail α_d satisfies $R(\alpha_d) = \emptyset$, and so Λ_d is Gorenstein of dimension $\leq d$ by Theorem 1.6. However $\alpha_1\ldots\alpha_{d-1}$ is an $(d-2)$-chain with tail α_{d-1} and $R(\alpha_{d-1}) = \{\alpha_d\}$ is neither empty nor is α_d perfect, and so Λ_d is not Gorenstein of dimension $\leq d-1$. Finally, note that $\alpha_d\Lambda_d$ is a non-trivial projective summand in $\Omega^d\Bbbk$, and that such projective summands do not occur in higher degrees as all further Anick chains are given by $\beta_1\beta_2\beta_1\ldots$ and $\beta_2\beta_1\beta_2\ldots$ with tails β_i.

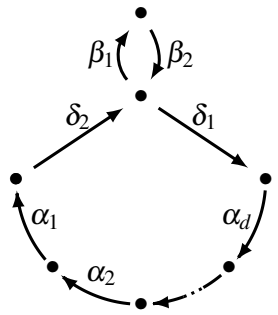


Figure 5.2: The algebra Λ_d.

1.4 Perfect walks In analysing Anick chains of sufficiently high degree over Gorenstein algebras, it will turn out useful to introduce paths which consist of walking along a perfect cycle.

Definition 12. A sequence of paths $(p_0, p_1, \dots, p_{l-1})$ is called a perfect walk if it can be extended to a perfect cycle $\mathbf{p} = (p_0, p_1, \dots, p_{l-1}, p_l, \dots, p_{s-1})$. We call $l \geq 1$ the length of the walk.

Note that the next and previous paths in a perfect walk are uniquely determined as we have $R(p_i) = \{p_{i+1}\}$ and $L(p_i) = \{p_{i-1}\}$, and in particular any perfect path $p = p_0$ can be extended in both directions to a perfect walk of any length

$$\dots, p_{-2}, p_{-1}, p_0, p_1, p_2, \dots \tag{5.15}$$

By abuse of notation we will often say that a path $\alpha = p_0 p_1 \dots p_l$ is given by a perfect walk if $(p_0, p_1, \dots, p_l)$ forms a perfect walk. For instance, the paths $\beta_1\beta_2\beta_1\dots$ and $\beta_2\beta_1\beta_2\dots$ of example in Section 1.3 are given by perfect walks. We note that the decomposition of a path as a perfect walk is not in general unique.

Example 1. Let $\Lambda = k[t]/(t^n)$ for $n \geq 2$. Then t^{kn} can be written as a perfect walk in two ways:

$$t^{kn} = t \cdot t^{n-1} \cdots t \cdot t^{n-1} = t^{n-1} \cdot t \cdots t^{n-1} \cdot t \tag{5.16}$$

corresponding to the perfect walks $(t, t^{n-1}, \dots, t, t^{n-1})$ and $(t^{n-1}, t, \dots, t^{n-1}, t)$.

In the above example, the path t^{kn} forms a $(2k-1)$-chain, and one might be tempted to think that all perfect walks will give rise to Anick chains. This is not so:

Example 2. Let $\Lambda = \Bbbk[t]/(t^n)$ for $n \geq 3$. Consider the perfect cycle (t^{n-1}, t). Then (t^{n-1}, t, t^{n-1}) is a perfect walk as it can be extended to (t^{n-1}, t, t^{n-1}, t) but $t^{2n-1} = t^{n-1} \cdot t \cdot t^{n-1}$ is not an Anick chain.

We will be particularly interested in Anick chains that can be (partially) written as perfect walks, and so we will derive a few properties of such chains. The following lemma easily follows from the definition, we leave the proof to the reader.

Lemma 1.9 (Extending Anick chains)**.** *Let γ_n be an n-chain. Then:*

i) $\gamma_{n+1} = \gamma_n p$ *is an* $(n+1)$*-chain with tail* $t_{\gamma_{n+1}} = p$ *if and only if* $p \in R(t_{\gamma_n})$.

ii) $\gamma_{n+1} = q\gamma_n$ *is an* $(n+1)$*-chain with head* $h_{\gamma_{n+1}} = q$ *if and only if* $q \in L(h_{\gamma_n})$.

Proposition 1.10 (Weak unique extension property)**.** *Let γ_n be an n-chain. Then:*

i) If $t_{\gamma_n} = p_0$ *is perfect, then* $\gamma_{n+1} = \gamma_n p_1$ *is the unique right extension of* γ *as an* $(n+1)$*-chain, then with tail* p_1.

ii) If $h_{\gamma_n} = q_0$ *is perfect, then* $q_{-1}\gamma_n$ *is the unique left extension of* γ *as an* $(n+1)$*-chain, then with head* q_{-1}.

Proof. This follows from Lemma 1.9 as $R(p_0) = \{p_1\}$ and $L(q_0) = \{q_{-1}\}$. □

Corollary 1.11 (Unique extension property)**.** *Let γ be an n-chain and α some non-trivial path in Q.*

i) If $t_\gamma = p_0$ *is perfect, then* $\gamma\alpha$ *is an Anick* $(n+k)$*-chain for some* $k \geq 1$ *if and only if* $\alpha = p_1 \dots p_k$*, in which case* $t_{\gamma\alpha} = p_k$.

ii) If $h_\gamma = q_0$ *is perfect, then* $\alpha\gamma$ *is an Anick* $(n+k)$*-chain for some* $k \geq 1$ *if and only if* $\alpha = q_{-k} \dots q_{-1}$*, in which case* $h_{\alpha\gamma} = q_{-k}$.

Proof. We only prove the first claim as the proof of the second one is similar. The if direction follows from iterating Proposition 1.10, which shows $\alpha = p_1 \dots p_k$ has the required property. We prove the converse by induction starting with $k = 1$. Clearly if $\gamma\alpha$ is an $(n+1)$-chain then $\alpha = p_1$ by Proposition 1.10, and then $t_{\gamma\alpha} = p_1$.
For $k > 1$ we can write $\gamma\alpha = \eta t_{\gamma\alpha}$ for an $(n+k-1)$-chain η. Then one of γ and η divide the other, which forces $\eta = \gamma\alpha'$ as $n < n+k-1$ by minimality of Anick chains. By induction $\alpha' = p_1 \dots p_{k-1}$ and $t_{\gamma\alpha'} = p_{k-1}$. Then $\gamma\alpha = (\gamma\alpha')t_{\gamma\alpha}$ is an $(n+k)$-chain extending the $(n+k-1)$-chain $\gamma\alpha' = \gamma p_1 \dots p_{k-1}$ and so $t_{\gamma\alpha} = p_{n+k}$ by the base case. Thus $\alpha = \alpha' t_{\gamma\alpha} = p_1 \dots p_k$ and $t_{\gamma\alpha} = p_k$ as claimed. □

From this, we obtain a description of Anick chains of high degree over a Gorenstein monomial algebra.

Corollary 1.12. *Let Λ be a Gorenstein monomial algebra, let $d = \dim\Lambda$ and let γ_n be an n-chain for $n \geq d$. Then γ_n is extended from a $(d-1)$-chain by a perfect walk. More precisely:*

i) There is a $(d-1)$*-chain* γ_{d-1} *and a perfect walk* $(p_d, p_{d+1}, \dots, p_n)$ *such that* $\gamma_{d-1}p_d p_{d+1} \dots p_i$ *is an* i*-chain with tail* p_i *for* $i = d, d+1, \dots, n$ *and* $\gamma_n = \gamma_{d-1}p_d p_{d+1} \dots p_n$.

ii) There is a $(d-1)$-chain γ_{d-1} and a perfect walk $(q_{-n},\dots,q_{-d})$ such that $q_{-i}q_{-i+1}\dots q_{-d}\gamma_{d-1}$ is an i-chain with head q_{-i} for $i=d,d+1,\dots,n$ and $\gamma_n = q_{-n}q_{-n+1}\dots q_d\gamma_{d-1}$.

Proof. We prove i) as ii) is dual. Write $\gamma_n = \gamma_{n-1}t_{\gamma_n} = \dots = \gamma_{d-1}t_{\gamma_d}t_{\gamma_{d+1}}\dots t_{\gamma_n}$ for a $(d-1)$-chain γ_{d-1} and a suitable sequence of tails. Now the d-chain γ_d must have a perfect tail by Theorem 1.6 since $d+1 > d = \dim\Lambda$ and so we may set $t_{\gamma_d} = p_d$. The unique extension property then gives $t_{\gamma_{d+1}} = p_{d+1},\dots,t_{\gamma_n} = p_n$. □

In other words, this result shows that the structure of n-chains over Gorenstein monomial algebras eventually stabilises for $n \geq \dim\Lambda$ and becomes predictable, with the subchain γ_{d-1} of γ_n consisting of noise which we will be able to ignore.

2 The FG conditions and higher structures

2.1 Hochschild cohomology and the FG conditions. Let $\Lambda^{\mathsf{e}} := \Lambda^{\mathrm{op}} \otimes_k \Lambda$ be the enveloping algebra of Λ, and consider $\Lambda \in \operatorname{mod}\Lambda^{\mathsf{e}}$ with its natural module structure. Recall that the Hochschild cohomology ring $\mathrm{HH}^*(\Lambda,\Lambda) := \mathrm{Ext}^*_{\Lambda^{\mathsf{e}}}(\Lambda,\Lambda)$ is a graded-commutative algebra, and we let $\mathrm{HH}^{\mathsf{ev}}(\Lambda,\Lambda)$ denote the subalgebra of elements of even degree.
For any $M \in \operatorname{mod}\Lambda$, we have an algebra homomorphism

$$\varphi_M := M\otimes_\Lambda - : \mathrm{HH}^*(\Lambda,\Lambda) \to \mathrm{Ext}^*_\Lambda(M,M). \tag{5.17}$$

The map φ_M induces on $\mathrm{Ext}^*_\Lambda(M,N)$ for any $N \in \operatorname{mod}\Lambda$ the structure of a right graded module over Hochschild cohomology; moreover this is compatible with the left action coming from φ_N as these satisfy $\varphi_N(x)\cdot\theta = (-1)^{|x||\theta|}\theta\cdot\varphi_M(x)$ for all elements $\theta \in \mathrm{Ext}^*_\Lambda(M,N)$ and $x \in \mathrm{HH}^*(\Lambda,\Lambda)$. This further restricts to a right module structure over any subalgebra $\mathrm{H} \subseteq \mathrm{HH}^*(\Lambda,\Lambda)$. The finite generation (**FG**) conditions of Snashall–Solberg are then stated as follows:

FG1. The commutative algebra $\mathrm{HH}^{\mathsf{ev}}(\Lambda,\Lambda)$ contains a Noetherian graded subalgebra H with $\mathrm{H}^0 = \mathrm{HH}^0(\Lambda,\Lambda)$.
FG2. $\mathrm{Ext}^*_\Lambda(M,N)$ is a finitely generated H-module for all $M,N \in \operatorname{mod}\Lambda$.

We say that Λ satisfies **FG** if both conditions above hold. The **FG** conditions imply that $\mathrm{HH}^*(\Lambda,\Lambda)$ is module-finite over H, and so that $\mathrm{HH}^*(\Lambda,\Lambda)$ is itself a Noetherian algebra. Moreover, in the presence of **FG1**, by Jordan–Hölder filtration arguments it is enough to establish **FG2** in the case $M = N = \Bbbk$.

2.2 A_∞-centres of minimal A_∞-algebras. In [32] the image of the characteristic homomorphism

$$\varphi_M : \mathrm{HH}^*(\Lambda,\Lambda) \to \mathrm{Ext}^*_\Lambda(M,M) \tag{5.18}$$

was related for any $M \in \operatorname{mod}\Lambda$ to the A_∞-structure on $\operatorname{Ext}^*_\Lambda(M,M)$, where it was shown to consist of A_∞-central classes. Recall that a class $a \in E = (E, m_2)$ in a graded algebra is graded-central if $m_2(a,x) = (-1)^{|a||x|} m_2(x,a)$ for all $x \in E$. Equivalently, the inner derivation ad_a vanishes:

$$\operatorname{ad}_a(x) = [a,x] = m_2(a,x) - (-1)^{|a||x|} m_2(x,a) = 0. \tag{5.19}$$

We denote by $\mathscr{Z}_{gr}E = \{a \in E \mid \operatorname{ad}_a = 0\}$ the graded centre.
Generalising to $E = (E, \{m_n\}_{n\geq 2})$ a minimal A_∞-algebra, given $a \in E$, for each $n \geq 1$ we form the n-ary higher commutator

$$[a; x_1, \ldots, x_n]_{1,n} := \sum_{i=0}^{n} (-1)^i (-1)^{|a|(|x_1|+\cdots+|x_i|)}\, m_{n+1}(x_1, \ldots, x_i, a, x_{i+1}, \ldots, x_n). \tag{5.20}$$

In other words we apply m_{n+1} to the (signed) shuffle product $a \cdot_{\mathrm{sh}} (x_1 \otimes \cdots \otimes x_n)$. We then define the homotopy inner derivation $\operatorname{ad}_a = \{\operatorname{ad}_{a,n}\}_{n\geq 1}$ to be a collection of n-ary operations $\operatorname{ad}_{a,n} : E^{\otimes n} \to E$ given by $\operatorname{ad}_{a,n}(x_1, \ldots, x_n) := [a; x_1, \ldots, x_n]_{1,n}$. The collection ad_a forms a cocycle in the complex of *homotopy derivations*

$$\operatorname{ad}_a \in \operatorname{hoder}(E) := \left(\operatorname{Hom}_{\Bbbk}(\oplus_{n\geq 1} \overline{E}^{\otimes n}, E), \partial\right) \tag{5.21}$$

which is a subcomplex of the Hochschild cochain complex $C^*(E,E)$ of the A_∞-algebra E, see [32] for details.

Definition 13. The A_∞-centre of E is the space

$$\mathscr{Z}_\infty E := \{a \in E \mid [\operatorname{ad}_a] = 0 \text{ in } \mathrm{H}^*(\operatorname{hoder}(E))\}.$$

The A_∞-centre $\mathscr{Z}_\infty E \subseteq E$ is a graded subalgebra of the underlying graded algebra $E = (E, m_2)$, and A_∞-central classes are always graded-central, so that we have containment $\mathscr{Z}_\infty E \subseteq \mathscr{Z}_{gr} E$ with equality whenever $m_n = 0$ for all $n \geq 3$.
The image of the characteristic homomorphism $\varphi_{\underline{\Bbbk}} : \mathrm{HH}^*(\Lambda,\Lambda) \to \operatorname{Ext}^*_\Lambda(\underline{\Bbbk},\underline{\Bbbk})$ is always contained in the graded centre

$$\operatorname{im}(\varphi_{\underline{\Bbbk}}) \subseteq \mathscr{Z}_{gr}\operatorname{Ext}^*_\Lambda(\underline{\Bbbk},\underline{\Bbbk}) \tag{5.22}$$

with equality in the case Λ is Koszul by a theorem of Buchweitz–Green–Snashall–Solberg, but with proper inclusion in general. Since the Koszul case corresponds to the situation where one can take $m_n = 0$ for all $n \geq 3$ for the A_∞-structure on $\operatorname{Ext}^*_\Lambda(\underline{\Bbbk},\underline{\Bbbk})$, this result was refined in [32] as follows:

Theorem 2.1 (See [32])**.** *The image of the characteristic homomorphism is precisely the A_∞-centre:*

$$\operatorname{im}(\varphi_{\underline{\Bbbk}}) = \mathscr{Z}_\infty \operatorname{Ext}^*_\Lambda(\underline{\Bbbk},\underline{\Bbbk}). \tag{5.23}$$

On the level of the cohomology $\mathrm{H}^*(\mathrm{hoder}(E,E))$, reading off the A_∞-central condition $[\mathrm{ad}_a]=0$ can be subtle, and so in practice one may focus on the simpler sufficient condition that ad_a vanishes on the nose. Equivalently, for $a\in\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$, we are interested in the vanishing of higher commutators

$$\sum_{i=0}^{n}(-1)^i(-1)^{|a|(|x_1|+\cdots+|x_i|)}\,m_{n+1}(x_1,\ldots,x_i,a,x_{i+1},\ldots,x_n)=0 \tag{5.24}$$

for all $x_1,\ldots,x_n\in\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ and all $n\geq 1$. We record this as a corollary:

Corollary 2.2. *Let $a\in\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ be such that all higher commutators (5.24) vanish. Then a is in the image of $\varphi_\Bbbk:\mathrm{HH}^*(\Lambda,\Lambda)\to\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$.*

Finally, we can recast the **FG** conditions in terms of $\mathscr{Z}_\infty\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$. For that, we introduce another set of conditions **FG′**:

FG'1. $\mathscr{Z}_\infty:=\mathscr{Z}_\infty\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is a Noetherian algebra.

FG'2. $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is module-finite over $\mathscr{Z}_\infty$.

Proposition 2.3. *The conditions* **FG** *and* **FG′** *are equivalent.*

Proof. If Λ satisfies **FG** with regards to $\mathrm{H}\subseteq\mathrm{HH}^*(\Lambda,\Lambda)$ then $\varphi_\Bbbk(\mathrm{H})\subseteq\mathscr{Z}_\infty$ is a Noetherian algebra over which $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is module-finite and so **FG′** holds.
Conversely if Λ satisfies **FG′**, then $\mathscr{Z}_\infty$ is Noetherian and must be finitely generated as it is positively graded. Moreover, so must be its even subalgebra $\mathscr{Z}_\infty^{\mathsf{ev}}$. Picking lifts of said generators, we can form a Noetherian subalgebra $\mathrm{H}\subseteq\mathrm{HH}^{\mathsf{ev}}(\Lambda,\Lambda)$ satisfying **FG1**, possibly after adding $\mathrm{H}^0=\mathrm{HH}^0(\Lambda,\Lambda)$. Since $\varphi_\Bbbk(\mathrm{H})=\mathrm{Z}_\infty^{\mathsf{ev}}$, one sees from **FG′2** that $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is module-finite over H and the same then holds for $\mathrm{Ext}^*_\Lambda(M,N)$ for all $M,N\in\mathsf{mod}\,\Lambda$ by Jordan–Hölder filtration arguments. Hence **FG** holds. □

Let us remark that this approach to studying the **FG** condition is not new, see [60, Th. 1.3] for the Koszul case where $\mathscr{Z}_{gr}\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)=\mathscr{Z}_\infty\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$.

2.3 Combinatorics of the A_∞-centre. In the case of monomial algebras, it is possible to use Proposition 3.3.5 to simplify the general formula for higher commutators.

Lemma 2.4. *Suppose $a\in\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is of even degree. Then for any $x_1,\ldots,x_n$ in $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$, the higher commutators simplify to*

$$[a;x_1,\ldots,x_n]_{1,n}=m_{n+1}(a,x_1,\ldots,x_n)+(-1)^n m_{n+1}(x_1,\ldots,x_n,a). \tag{5.25}$$

Proof. Indeed, if the degree of a is even, we know that $m_{n+1}(x_1,\ldots,x_i,a,x_{i+1},\ldots,x_n)$ vanishes for each $i\in\{1,\ldots,n-1\}$. □

Using this lemma, we obtain a combinatorial criterion for A_∞-centrality. Given elements $a_1,\dots,a_m$ with $a_i \in \mathrm{Ext}^{r_i+1}_\Lambda(\Bbbk,\Bbbk) = kC^\vee_{r_i}$ with $r_i \geq 0$, we interpret the expression $a_1\dots a_m \in kC^\vee_{r_1+\dots+r_m+1}$ as denoting the sum of terms of $a_1\dots a_m \in kQ^{op}$ which are in $kC^\vee_{r_1+\dots+r_m+1}$, that is as the projection on the natural summand. Next, we say that an element $a \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is symmetric if $a\cdot e = e\cdot a$ for all $e \in \Bbbk$, i.e. a is a linear combination of closed oriented cycles in the quiver for Ext. The combinatorial criterion then states:

Proposition 2.5 (Combinatorial criterion)**.** *Suppose $a \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is a symmetric element of even degree $|a| = r_0+1 \geq 2$. Assume that for each tuple of chains $c_1,\dots,c_n$ with $c_i \in C^\vee_{r_i}$, we have equalities*

$$ac_1\dots c_n = c_1\dots c_n a \tag{5.26}$$

in $kC^\vee_{r_0+r_1+\dots+r_n+1}$. Then a is ∞-central and so lies in the image of Hochschild cohomology.

Proof. By Corollary 2.2 it's enough to show that $[a;x_1,\dots,x_n]_{1,n} = 0$ for all $x_1,\dots,x_n$ in $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$, and by linearity it's enough to do this for chains $c_1,\dots,c_n$. Since $ac_1\dots c_n = c_1\dots c_n a$, we deduce that $m_{n+1}(a,c_1,\dots,c_n)$ and $m_{n+1}(c_1,\dots,c_n,a)$ differ by a sign, and so the two products are zero or non-zero simultaneously. If both are zero we are done by Lemma 2.4, and if both are nonzero the elements $c_1,\dots,c_n \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ have odd degree by Proposition 3.3.5. Since $kC^\vee_{r_i} = \mathrm{Ext}^{r_i+1}_\Lambda(\Bbbk,\Bbbk)$, we deduce that $r_1,\dots,r_n$ are even. We can then compute the sign precisely:

$$\begin{aligned} m_{n+1}(a,c_1,\dots,c_n) &= (-1)^{(n+1)r_0+r_0r_n+r_0+r_n}\, ac_1\dots c_n = (-1)^{(n+1)r_0+r_0}\, ac_1\dots c_n \\ &= (-1)^{(n+1)+r_0}\, c_1\dots c_n a = (-1)^{n+1}(-1)^{r_0}\, c_1\dots c_n a \\ &= (-1)^{n+1}(-1)^{(n+1)r_1+r_1r_0+r_1+r_0}\, c_1\dots c_n a = (-1)^{n+1} m_{n+1}(c_1,\dots,c_n,a). \end{aligned}$$

Lemma 2.4 then gives

$$[a;c_1,\dots,c_n]_{1,n} = m_{n+1}(a,c_1,\dots,c_n) + (-1)^n m_{n+1}(c_1,\dots,c_n,a) = 0,$$

which is what we wanted. □

3 Periodicity operators for Gorenstein algebras

We now turn to the construction of A_∞-central operators $\chi_\gamma \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ associated to special Anick chains γ which we will call *stable relation cycles*. In this subsection we will always assume that Λ is Gorenstein.

We define the *period* of a Gorenstein monomial algebra Λ as follows. The algebra Λ contains finitely many perfect paths and so there are finitely many indecomposable Gorenstein-projectives

M over Λ, each with a periodic minimal projective resolution. We let p_M denote the minimal period of this resolution.

Definition 14. The period of Λ is the least common multiple of all periods p_M, as M runs over indecomposable non-projective Gorenstein-projectives. We denote that number by ℓ.

One notes that this is unchanged if we replace right modules by left modules as the duality $M \mapsto M^*$ preserves the period of Gorenstein-projectives.

Definition 15 (Stable relation cycles)**.** Let γ be an $(s-1)$-chain. We say that γ is a stable relation cycle if the following conditions hold:

i) s is even and at most $\dim\Lambda + 1$.

ii) s is a multiple of ℓ.

iii) $\gamma = p_0 p_1 \dots p_{s-1}$ for a perfect cycle $\mathbf{p} = (p_0, p_1, \dots, p_{s-1})$ with tail $t_\gamma = p_{s-1}$.

iv) $\gamma = q_0 q_1 \dots q_{s-1}$ for a perfect cycle $\mathbf{q} = (q_0, q_1, \dots, q_{s-1})$ with head $h_\gamma = q_0$.

Let us unpack this definition.

Properties iii) and iv) show that stable relation cycles enjoy the unique extension property of Corollary 1.11. Specifically, for any $k \geq 1$, the path $\gamma p_s \dots p_{s-1+k}$ is the unique extension of γ to the right as an $(s-1+k)$-chain, then with tail $t_{\gamma p_s \dots p_{s-1+k}} = p_{s-1+k}$, and dually $q_{-k} \dots q_{-1}\gamma$ is the unique extension to the left as an $(s-1+k)$-chain, then with head $h_{q_{-k}\dots q_{-1}\gamma} = q_{-k}$. Note that the decomposition iii) and iv) can differ as seen in Example 1.
The unique extension property of stable relation cycles γ will give us control over multiplication against $\gamma^\vee$ in $\mathrm{Ext}^*_\Lambda(\Bbbk, \Bbbk)$. However $\gamma^\vee$ is not a central class in general, and the additional constraints i-ii) will allow us to "rotate" γ without breaking the constraints iii)-iv), to obtain new stable relation cycles whose total sum will form an A_∞-central class χ_γ in $\mathrm{Ext}^*_\Lambda(\Bbbk, \Bbbk)$.
Green, Snashall and Solberg have introduced in [76] a closely related notion of stability for relation cycles which is weaker than what we consider here; theirs is concerned with the stability of tails and heads under taking power, which also follows from our definition. However the precise decomposition in terms of perfect paths is what gives us the additional leverage needed to study higher commutators and the finite generation conditions. The whole apparatus requires a careful balancing act, and so we begin with studying properties of perfect walks.

3.1 Perfect walks of even length. Now, in general if $(w_0, w_1, \dots, w_{n-1})$ is a perfect walk of even length n, then the path $w_0 w_1 \dots w_{n-2} w_{n-1}$ is a concatenation of relations $r_0 = w_0 w_1$, $r_2 = w_2 w_3$, $\dots$, $r_{n-2} = w_{n-2} w_{n-1}$, and so we have $w_0 w_1 \dots w_{n-1} = r_0 r_2 \dots r_{n-2}$. In particular stable relation cycles are concatenations of relations. Our next aim will be to recognise stable relation cycles amongst perfect walks of even length. This will come through a series of lemmas. We remark that Lemma 3.1 below is closely related to [76, Prop. 2.3].

Lemma 3.1. *Let $(w_0, w_1, \dots, w_{n-1})$ be a perfect walk of even length n. Then $\gamma = w_0 w_1 \dots w_{n-1}$ is an $(n-1)$-chain and w_{n-1} is a right divisor of t_γ.*

Proof. Write $n = 2k$, we work by induction on $k \geq 1$. For $k = 1$, $\gamma = w_0 w_1 \in R = C_1$ is a relation and $w_0 w_1 = \gamma = a t_\gamma$ for some arrow a, so that w_1 right divides t_γ for length reasons. Next, consider $\gamma' = w_0 w_1 \dots w_{2k-2}$ which is a chain by induction, and for which w_{2k-2} right divides $t_{\gamma'}$. This gives $t_{\gamma'} w_{2k-1} = 0$ in Λ, so there is a left divisor w' of $w_{2k-1} = w'w''$ such that $\gamma' w'$ is a chain with tail w'. Consider $w''w_{2k}$, and note that $w'w''w_{2k} = w_{2k-1}w_{2k} = 0$ in Λ. Hence we can find a left divisor u of $w''w_{2k}$ such that $w'u = 0$ in Λ and such that $\gamma' w' u$ forms a chain. Since $w'w'' = w_{2k-1} \neq 0$ in Λ, we have $u = w''u'$ and in particular u' left divides w_{2k}. But we have $w_{2k-1}u' = w'w''u' = w'u = 0$, which shows that w_{2k} left divides u', thus forcing $w_{2k} = u'$.
Putting it together, this gives a chain $\gamma' w' u = \gamma' w' w'' u' = \gamma' w_{2k-1} w_{2k}$ with tail u right divisible by w_{2k}, as claimed. □

Lemma 3.2. *Let $\gamma = w_0 w_1 \dots w_{n-1}$ be an $(n-1)$-chain given by a perfect walk of even length n. Then:*

i) The relations $r_0, r_2, \dots, r_{n-2}$ are independent of the choice of perfect walk $(w_0, w_1, \dots, w_{n-1})$ with $\gamma = w_0 w_1 \dots w_{n-1}$.

ii) If $\gamma' = w'_0 w'_1 \dots w'_{n-1}$ is another perfect walk of even length n with $r'_0 = r_0$ or $r'_{n-2} = r_{n-2}$, then $\gamma' = \gamma$.

Proof. i). If $\gamma = r_0 r_2 \dots r_{s-2} = \widetilde{r}_0 \widetilde{r}_2 \dots \widetilde{r}_{s-2}$ is written in two ways as a concatenation of relations, then one of $r_0, \widetilde{r}_0$ divides the other and so $r_0 = \widetilde{r}_0$. Applying the same argument inductively to the substrings $r_{2k} \dots r_{s-2} = \widetilde{r}_{2k} \dots \widetilde{r}_{s-2}$ gives the result.
ii). We assume that $r'_{n-2} = r_{n-2}$, the case $r'_0 = r_0$ is similar. We will show that the previous relation in a perfect walk of even length is uniquely determined.
The hypothesis gives $w'_{n-2} w'_{n-1} = r'_{n-2} = r_{n-2} = w_{n-2} w_{n-1}$, so one of w'_{n-2}, w_{n-2} left divides the other, say $w_{n-2} = w'_{n-2} x$. Then

$$w'_{n-3} w_{n-2} = w'_{n-3} w'_{n-2} x = 0$$

in Λ. As $w'_{n-3} \neq 0$ in Λ, we must have $w'_{n-3} = y w_{n-3}$ since (w_{n-3}, w_{n-2}) forms a perfect pair. Continuing, we have $w'_{n-4} w'_{n-3} = w'_{n-4} y w_{n-3} = 0$ in Λ and so $w'_{n-4} y = z w_{n-4}$. We obtain the equality $w'_{n-4} w'_{n-3} = w'_{n-4} y w_{n-3} = z w_{n-4} w_{n-3}$, so that z is the empty word and we have equality of relations $r'_{n-4} = w'_{n-4} w'_{n-3} = w_{n-4} w_{n-3} = r_{n-4}$. Iterating, we get $r'_{2k} = r_{2k}$ for all k and so $\gamma' = \gamma$. □

The next result allows us to rewrite perfect walks of even length in normal forms.

Lemma 3.3 (Perfect rewriting)**.** *Let $(w_0, w_1, \dots, w_{n-1})$ be a perfect walk of even length n and let $\gamma = w_0 w_1 \dots w_{n-1}$.*

i) If t_γ is perfect, then $\gamma = p_0 \dots p_{n-1}$ is given by the perfect walk with $p_{n-1} = t_\gamma$.

ii) If h_γ is perfect, then $\gamma = q_0 \dots q_{n-1}$ is given by the perfect walk with $q_0 = h_\gamma$.

Proof. We prove i) as ii) is dual. Assuming $t_\gamma = p_{n-1}$ perfect, since both p_{n-1} and w_{n-1} right divide γ, one must divide the other, say $p_{n-1} = xw_{n-1}$ for some x (the other case is similar). Then $p_{n-2}p_{n-1} = p_{n-2}xw_{n-1} = 0$ in Λ. Since $p_{n-2}x \neq 0$ in Λ as it is a proper divisor of $p_{n-2}p_{n-1}$, we must have $p_{n-2}x = yw_{n-2}$ for some y, which gives $p_{n-2}p_{n-1} = p_{n-2}xw_{n-1} = yw_{n-2}w_{n-1}$. It follows that y is the empty word and $p_{n-2}p_{n-1} = w_{n-2}w_{n-1}$.
Now $\gamma' = p_0 \dots p_{n-2}p_{n-1}$ and $\gamma = w_0 \dots w_{n-2}w_{n-1}$ share the last relation $p_{n-2}p_{n-1} = w_{n-2}w_{n-1}$, and so $\gamma' = \gamma$ by Lemma 3.2 ii). □

We now obtain a recognition lemma for stable relation cycles.

Lemma 3.4 (Recognition Lemma)**.** *Let $(w_0, w_1, \dots, w_{n-1})$ be a perfect walk, and assume that $n \geq \dim\Lambda + 1$, n is even and divisible by ℓ. Then $\gamma = w_0 w_1 \dots w_{n-1}$ is a stable relation cycle.*

Proof. First γ is an $(n-1)$-chain by Lemma 3.1. Next, since $n - 1 \geq \dim\Lambda$ we see that $p_{n-1} := t_\gamma$ and $q_0 := h_\gamma$ are perfect by Theorem 1.6, and Lemma 3.3 shows that $\gamma = p_0 p_1 \dots p_{n-1} = q_0 q_1 \dots q_{n-1}$. Lastly, since n is a multiple of the period ℓ, it is then a multiple of the minimal period of every perfect cycle and so we see that $\mathbf{p} = (p_0, p_1, \dots, p_{n-1})$ and $\mathbf{q} = (q_0, q_1, \dots, q_{n-1})$ are in fact perfect cycles. □

Corollary 3.5. *Let p be a perfect path in Λ. Then $p = p_0$ (resp. $p = p_{s-1}$) can be extended to a stable relation cycle $\gamma = p_0 p_1 \dots p_{s-1}$.*

Proof. Simply take the perfect walk $(p_0, p_1, \dots, p_{s-1})$ of even length s, satisfying $s \geq \dim\Lambda + 1$ and with s some multiple of ℓ. □

3.2 Rotating stable relation cycles. Now let $\gamma = a_0 a_1 \dots a_{n-1} \in C_{s-1}$ be a stable relation cycle, which we write as a path of length n with $a_i \in Q_1$. We now want to "rotate" γ to produce new stable relation cycles, also of path length n. We can think of γ as a substring of a periodic string, infinite in both directions:

$$\underline{a} : \dots a_0 a_1 \dots a_{n-1}(a_0 a_1 \dots a_{n-1})a_0 a_1 \dots a_{n-1} \dots \tag{5.27}$$

One can recover this string by periodicity from any length n substring. Setting $\gamma_0 = \gamma$, we define the set $S = \{\gamma_0, \gamma_1, \dots, \gamma_{n_\gamma - 1}\}$ by the property that:

i) $\gamma_i = a_{j_i} a_{j_i+1} \dots a_{j_i+n-1}$ is an $(s-1)$-chain which is also a stable relation cycle.

ii) $0 = j_0 < j_1 < \dots < j_{n_\gamma - 1} \leq n - 1$.

iii) For $\gamma_i = a_{j_i} a_{j_i+1} \dots a_{j_i+n-1} \in S$, we have $\gamma_i \neq a_j a_{j+1} \dots a_{j+n-1}$ if $0 \leq j < j_i$.

iv) $\{j_0, j_1, \dots, j_{n_\gamma - 1}\} \subseteq \{0, 1, \dots, n-1\}$ is maximal with these property.

We think of n_γ as the number of possible distinct rotations of γ as a stable relation cycle. Moreover, all γ_i are length n substrings of $\underline{a}$, and it is clear that the same set is produced by this construction starting from any other $\gamma_i \in \{\gamma_0, \gamma_1, \dots, \gamma_{n_\gamma - 1}\}$. We call this set S the associated set of γ, which is also the equivalence class for an equivalence relation on stable relation cycles which we denote $\gamma \sim \gamma'$.

Lemma 3.6 (Completeness)**.** *Let $\gamma \in C_{s-1}$ be a stable relation cycle, and write $\gamma = p_0 \dots p_{s-1} = q_0 \dots q_{s-1}$ for the corresponding perfect walks. Then for any $k \in \mathbb{Z}$, both $p_k p_{k+1} \dots p_{k+s-1}$ and $q_k q_{k+1} \dots q_{k+s-1}$ are in $\{\gamma_0, \gamma_1, \dots, \gamma_{n_\gamma - 1}\}$.*

Proof. Both strings are substrings of $\underline{a}$, and the periodicities $p_{i+s} = p_i$ and $q_{i+s} = q_i$ shows that if γ is a substring of length n in $\underline{a}$, then so are $p_k p_{k+1} \dots p_{k+s-1}$ and $q_k q_{k+1} \dots q_{k+s-1}$ as these differ by cyclic shifts, which preserves path length. Finally both strings are given by perfect walks of even lengths s satisfying the hypotheses of Lemma 3.4, and so $p_k p_{k+1} \dots p_{k+s-1}$ and $q_k q_{k+1} \dots q_{k+s-1}$ are both relation cycles. □

Lemma 3.7 (Rigidity Lemma)**.** *Let α be a path in Q and let $k \geq 1$. Then $\gamma_i \alpha$ (resp. $\alpha \gamma_i$) is an $(s-1+k)$-chain for at most one choice of $i \in \{0, 1, \dots, n_\gamma - 1\}$.*

Proof. Let $\gamma_i, \gamma_{i'}$ be two stable relation cycles in the associated set, and write $\gamma_i = p_0 p_1 \dots p_{s-1}$ with tail $t_{\gamma_i} = p_{s-1}$ for the corresponding perfect cycle (resp. $\gamma_{i'} = p'_0 p'_1 \dots p'_{s-1}$ with tail $t_{\gamma_{i'}} = p'_{s-1}$). If $\gamma_i \alpha$ and $\gamma_{i'} \alpha$ are both $(s-1+k)$-chains, then the unique extension property shows that $\alpha = p_s p_{s+1} \dots p_{s-1+k} = p'_s p'_{s+1} \dots p'_{s-1+k}$, and by s-periodicity we can rewrite this as $\alpha = p_0 p_1 \dots p_{k-1} = p'_0 p'_1 \dots p'_{k-1}$.
If $k = 1$ we are done as $p_0 = \alpha = p'_0$, and the perfect walks $(p_0, p_1, \dots, p_{s-1})$ and $(p'_0, p'_1, \dots, p'_{s-1})$ are equal the moment $p_k = p'_k$ for any k, so that $\gamma_i = \gamma_{i'}$.
If $k \geq 2$, then the equality $\alpha = p_0 p_1 \dots p_{k-1} = p'_0 p'_1 \dots p'_{k-1}$ shows that one of the two relations $r_0 = p_0 p_1$, $r'_0 = p'_0 p'_1$ divides the other, so that $r_0 = r'_0$. Since stable relation cycles are determined by their first relation (Lemma 3.2) then $\gamma_i = \gamma_{i'}$. □

3.3 Periodicity operators We can now construct the operators χ_γ associated to stable relation cycles.

Definition 16 (Periodicity operators)**.** Let $\gamma \in C_{s-1}$ be a stable relation cycle and $\{\gamma_0, \gamma_1, \dots, \gamma_{n_\gamma - 1}\}$ the associated set. Then

$$\chi_\gamma = \sum_{i=0}^{n_\gamma - 1} \gamma_i^\vee \tag{5.28}$$

is the periodicity operator $\chi_\gamma \in \mathrm{Ext}^s_\Lambda(\underline{\Bbbk}, \underline{\Bbbk})$ attached to γ.

Example 3. Let $\Lambda = k[t]/(t^n)$ for $n \geq 2$. Then $\gamma = t^n \in C_1$ is the unique stable relation cycle of minimal length. The periodicity operator $\chi_\gamma = (t^n)^\vee \in \mathrm{Ext}^2_\Lambda(\underline{\Bbbk}, \underline{\Bbbk})$ is the classical periodicity operator of Gulliksen [85].

Finally, we can show that the operators $\chi_\gamma \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ lift to Hochschild cohomology.

Proposition 3.8. *Let Λ be a Gorenstein monomial algebra and let $\gamma \in C_{s-1}$ be a stable relation cycle. Then χ_γ is ∞-central, and therefore lies in the image of Hochschild cohomology.*

Proof. By the combinatorial criterion of Proposition 2.5, it's enough to show the equality

$$\chi_\gamma c_1 \dots c_n = c_1 \dots c_n \chi_\gamma \tag{5.29}$$

in $kC_N^\vee$ for all tuples $c_1, c_2, \dots, c_n$ with $c_j \in C_{k_j}^\vee$ and $N = (s-1) + k_1 + \dots + k_n + 1$. Let us write $c_j = b_j^\vee$ for the dual of a chain $b_j \in C_{k_j}$. Since $\gamma_i^\vee b_j^\vee = (b_j \gamma_i)^\vee$, we let $\chi_\gamma^\vee = \sum_{i=0}^{n_\gamma - 1} \gamma_i$ in kC_{s-1} and prove the dual equality

$$\chi_\gamma^\vee b_n \dots b_1 = b_n \dots b_1 \chi_\gamma^\vee$$

in kC_N. Let $\alpha = b_n \dots b_1$, considered as a path in Q.
If $\chi_\gamma^\vee \alpha \neq 0$ in kC_N then $\gamma_i \alpha$ is an N-chain for some i, in which case i is unique by rigidity (Lemma 3.7). Writing $\gamma_i = p_0 p_1 \dots p_{s-1}$ for the perfect walk with $t_{\gamma_i} = p_{s-1}$, the unique extension property gives $\alpha = p_s \dots p_N$. We then rotate the string $\gamma_i \alpha$:

$$\begin{aligned}
\gamma_i \alpha &= (p_0 p_1 \dots p_{s-1}) p_s \dots p_N \\
&= (p_0 p_1 \dots p_{s-1}) p_0 \dots p_{N-s} \\
&= p_0 p_1 \dots p_{N-s} (p_{N-s+1} \dots\dots p_{N-s}) \\
&= \alpha (p_{N-s+1} \dots\dots p_{N-s}).
\end{aligned}$$

By completeness (Lemma 3.6) we have $p_{N-s+1} \dots p_{N-s} = \gamma_j$ for some $\gamma_j \in \{\gamma_0, \gamma_1, \dots, \gamma_{n_\gamma - 1}\}$, and as $\alpha\gamma_j = \gamma_i \alpha \neq 0$ in kC_N, rigidity again gives γ_j as the unique stable relation cycle with $\alpha\gamma_j \neq 0$ in kC_N. Putting this together gives

$$\chi_\gamma^\vee \alpha = \gamma_i \alpha = \alpha \gamma_j = \alpha \chi_\gamma^\vee$$

in kC_N. More formally, this argument shows that $\chi_\gamma^\vee \alpha \neq 0$ in kC_N implies $\alpha \chi_\gamma^\vee \neq 0$ in kC_N, in which case $\chi_\gamma^\vee \alpha = \alpha \chi_\gamma^\vee$ in kC_N. As the argument is symmetrical (using the dual properties established instead), we also see that $\alpha \chi_\gamma^\vee \neq 0$ in kC_N implies $\chi_\gamma^\vee \alpha \neq 0$ in kC_N, in which case $\alpha \chi_\gamma^\vee = \chi_\gamma^\vee \alpha$ in kC_N, and so the claim holds. □

Remark 4. We thus see that periodicity operators lift to Hochschild cohomology, and any two lifts differ by an element of $\ker(\varphi_\Bbbk)$, which is nilpotent. However our proof shows the stronger statement that all operators $\chi_\gamma \in \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ have an unambiguous lift to an operator $\chi_\gamma \in \mathrm{HH}^*(\Lambda,\Lambda)$, which we denote by the same letter by abuse of notation.
This lift is described as follows [32]: the invariance of Hochschild cohomology under Koszul duality gives an isomorphism $\mathrm{HH}^*(\Lambda,\Lambda) \cong \mathrm{HH}^*(E,E)$, with $E = \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ treated as an A_∞-algebra.

Consider the cochain in $C^*(E,E)$ given by $1 \mapsto \chi_\gamma$ and sending everything else to zero. The condition $\mathrm{ad}_{\chi_\gamma} = 0$ is precisely the Hochschild cocycle condition and so this cocycle gives rise to a cohomology class $\chi_\gamma \in \mathrm{HH}^*(E,E) \cong \mathrm{HH}^*(\Lambda,\Lambda)$, which provides a section for the characteristic morphism. Hence to any stable relation cycle γ we attach a well-defined class $\chi_\gamma \in \mathrm{HH}^*(\Lambda,\Lambda)$.

3.4 The ring of periodicity operators. We now turn to the study of the multiplicative properties of the periodicity operators $\{\chi_\gamma\}_\gamma \subseteq \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$. In this subsection we assume that Λ is a Gorenstein monomial algebra of infinite global dimension. In this case Λ always has stable relation cycles by Corollary 3.5.

Let $\mathscr{R} := k\langle \chi_\gamma \mid \gamma \rangle \subseteq \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ be the A_∞-central graded subalgebra generated by the classes $\{\chi_\gamma\}_\gamma$. It will turn out that the graded connected algebra $\mathscr{R}$ is Noetherian, reduced and of Krull dimension one, and the structure theory of such rings dictates that $\mathscr{R}$ embeds in a finite product of polynomial algebras

$$\mathscr{R} \subseteq \prod_{i=1}^{b} k[t_i] \tag{5.30}$$

with $k[t_i]$ the graded normalisation of $\mathscr{R}/\mathfrak{p}_i$ for $\mathfrak{p}_i$ the i-th minimal homogeneous prime. Composing with the i-th projection gives a map $\pi_i : \mathscr{R} \to k[t_i]$, and so for each $\chi_\gamma \in \mathscr{R}$ we may write $\pi_i(\chi_\gamma) = t_i^{n_i}$ for some $n_i \in \mathbb{N} \cup \{-\infty\}$ (where we interpret $t_i^{-\infty} = 0$). Denoting this exponent by $n_i = \mathsf{ord}_i(\gamma)$, it is clear that the ring structure of $\mathscr{R}$ is determined by the values b and $\mathsf{ord}_i(\gamma)$ as $i = 1,2,\ldots,b$ and γ ranges over all stable relation cycles. Turning this around, we will first give a combinatorial definition of b and $\mathsf{ord}_i(\gamma)$ which we will use to establish the structure of $\mathscr{R}$ as claimed in the above paragraph. In fact we will obtain a slightly more precise description of the structure of $\mathscr{R}$, see Prop. 3.16 below.

Branch equivalence relation Let γ_1,γ_2 be stable relation cycles, and recall that $\gamma_1 \sim \gamma_2$ if they have the same associated set (in particular then $\chi_{\gamma_1} = \chi_{\gamma_2}$). It is easy to see that powers of stable relation cycles are also stable relation cycles and that $\gamma_1 \sim \gamma_2$ implies $\gamma_1^n \sim \gamma_2^n$ for any $n \geq 2$, and so this equivalence relation determines another coarser equivalence relation.

Definition 17 (Branch equivalence relation)**.** The stable relation cycles γ_1,γ_2 are branch equivalent if there exists $n_1,n_2 \geq 1$ such that $\gamma_1^{n_1} \sim \gamma_2^{n_2}$. We denote the branch equivalence relation by $\gamma_1 \approx \gamma_2$.

Lemma 3.9. *Let γ_1,γ_2 be stable relation cycles. If $t_{\gamma_1} = t_{\gamma_2}$ or $h_{\gamma_1} = h_{\gamma_2}$, then $\gamma_1 \approx \gamma_2$.*

Proof. We assume that $t_{\gamma_1} = t_{\gamma_2} = p$, the other case is dual. We can write

$$\gamma_1 = p_0 p_1 \ldots p_{s-1} \tag{5.31}$$
$$\gamma_2 = p'_0 p'_1 \ldots p'_{s'-1} \tag{5.32}$$

for perfect cycles with $p_{s-1} = p'_{s'-1} = p$. This implies $p_0 = p'_0$, $p_1 = p'_1$, …, and so on, so that $\mathbf{p} = (p_0,p_1,\ldots,p_{s-1})$ and $\mathbf{p}' = (p'_0,p'_1,\ldots,p'_{s'-1})$ are both powers of the same perfect cycle of minimal

length $s'' \mid s, s'$. Taking a common power, we see that there are $n_1, n_2 \geq 1$ such that $\gamma_1^{n_1} = \gamma_2^{n_2}$, and so $\gamma_1 \approx \gamma_2$. □

Lemma 3.10. *There are finitely many branch equivalence classes $\{\Gamma_i\}_{i\in I}$ of stable relation cycles.*

Proof. The map $\bigcup_{i\in I}\Gamma_i \to \{\text{perfect paths}\}$ sending γ to its tail t_γ sends distinct equivalence classes into disjoint non-empty subsets by Lemma 3.9, and so $|I|$ is bounded above by the number of perfect paths. □

It follows that we can identify $I = [b]$ and so list the branch equivalence classes as $\Gamma_1, \Gamma_2, \ldots, \Gamma_b$ for some $b \in \mathbb{N}$. We call b the number of branches.
Fixing an equivalence class Γ_i, consider the set of path lengths of elements of Γ_i which we denote $\widetilde{\mathbb{N}_{\Gamma_i}} = \{\mathsf{len}(\gamma) \mid \gamma \in \Gamma_i\} \subseteq \mathbb{N}$. Letting $\gcd_i := \gcd(\widetilde{\mathbb{N}_{\Gamma_i}})$, we normalise this to

$$\mathbb{N}_{\Gamma_i} := \frac{1}{\gcd_i}\widetilde{\mathbb{N}_{\Gamma_i}} \subseteq \mathbb{N}. \tag{5.33}$$

Definition 18. We define the function $\mathsf{ord}_i : \{\gamma\} \to \mathbb{N} \cup \{-\infty\}$ on the set of stable relation cycles as follows:

$$\mathsf{ord}_i(\gamma) := \begin{cases} \frac{1}{\gcd_i}\mathsf{len}(\gamma) & \text{if } \gamma \in \Gamma_i \\ -\infty & \text{if } \gamma \notin \Gamma_i. \end{cases}$$

Note in particular that $\mathbb{N}_{\Gamma_i} = \{\mathsf{ord}_i(\gamma) \mid \gamma \in \Gamma_i\} \subseteq \mathbb{N}$.

Lemma 3.11. *Let $\gamma_1, \gamma_2 \in \Gamma_i$. Then $\gamma_1 \sim \gamma_2$ if and only if $\mathsf{ord}_i(\gamma_1) = \mathsf{ord}_i(\gamma_2)$.*

Proof. Equivalently we show that $\gamma_1 \sim \gamma_2$ if and only if $\mathsf{len}(\gamma_1) = \mathsf{len}(\gamma_2)$. Since stable relation cycles in the same associated set have the same path length, the necessary implication holds by definition. For the converse, assume that $\mathsf{len}(\gamma_1) = \mathsf{len}(\gamma_2) = n$. Write $\gamma_1 = a_0a_1\ldots a_{n-1}$ and $\gamma_2 = a_0'a_1'\ldots a_{n-1}'$ with $a_i, a_i' \in Q_1$. Since $\gamma_1 \approx \gamma_2$, there are $n_1, n_2 \geq 1$ such that $\gamma_1^{n_1} \sim \gamma_2^{n_2}$, and so the two infinite periodic strings

$$\underline{a} : \quad \ldots(a_0a_1\ldots a_{n-1})^{n_1}(a_0a_1\ldots a_{n-1})^{n_1}(a_0a_1\ldots a_{n-1})^{n_1}\ldots$$

$$\underline{a}' : \quad \ldots(a_0'a_1'\ldots a_{n-1}')^{n_2}(a_0'a_1'\ldots a_{n-1}')^{n_2}(a_0'a_1'\ldots a_{n-1}')^{n_2}\ldots$$

must coincide up to some translation. But then γ_1, γ_2 are two length n substrings of the same infinite periodic string

$$\ldots a_0a_1\ldots a_{n-1}(a_0a_1\ldots a_{n-1})a_0a_1\ldots a_{n-1}\ldots$$

and so $\gamma_1 \sim \gamma_2$ by construction of the associated set. □

Multiplicative properties We next establish the multiplicative properties of $\{\chi_\gamma\}_\gamma$. We aim to prove the following:

Proposition (Prop. 3.15)**.** *Let γ and $\gamma_1, \gamma_2, \gamma_3$ denote stable relation cycles.*

i) If $\chi_{\gamma_1} \cdot \chi_{\gamma_2} \neq 0$, then $\gamma_1 \approx \gamma_2$.

ii) Assume that $\gamma_1 \approx \gamma_2$. Then there is a $\gamma_3 \approx \gamma_1, \gamma_2$ such that $\chi_{\gamma_1} \cdot \chi_{\gamma_2} = \chi_{\gamma_3}$, and then $\mathsf{ord}_i(\gamma_1) + \mathsf{ord}_i(\gamma_2) = \mathsf{ord}_i(\gamma_3)$.

iii) We have $\chi_\gamma^n = \chi_{\gamma^n}$ for any $n \geq 1$.

The proof will be done in a series of steps. Let us first collect some comments.
By property i) it is enough to understand the product of $\chi_{\gamma_1}, \chi_{\gamma_2}$ for $\gamma_1, \gamma_2 \in \Gamma_i$ within the same branch equivalence class. Property ii) implies that $\{\chi_\gamma\}_{\gamma \in \Gamma_i}$ forms a multiplicative basis for the subalgebra $\mathscr{R}_i := k\langle \chi_\gamma \mid \gamma \in \Gamma_i \rangle \subseteq \mathscr{R}$. To analyse the ring structure, we may extend the notation $\mathsf{ord}_i(\chi_\gamma) := \mathsf{ord}_i(\gamma)$ to the class χ_γ, which is independent of choice of representative γ by Lemma 3.11, and in fact Lemma 3.11 tells us that any value $n \in \mathbb{N}_{\Gamma_i}$ can be represented by a unique class χ_γ for $\gamma \in \Gamma_i$. Hence the function ord_i sets up a bijection $\{\chi_\gamma\}_{\gamma \in \Gamma_i} \leftrightarrow \mathbb{N}_{\Gamma_i}$. Since $\mathsf{ord}_i(-)$ is additive on products, that the set $\{\chi_\gamma\}_{\gamma \in \Gamma_i}$ forms a multiplicative basis for $\mathscr{R}_i$ by property ii) translates into $\mathbb{N}_{\Gamma_i} \subseteq \mathbb{N}$ being a subsemigroup, and we see that the above bijection induces an algebra isomorphism onto the semigroup algebra

$$\mathscr{R}_i = k\langle \chi_\gamma \mid \gamma \in \Gamma_i \rangle \xrightarrow{\cong} k[t^{\mathbb{N}_{\Gamma_i}}] := k\langle t^n \mid n \in \mathbb{N}_{\Gamma_i} \rangle \subseteq k[t]$$

sending $\chi_\gamma \mapsto t^{\mathsf{ord}_i(\gamma)}$.
Keeping this in mind, we see that many conditions on the behavior of $\mathsf{ord}_i(-)$ and $\{\chi_\gamma\}_\gamma$ have to be verified. First, assuming the above isomorphism we see that the ring $\mathscr{R}_i \cong k[t^{\mathbb{N}_{\Gamma_i}}]$ is bigraded by $\mathsf{ord}_i(\chi_\gamma)$ and cohomological degree $|\chi_\gamma|$, and either degree determines the other. This translates into the next simple combinatorial lemma.
If γ is an n-chain, let us call n the Anick degree of γ.

Lemma 3.12. *Let $\gamma_1, \gamma_2 \in \Gamma_i$. Then γ_1, γ_2 have the same Anick degree if and only if $\gamma_1 \sim \gamma_2$.*

Proof. If $\gamma_1 \sim \gamma_2$ then the implication holds by definition. Conversely assume that γ_1, γ_2 are both n-chains. Since $\gamma_1 \approx \gamma_2$, there exists $n_1, n_2 \geq 1$ such that $\gamma_1^{n_1} \sim \gamma_2^{n_2}$. The relation $\sim$ preserves both the Anick degree and the path length, and so we have $n_1 n = n_2 n$ and $n_1 \mathsf{len}(\gamma_1) = n_2 \mathsf{len}(\gamma_2)$. Cancellation first gives $n_1 = n_2$ and then $\mathsf{len}(\gamma_1) = \mathsf{len}(\gamma_2)$. Thus $\mathsf{ord}_i(\gamma_1) = \mathsf{ord}_i(\gamma_2)$ and so $\gamma_1 \sim \gamma_2$ by Lemma 3.11. □

Next, understanding the product $\chi_{\gamma_1} \cdot \chi_{\gamma_2}$ will quickly reduce to understanding triples of stable relation cycles $\gamma_1, \gamma_2, \gamma_3$ such that $s(\gamma_2) = t(\gamma_1)$ and $\gamma_3 = \gamma_2 \gamma_1$. In particular establishing identities of the form $\chi_{\gamma_1} \cdot \chi_{\gamma_2} = \chi_{\gamma_3}$ will require understanding the behavior of associated sets under products of stable relation cycles. This is done in the next two lemma.

Lemma 3.13 (Product lemma)**.** *Let γ_1, γ_2 be stable relation cycles of Anick degrees $s_1 - 1$ and $s_2 - 1$, respectively. Assume that $s(\gamma_2) = t(\gamma_1)$ and $\gamma_2 \gamma_1$ is also a stable relation cycle.*

i) We have $\gamma_1 \approx \gamma_2 \approx \gamma_2\gamma_1$.

ii) If $\widetilde{\gamma}_2 \sim \gamma_2$ *and* $\widetilde{\gamma}_1 \sim \gamma_2$ *are equivalent stable relation cycles with* $s(\widetilde{\gamma}_2) = t(\widetilde{\gamma}_1)$ *and* $\widetilde{\gamma}_2\widetilde{\gamma}_1$ *also a stable relation cycle, then* $\widetilde{\gamma}_2\widetilde{\gamma}_1 \sim \gamma_2\gamma_1$.

Proof. i). Since γ_1, γ_2 and $\gamma_2\gamma_1$ are stable relation cycles, the unique extension property (Prop. 1.11) shows that $t_{\gamma_2\gamma_1} = t_{\gamma_2}$ and $h_{\gamma_2\gamma_1} = h_{\gamma_1}$. Lemma 3.9 then gives $\gamma_2 \approx \gamma_2\gamma_1 \approx \gamma_1$.
ii). Applying i) to $\widetilde{\gamma}_2, \widetilde{\gamma}_1$ gives $\widetilde{\gamma}_2\widetilde{\gamma}_1 \approx \widetilde{\gamma}_1 \sim \gamma_1 \approx \gamma_2\gamma_1$, and so $\widetilde{\gamma}_2\widetilde{\gamma}_1 \approx \gamma_2\gamma_1$. Since the stable relation cycles $\widetilde{\gamma}_2\widetilde{\gamma}_1$ and $\gamma_2\gamma_1$ both have Anick degree $s_3 - 1$ for $s_3 = s_2 + s_1$, Lemma 3.12 shows that $\widetilde{\gamma}_2\widetilde{\gamma}_1 \sim \gamma_2\gamma_1$. □

Lemma 3.14 (Product decompositions for associated sets)**.** *Let* $\gamma, \gamma_1, \gamma_2, \gamma_3$ *denote stable relation cycles and assume that* $\gamma_3 = \gamma_2\gamma_1$. *Let* $S_i = \{\widetilde{\gamma}_i \mid \widetilde{\gamma}_i \sim \gamma_i\}$ *be the associated sets of* γ_i, $i = 1, 2, 3$.

i) The associated set S_3 *decomposes as a product of elements of* S_2, S_1, *in that:*

a) For every $\widetilde{\gamma}_2 \in S_2$, *there is a unique* $\widetilde{\gamma}_1 \in S_1$ *such that* $\widetilde{\gamma}_2\widetilde{\gamma}_1 \in S_3$.

b) For every $\widetilde{\gamma}_1 \in S_1$, *there is a unique* $\widetilde{\gamma}_2 \in S_2$ *such that* $\widetilde{\gamma}_2\widetilde{\gamma}_1 \in S_3$.

c) Every $\widetilde{\gamma}_3 \in S_3$ *is of the form* $\widetilde{\gamma}_3 = \widetilde{\gamma}_2\widetilde{\gamma}_1$ *for unique* $\widetilde{\gamma}_2 \in S_2$, $\widetilde{\gamma}_1 \in S_1$.

Hence we have $S_3 = S_2 \cdot S_1 := \{\widetilde{\gamma}_2\widetilde{\gamma}_1 \mid \widetilde{\gamma}_i \in S_i \text{ and } (\widetilde{\gamma}_2, \widetilde{\gamma}_1) \text{ compatible}\}$, *with the compatibility in the sense of a)-b).*

ii) For any $n \geq 1$ *the associated set of* γ^n *is* $\{\widetilde{\gamma}^n \mid \widetilde{\gamma} \sim \gamma\}$.

Proof. i). Let $s_i - 1$ be the Anick degree of γ_i, so that $s_3 = s_2 + s_1$. We prove 1). Let $\widetilde{\gamma}_2 \in S_2$ and write $\widetilde{\gamma}_2 = p_0p_1\dots p_{s_2-1}$ for the perfect cycle with $t_{\widetilde{\gamma}_2} = p_{s_2-1}$, and extend this to a perfect walk $p_0p_1\dots p_{s_2-1}p_0p_1\dots p_{s_1-1}$ of length $s_2 + s_1 = s_3$. Set $\widetilde{\gamma}_1 := p_0p_1\dots p_{s_1-1}$ and write $\widetilde{\gamma}_3 := \widetilde{\gamma}_2\widetilde{\gamma}_1$. Note that $\widetilde{\gamma}_1, \widetilde{\gamma}_3$ are given by perfect walks of appropriate length and so are stable relation cycles by the recognition lemma (Lemma 3.4).
Part i) of the product lemma (Lemma 3.13) gives $\widetilde{\gamma}_1 \approx \widetilde{\gamma}_2 \sim \gamma_2 \approx \gamma_1$, and so $\widetilde{\gamma}_1 \approx \gamma_1$. Since $\widetilde{\gamma}_1$ and γ_1 are both $(s_1 - 1)$-chains, Lemma 3.12 then gives $\widetilde{\gamma}_1 \sim \gamma_1$, and finally part ii) gives $\widetilde{\gamma}_3 = \widetilde{\gamma}_2\widetilde{\gamma}_1 \sim \gamma_2\gamma_1 = \gamma_3$. Finally, the uniqueness follows from the unique extension property (Prop. 1.11), and 1) follows. The proof of 2) is dual.
For 3), let $\widetilde{\gamma}_3 \in S_3$ and write $\widetilde{\gamma}_3 = p_0p_1\dots p_{s_3-1}$ for the perfect cycle with tail $t_{\widetilde{\gamma}_3} = p_{s_3-1}$. Since $s_3 = s_2 + s_1$ and both s_1, s_2 are multiples of ℓ, and therefore of the minimal period of this perfect cycle, we can break it down further as $\widetilde{\gamma}_3 = p_0p_1\dots p_{s_3-1} = (p_0p_1\dots p_{s_1-1})(p_0p_1\dots p_{s_2-1})$. Letting $\widetilde{\gamma}_i := p_0p_1\dots p_{s_i-1}$, since the length s_i is appropriate the recognition lemma (Lemma 3.4) again shows that $\widetilde{\gamma}_i$ are stable relation cycles. Part i) of the product lemma (Lemma 3.13) then gives $\widetilde{\gamma}_i \approx \widetilde{\gamma}_3 \sim \gamma_3 \approx \gamma_i$ for $i = 1, 2$, so that $\widetilde{\gamma}_i \approx \gamma_i$, and Lemma 3.12 then gives $\widetilde{\gamma}_i \sim \gamma_i$. Hence $\widetilde{\gamma}_3 = \widetilde{\gamma}_2\widetilde{\gamma}_1$ with $\widetilde{\gamma}_2 \in S_2$, $\widetilde{\gamma}_1 \in S_1$ as claimed. Finally, to see uniqueness assume that $\widetilde{\gamma}_3 = \widetilde{\gamma}_2\widetilde{\gamma}_1 = \widetilde{\gamma}_2'\widetilde{\gamma}_1'$ for some possibly different $\widetilde{\gamma}_1' \in S_1, \widetilde{\gamma}_2' \in S_2$. The unique extension property (Prop. 1.11) implies that

$t_{\widetilde{\gamma}_1} = t_{\widetilde{\gamma}_3} = t_{\widetilde{\gamma}_1}$. Lemma 3.9 then gives $\widetilde{\gamma}_1' \approx \widetilde{\gamma}_1$, and since they have the same Anick degree Lemma 3.12 gives $\widetilde{\gamma}_1' \sim \widetilde{\gamma}_1$. In particular $\mathsf{len}(\widetilde{\gamma}_1') = \mathsf{len}(\widetilde{\gamma}_1)$, and so $\widetilde{\gamma}_1' = \widetilde{\gamma}_1$. The equality $\widetilde{\gamma}_2' = \widetilde{\gamma}_2$ then follows from 2), and we have shown 3).
ii). This follows from i) by setting $\gamma_1 = \gamma$ and $\gamma_2 = \gamma^{n-1}$ and induction on n. □

We can now prove the claim.

Proposition 3.15. *Let γ and $\gamma_1, \gamma_2, \gamma_3$ denote stable relation cycles.*

i) If $\chi_{\gamma_1} \cdot \chi_{\gamma_2} \neq 0$, then $\gamma_1 \approx \gamma_2$.

ii) Assume that $\gamma_1 \approx \gamma_2$. Then there is a $\gamma_3 \approx \gamma_1, \gamma_2$ such that $\chi_{\gamma_1} \cdot \chi_{\gamma_2} = \chi_{\gamma_3}$, and then $\mathsf{ord}_i(\gamma_1) + \mathsf{ord}_i(\gamma_2) = \mathsf{ord}_i(\gamma_3)$.

iii) We have $\chi_\gamma^n = \chi_{\gamma^n}$ for any $n \geq 1$.

Proof. i). If $\chi_{\gamma_1} \cdot \chi_{\gamma_2} \neq 0$ then there exists $\widetilde{\gamma}_1 \sim \gamma_1$ such that $\widetilde{\gamma}_1^\vee \cdot \gamma_2^\vee = (\gamma_2 \widetilde{\gamma}_1)^\vee \neq 0$. In this case the Anick chain $\gamma_2 \widetilde{\gamma}_1$ must be given by a perfect walk by the unique extension property (Prop. 1.11), and $\gamma_2 \widetilde{\gamma}_1$ is then a stable relation cycle by the recognition lemma (Lemma 3.4) as this perfect walk has the required length. The product lemma (Lemma 3.13) then gives $\gamma_1 \sim \widetilde{\gamma}_1 \approx \gamma_2$ and so $\gamma_1 \approx \gamma_2$.

ii). Let $n_1, n_2 \geq 1$ be such that $\gamma_1^{n_1} \sim \gamma_2^{n_2}$. Then $\chi_{\gamma_1^{n_1}} = \chi_{\gamma_2^{n_2}}$ and so

$$\chi_{\gamma_1}^{n_1} \cdot \chi_{\gamma_2}^{n_2} = \chi_{\gamma_1^{n_1}} \cdot \chi_{\gamma_2^{n_2}} = \chi_{\gamma_1^{n_1}} \cdot \chi_{\gamma_1^{n_1}} = \chi_{\gamma_1^{n_1+n_1}} \neq 0.$$

Hence $\chi_{\gamma_1} \cdot \chi_{\gamma_2} \neq 0$. Possibly replacing γ_1 by $\widetilde{\gamma}_1 \sim \gamma_1$, we can assume that $\gamma_1^\vee \gamma_2^\vee = (\gamma_2 \gamma_1)^\vee \neq 0$; in particular $\gamma_2 \gamma_1$ is an Anick chain, and the recognition lemma again shows that $\gamma_3 := \gamma_2 \gamma_1$ is a stable relation cycle. We are then in the setting of Lemma 3.14.
The decomposition of the associated set $S_3 = S_2 \cdot S_1$ of Lemma 3.14 i) then gives

$$\begin{aligned}
\chi_{\gamma_3} = \sum_{\widetilde{\gamma}_3 \in S_3} \widetilde{\gamma}_3^\vee &= \sum_{\widetilde{\gamma}_2 \widetilde{\gamma}_1 \in S_3} (\widetilde{\gamma}_2 \widetilde{\gamma}_1)^\vee \\
&= \sum_{\widetilde{\gamma}_2 \widetilde{\gamma}_1 \in S_3,} \widetilde{\gamma}_1^\vee \widetilde{\gamma}_2^\vee \\
&= \sum_{\widetilde{\gamma}_1 \in S_1} \widetilde{\gamma}_1^\vee \cdot \sum_{\widetilde{\gamma}_2 \in S_2} \widetilde{\gamma}_2^\vee \\
&= \chi_{\gamma_1} \cdot \chi_{\gamma_2}
\end{aligned}$$

where the second equality writes $\widetilde{\gamma}_3 = \widetilde{\gamma}_2 \widetilde{\gamma}_1$ in terms of the unique decomposition of Lemma 3.14 and the fourth equality follows from the Rigidity Lemma (Lemma 3.7). This proves the main claim, and $\mathsf{ord}_i(\gamma_1) + \mathsf{ord}_i(\gamma_2) = \mathsf{ord}_i(\gamma_3)$ simply follows from additivity of path lengths.

iii). For $n = 1$ there is nothing to prove and $n \geq 2$ follows from specialising the argument of ii) to $\gamma_1 = \gamma^{n-1}$ and $\gamma_2 = \gamma$, giving $\gamma_3 = \gamma^n$. □

We can finally describe the ring structure of $\mathscr{R} = k\langle \chi_\gamma \mid \gamma \rangle \subseteq \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$. First, we observed at the beginning of this subsection that Prop. 3.15 implies that $\mathbb{N}_{\Gamma_i} \subseteq \mathbb{N}$ is a subsemigroup for each branch equivalence class Γ_i. Since $\gcd(\mathbb{N}_{\Gamma_i}) = 1$ by construction, we see that $|\mathbb{N} \setminus \mathbb{N}_{\Gamma_i}| < \infty$ by [159, Lemma 2.1], so that $\mathbb{N}_{\Gamma_i}$ is a numerical semigroup. More importantly for us, $\mathbb{N}_{\Gamma_i}$ is a finitely generated semigroup [159, Corollary 2.8]. It follows that the (semigroup) algebra

$$k[t_i^{\mathbb{N}_{\Gamma_i}}] = k\langle t_i^n \mid n \in \mathbb{N}_{\Gamma_i} \rangle \subseteq k[t_i]. \tag{5.34}$$

is a finitely generated k-algebra, and in particular is Noetherian.

Let us write $\varepsilon : k[t_i^{\mathbb{N}_{\Gamma_i}}] \to k$ for the augmentation with $\varepsilon(t_i^n) = 0$ for all $n \in \mathbb{N}_{\Gamma_i}$. Lastly, we grade the algebra above by setting $|t_i| = \gcd\{|\chi_\gamma| \mid \gamma \in \Gamma_i\}$.

Proposition 3.16. *The graded algebra $\mathscr{R} = k\langle \chi_\gamma \mid \gamma \rangle$ is Noetherian, reduced and of Krull dimension one. There is an embedding of graded algebras*

$$\phi : \mathscr{R} \hookrightarrow \prod_{i=1}^{b} k[t_i]$$

inducing an isomorphism onto the fibre product $\mathscr{R} \cong k[t_1^{\mathbb{N}_{\Gamma_1}}] \times_\varepsilon \cdots \times_\varepsilon k[t_b^{\mathbb{N}_{\Gamma_b}}]$.

Proof. It is clear that the graded algebra $k[t_1^{\mathbb{N}_{\Gamma_1}}] \times_\varepsilon \cdots \times_\varepsilon k[t_b^{\mathbb{N}_{\Gamma_b}}]$ is Noetherian, reduced and of Krull dimension one, and so it suffices to prove the second claim. Define the k-algebra morphism $\phi : \mathscr{R} \to \prod_{i=1}^b k[t_i]$ on generators by

$$\phi(\chi_\gamma) = (0,\ldots,0,t_i^{\mathsf{ord}_i(\gamma)},0,\ldots,0) \text{ for } \gamma \in \Gamma_i$$

where one may interpret the zero coefficients as $t_j^{\mathsf{ord}_j(\gamma)} = t_j^{-\infty} = 0$ for $j \neq i$.

Then ϕ is a well-defined k-algebra morphism by Lemma 3.11 and Prop. 3.15. Moreover Prop. 3.15 shows that $\{\chi_\gamma\}_\gamma$ is a multiplicative basis for $\mathscr{R}$, which ϕ sends bijectively onto a multiplicative basis for $k[t_1^{\mathbb{N}_{\Gamma_1}}] \times_\varepsilon \cdots \times_\varepsilon k[t_b^{\mathbb{N}_{\Gamma_b}}] \subseteq \prod_{i=1}^b k[t_i]$, and so ϕ induces an algebra isomorphism of $\mathscr{R}$ onto its image.

Finally, the grading on $\mathscr{R}$ induces a grading onto $k[t_1^{\mathbb{N}_{\Gamma_1}}] \times_\varepsilon \cdots \times_\varepsilon k[t_b^{\mathbb{N}_{\Gamma_b}}]$ by transport of structure, and since $\mathbb{N}_{\Gamma_i} \subseteq \mathbb{N}$ are numerical semigroups (in particular $\gcd(\mathbb{N}_{\Gamma_i}) = 1$) this forces $|t_i| = \gcd\{|\chi_\gamma| \mid \gamma \in \Gamma_i\}$. Since this is the grading we imposed on the larger algebra $\prod_{i=1}^b k[t_i]$, we see that ϕ was in fact an embedding of graded subalgebras and we are done. □

3.5 Computation of a ring of periodicity operators. Let $\Lambda = kQ/I$ for the quiver given by an oriented cycle on seven vertices with arrows a,b,c,d,e,f,g and relations

$$abcd, bcde, def, efg, fgab, gabc.$$

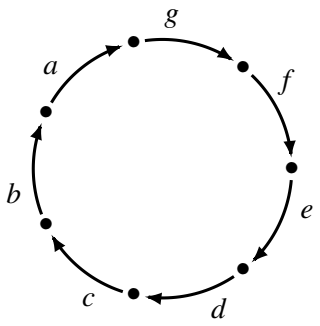


Figure 5.3: A crown.

This example was considered by Green, Snashall and Solberg in [76, Example 4.1, 7.5]. The algebra Λ is Gorenstein of dimension 6, and has a single branch Γ with numerical semigroup $\mathbb{N}_\Gamma = \{2,3,\dots\}$. We then have $\mathscr{R} \cong k[t^2,t^3] \subset k[t]$ with $|t| = 4$, with t^2, t^3 corresponding to $\chi_{\gamma_2}, \chi_{\gamma_3}$ where $\gamma_2 = (abcdefg)^2$ and $\gamma_3 = (abcdefg)^3$. In [76, Example 7.5] the authors compute the Hochschild cohomology ring modulo nilpotents as $\mathrm{HH}^*(\Lambda,\Lambda)/\mathscr{N} \cong k[t]$ with $|t| = 4$, and the embedding above is precisely the natural map $\mathscr{R} \hookrightarrow \mathrm{HH}^*(\Lambda,\Lambda)/\mathscr{N}$, with agreement in all degrees past the Gorenstein dimension. Note that the ring $\mathscr{R}$ does not contain elements of lower degree by design.

4 Main theorem

Using the machinery developed in the previous sections we are now able to characterise those monomial algebras satisfying the **FG** conditions of Snashall–Solberg. We begin with some minor setup.

Consider a Gorenstein monomial algebra Λ. Without loss of generality, we may assume that $\operatorname{gldim}\Lambda = \infty$, as all results below will immediately reduce to this case. To prove that **FG** holds for Λ, it is enough to show that $\mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is module-finite over its A_∞-central subalgebra $\mathscr{R} \subseteq \mathrm{Ext}^*_\Lambda(\Bbbk,\Bbbk)$. We will prove a slightly stronger statement with respect to a Noether normalisation of $\mathscr{R}$.

Let $\Gamma_1, \Gamma_2, \dots, \Gamma_b$ be all branch equivalence classes of stable relation cycles, with accompanying numerical semigroups $\mathbb{N}_{\Gamma_1}, \mathbb{N}_{\Gamma_2}, \dots, \mathbb{N}_{\Gamma_b}$. We have seen in Prop. 3.16 that $\mathscr{R}$ depends only on the structure of these numerical semigroups, and this makes it easy to construct a Noether normalisation.

Let $n_i := \min \mathbb{N}_{\Gamma_i}$ and let $\gamma_i \in \Gamma_i$ be a stable relation cycle with $\mathsf{ord}_i(\gamma_i) = n_i$. Consider the class $\chi_i := \chi_{\gamma_i}$, which is independent of the choice of γ_i above by Lemma 3.11. Introduce the following class $\chi \in \mathscr{R}$:

$$\chi := \sum_{i=1}^{b} \chi_i^{m_i} \tag{5.35}$$

where $m_i \geq 1$ are such that $m_i|\chi_i| = m_j|\chi_j|$ for all i, j and $\gcd\{m_i\} = 1$. From Prop. 3.16 it is clear that the polynomial subalgebra $k[\chi] \subseteq \mathscr{R}$ is a Noether normalisation, and we let $p := |\chi| = m_i|\chi_i|$ denote its degree. We note that p is a multiple of the period ℓ of Λ.

Next, recall that by Corollary 3.5, any perfect path $w \in \Lambda$ over a Gorenstein monomial algebra can be extended to a perfect cycle $(w_0, w_1, \ldots, w_{s-1})$ with $w = w_{s-1}$ such that $\gamma = w_0 w_1 \ldots w_{s-1}$ is a stable relation cycle.

Lemma 4.1. *Let Λ be a Gorenstein monomial algebra and $w \in \Lambda$ be a perfect path. Let $\gamma = w_0 w_1 \ldots w_{s-1}$ be a stable relation cycle with $w = w_{s-1}$, and γ has minimal path length amongst stable relation cycles with this property. If $\gamma \in \Gamma_i$, then $\mathsf{ord}_i(\gamma) \in \mathbb{N}_{\Gamma_i}$ takes on the minimal value.*

Proof. Let $\gamma_i \in \Gamma_i$ be a class with $\mathsf{ord}_i(\gamma_i)$ minimal; we will show that $\gamma \sim \gamma_i$, so that $\mathsf{ord}_i(\gamma) = \mathsf{ord}_i(\gamma_i)$. Since $\gamma \approx \gamma_i$, there are $n, n_i \geq 1$ such that $\gamma^n \sim \gamma_i^{n_i}$. We have seen in Lemma 3.14 that the associated set of $\gamma_i^{n_i}$ is of the form $\{\widetilde{\gamma}_i^{n_i} \mid \widetilde{\gamma}_i \sim \gamma_i\}$, and so $\gamma^n = \widetilde{\gamma}_i^{n_i}$ for some $\widetilde{\gamma}_i \sim \gamma_i$. The unique extension property (Prop. 1.11) then gives $t_\gamma = t_{\gamma^n} = t_{\widetilde{\gamma}_i^{n_i}} = t_{\widetilde{\gamma}_i}$, and so $\gamma, \widetilde{\gamma}_i$ are stable relation cycles with the same tails.

Let $p = t_\gamma = t_{\widetilde{\gamma}_i}$ be the common tail, and note that we may have $p \neq w_{s-1}$ as we did not assume w_{s-1} was the tail of $\gamma = w_0 w_1 \ldots w_{s-1}$. We can then rewrite the $(s-1)$-chain γ as $\gamma = p_0 p_1 \ldots p_{s-1}$ for the perfect cycle $(p_0, p_1, \ldots, p_{s-1})$ ending in $p = p_{s-1}$. While we may have $w_i \neq p_i$ in general, note that the relations $w_0 w_1 = p_0 p_1, \ldots, w_{s-2} w_{s-1} = p_{s-2} p_{s-1}$ are unique by Lemma 3.2, and therefore we have equalities of proper right subtrings γ_{2k} of γ:

$$\gamma_{2k} := p_{2k} p_{2k+1} \ldots p_{s-2} p_{s-1} = w_{2k} w_{2k+1} \ldots w_{s-2} w_{s-1}. \tag{5.36}$$

Our assumption on γ shows that γ_{2k} is not a stable relation cycle for any $k > 0$. It follows that γ also has the smallest path length amongst stable relation cycles with tail $t_\gamma = p$. Since $\widetilde{\gamma}_i$ has minimal value of $\mathsf{ord}_i(\widetilde{\gamma}_i) = \frac{1}{\gcd_i}\mathsf{len}(\widetilde{\gamma}_i)$ in $\mathbb{N}_{\Gamma_i}$ and tail $t_{\widetilde{\gamma}_i} = p$, we conclude that $\gamma = \widetilde{\gamma}_i$ and so finally $\gamma \sim \gamma_i$. □

By the lemma we see that any perfect path w in Λ can be extended to a stable relation cycle $\gamma = w_0 w_1 \ldots w_{s-1}$, not necessarily with $t_\gamma = w_{s-1}$, such that $\chi_\gamma = \chi_i$ agrees with a term of χ in (5.35), where $\gamma \in \Gamma_i$ belongs to the i-th branch equivalence class. This will let us control multiplication against $\chi \in \mathrm{Ext}^*_\Lambda(\Bbbk, \Bbbk)$, as in the next proposition:

Proposition 4.2. *Let Λ be a Gorenstein monomial algebra and χ be the class defined in (5.35). Then left multiplication by χ*

$$\chi \cdot - : \mathrm{Ext}^n_\Lambda(\Bbbk, \Bbbk) \to \mathrm{Ext}^{n+p}_\Lambda(\Bbbk, \Bbbk)$$

is an isomorphism for all $n \geq \dim \Lambda + 1$ and an epimorphism for $n = \dim \Lambda$.

Proof. If $\operatorname{gldim}\Lambda < \infty$ the claim is trivial and so assume that $\operatorname{gldim}\Lambda = \infty$. We consider the action of $\chi \cdot -$ on the basis of chains $kC_{n-1}^{\vee} \cong \operatorname{Ext}^n_{\Lambda}(\Bbbk,\Bbbk)$ and prove that $\chi \cdot -$ is injective for $n \geq \dim\Lambda + 1$ and surjective for $n \geq \dim\Lambda$.

Let $n \geq \dim\Lambda + 1$. Then by Theorem 1.6 any $\gamma_{n-1} \in C_{n-1}$ can be written as $\gamma_{n-1} = \gamma_{n-2}p$ for $p = t_{\gamma_{n-1}}$ perfect (using $n > \dim\Lambda$). Writing $p = p_{s-1}$, we let $\mathbf{p} = (p_0, p_1, \ldots, p_{s-1})$ be the perfect cycle of minimal length such that $\gamma = p_0 p_1 \ldots p_{s-1}$ is a stable relation cycle. Letting Γ_i be the branch equivalence class of γ, Lemma 4.1 shows that $\operatorname{ord}_i(\gamma) \in \mathbb{N}_{\Gamma_i}$ attains the minimal value. In particular $\gamma \sim \gamma_i$ for any other $\gamma_i \in \Gamma_i$ with $\operatorname{ord}_i(\gamma_i)$ also attaining the minimum value.

Writing $\chi = \sum_{i=1}^{b} \chi_i^{m_i}$ as in (5.35) where $\chi_i := \chi_{\gamma_i}$ for $\gamma_i \in \Gamma_i$ a class with $\operatorname{ord}_i(\gamma_i)$ minimal, this shows that each $(n-1)$-chain γ_{n-1} has tail p given by a perfect path occurring as tail of a stable relation cycle γ with $\gamma \sim \gamma_i$ for some i. The unique extension property (Prop. 1.11) and the rigidity lemma (Lemma 3.7) then give

$$\begin{aligned} \chi \cdot \gamma_{n-1}^{\vee} &= \chi_i^{m_i} \cdot \gamma_{n-1}^{\vee} && (5.37)\\ &= (\gamma^{m_i})^{\vee} \cdot \gamma_{n-1}^{\vee} && (5.38)\\ &= (\gamma_{n-1}\gamma^{m_i})^{\vee} && (5.39) \end{aligned}$$

where $\gamma_{n-1}\gamma^{m_i}$ is the unique right extension of γ_{n-1} as an $(n-1+p)$-chain. Since γ_{n-1} can be recovered from the $(n-1+p)$-chain $\gamma_{n-1}\gamma^{m_i}$ by successively removing tails, this shows that $\chi \cdot - : kC_{n-1}^{\vee} \to kC_{n-1+p}^{\vee}$ is injective.

Now let $n \geq \dim\Lambda$. We prove that $\chi \cdot - : kC_{n-1}^{\vee} \to kC_{n-1+p}^{\vee}$ is surjective analogously. Corollary 1.12 shows that every $(n-1+p)$-chain is of the form $\gamma_{n-1} w_n w_{n+1} \ldots w_{n-1+p}$ for a perfect walk $(w_n, w_{n+1}, \ldots, w_{n-1+p})$ and an $(n-1)$-chain γ_{n-1}. Since the perfect walk $(w_n, w_{n+1}, \ldots, w_{n-1+p})$ has length $p \geq \dim\Lambda + 1$, with p even and a multiple of ℓ, the recognition lemma (Lemma 3.4) shows that $w_n w_{n+1} \ldots w_{n-1+p}$ is a stable relation cycle, and we may write $w_n w_{n+1} \ldots w_{n-1+p} = \gamma^m$ for some stable relation cycle γ of minimal path length and some $m \geq 1$. Letting Γ_i again be the class of γ, Lemma 4.1 shows that $\operatorname{ord}_i(\gamma)$ takes on minimal value so that $\gamma \sim \gamma_i$; in particular $\chi_\gamma = \chi_{\gamma_i}$ and the equality $m|\chi_\gamma| = p = m_i|\chi_{\gamma_i}|$ forces $m = m_i$. It follows that every $(n-1+p)$-chain has the form $\gamma_{n-1}\gamma^{m_i}$, and so every basis element of $kC_{n-1+p}^{\vee}$ can be written as

$$(\gamma_{n-1}\gamma^{m_i})^{\vee} = \chi \cdot \gamma_{n-1}^{\vee} \tag{5.40}$$

for some $\gamma_{n-1}^{\vee} \in C_{n-1}^{\vee}$. This proves surjectivity, and we are done. □

Remark 5. The bound $n \geq \dim\Lambda + 1$ cannot be improved to $n \geq \dim\Lambda$ in general. This is clear if $\operatorname{gldim}\Lambda < \infty$ as $\chi = 0$ and $\operatorname{Ext}^{\dim\Lambda}_{\Lambda}(\Bbbk,\Bbbk) \neq 0$ then, but even when $\operatorname{gldim}\Lambda = \infty$ this can fail due to non-trivial projective summands in $\Omega^{\dim\Lambda}\Bbbk$ as seen in Section 1.3.

We now obtain the main result of this chapter.

Theorem 4.3. *Let Λ be a monomial algebra. Then Λ satisfies **FG** if and only if Λ is Gorenstein.*

Proof. As we mentioned before, the necessary implication is due to Erdmann-Holloway-Snashall-Solberg-Taillefer in [59, Th. 1.5]. Conversely assume that Λ is Gorenstein, and that $\operatorname{gldim}\Lambda = \infty$ as the result is trivial otherwise. In this case Proposition 4.2 shows that $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$, as a module over its polynomial subalgebra $\Bbbk[\chi] \subseteq \mathscr{R} \subseteq \mathscr{Z}_\infty\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$, is generated by elements of degree at most $\dim\Lambda + p - 1$. It follows that $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ and $\mathscr{Z}_\infty = \mathscr{Z}_\infty\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk) \subseteq \operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ are module-finite over $\Bbbk[\chi]$. In particular $\mathscr{Z}_\infty$ is Noetherian and the algebra $\operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ is module-finite over $\mathscr{Z}_\infty$, so that **FG** holds by Proposition 2.3. □

5 Further results

5.1 Combinatorial characterisation of finite generation. Let Λ be a monomial algebra. We have just established that Λ satisfies **FG** if and only if it is Gorenstein, and we would like to have a simple combinatorial criterion for testing the Gorenstein property of Λ starting from the quiver and relations.

For any d with $1 \leq d < \infty$, we have seen in Theorem 1.6 a combinatorial characterisation for the condition that Λ is Gorenstein of dimension $\dim\Lambda \leq d$ in terms of the structure of $(d-1)$-chains. More precisely, when this holds one sees a periodicity occurring in the structure n-chains for all $n \geq d$, in that all subsequent chains are obtained by appending the next path in a perfect cycle, which are periodic.

Starting with an arbitrary monomial algebra Λ, this eventual periodicity of n-chains, if it occurs, could a priori begin in arbitrarily large degree $n \gg 0$, and so it is not clear that one can rule out the **FG** property for Λ by inspecting finitely many chains.

As it turns out, there is a simple upper bound n_Λ such that this periodicity either occurs for n-chains for all $n \geq n_\Lambda$ or never occurs at all. Recall that the (little) right finitistic dimension of Λ is defined as

$$\operatorname{rfindim}\Lambda := \sup\{\operatorname{projdim} M \mid M \in \operatorname{mod}\Lambda \text{ and } \operatorname{projdim} M < \infty\}. \tag{5.41}$$

The left finitistic dimension is defined analogously via left modules, or simply via $\operatorname{lfindim}\Lambda = \operatorname{rfindim}\Lambda^{op}$. From the work of Igusa, Zacharia [97] and Green, Kirkman and Kuzmanovich [75], one knows that the finitistic dimension of a monomial algebra is always finite. Moreover, if Λ is any Gorenstein algebra then the finitistic dimension is also known to be finite, and in fact agrees with the Gorenstein dimension $\operatorname{rfindim}\Lambda = \dim\Lambda$ by [98] (and then $\operatorname{lfindim}\Lambda = \dim\Lambda^{op} = \dim\Lambda = \operatorname{rfindim}\Lambda$). Letting n_Λ be any upper bound for the finitistic dimension of a monomial algebra Λ, it follows that Λ is either Gorenstein of dimension $\dim\Lambda \leq n_\Lambda$ or not Gorenstein at all.

Simple upper bounds are known by work of Igusa, Zacharia [97] and Zimmermann-Huisgen [184]. For instance, let $K \geq 0$ be the minimal integer such that $\mathsf{r}^{K+1} = 0$ and set

$$n_\Lambda := \min\{\dim_k \mathsf{r},\, \dim_k \Lambda/\mathsf{r}^K\}. \tag{5.42}$$

Then $\operatorname{rfindim}\Lambda \leq n_\Lambda$ by [184, Section 3]. Using this upper bound, we obtain the following decidable combinatorial criterion for **FG**:

Theorem 5.1. *Let Λ be a monomial algebra. The following are equivalent:*

*(1) Λ satisfies **FG**.*

(2) Λ is Gorenstein.

(3) Let $n = \operatorname{rfindim}\Lambda$. Then every n-chain γ has a perfect path t_γ for tail.

(4) Let $n = \operatorname{lfindim}\Lambda$. Then every n-chain γ has a perfect path h_γ for head.

(5) Let $n = n_\Lambda$. Then every n-chain γ has a perfect path t_γ for tail.

(6) Let $n = n_\Lambda$. Then every n-chain γ has a perfect path h_γ for head.

Proof. The equivalence of (1) and (2) follows from Theorem 4.3. Next, note that (3)-(4) and (5)-(6) are dual under $\Lambda \leftrightarrow \Lambda^{\mathrm{op}}$, and as (2) is self-dual it is enough to show that (3) is equivalent to (2) and (5) is equivalent to (2).
Set $N_\Lambda = \operatorname{rfindim}\Lambda$. Assuming (3), Theorem 1.6 shows that Λ is Gorenstein of dimension $\dim\Lambda \leq N_\Lambda + 1$, in particular Λ is Gorenstein and (2) holds. Conversely, by the remarks above Λ is Gorenstein if and only if Λ is Gorenstein of dimension $\dim\Lambda \leq N_\Lambda$. Assuming (2), we obtain that $\dim\Lambda < N_\Lambda + 1$ and so (3) follows from Theorem 1.6. Finally the equivalence of (2) and (5) follows verbatim by setting $N_\Lambda = n_\Lambda$ instead in the previous argument. □

Remark 6. In practice the bound n_Λ appears not to be very sharp and so periodicity may occur much faster; in any practical implementation better upper bounds should be used, see [184] for further details. Our choice of bound n_Λ was mainly for its simplicity to state.

5.2 The structure of Hochschild cohomology. We conclude with some interesting corollaries on the structure of Hochschild cohomology. Let $\chi \in \operatorname{Ext}^p_\Lambda(\Bbbk,\Bbbk)$ be the A_∞-central class defined in (5.35), and we also write $\chi \in \operatorname{HH}^p(\Lambda,\Lambda)$ for the corresponding lift.

Theorem 5.2. *Let Λ be a Gorenstein monomial algebra. Then $\operatorname{HH}^*(\Lambda,\Lambda)$ is eventually periodic; more precisely, taking cup product with χ gives an isomorphism*

$$\chi \smile - : \operatorname{HH}^n(\Lambda,\Lambda) \xrightarrow{\cong} \operatorname{HH}^{n+p}(\Lambda,\Lambda) \quad \text{for all } n \geq \dim\Lambda + 1.$$

Proof. We give a quantitative version of a standard filtration argument [59, Prop. 1.4] . First, recall that for any $M,N \in \operatorname{mod}\Lambda$ there are natural isomorphisms

$$\operatorname{HH}^*(\Lambda, \operatorname{Hom}_k(M,N)) \cong \operatorname{Ext}^*_\Lambda(M,N). \tag{5.43}$$

Moreover one has $\operatorname{Hom}_k(\Bbbk,\Bbbk) \cong \mathrm{D}(\Bbbk) \otimes_k \Bbbk \cong \Lambda^{\mathsf{e}}/\operatorname{rad}\Lambda^{\mathsf{e}}$ for $\mathrm{D} = \operatorname{Hom}_k(-,k)$. For any bimodule $B \in \operatorname{mod}\Lambda^{\mathsf{e}}$ the vector space $\operatorname{HH}^*(\Lambda,B)$ is a right-module over $\operatorname{HH}^*(\Lambda,\Lambda)$, and this module structure on $\operatorname{HH}^*(\Lambda, \operatorname{Hom}_k(\Bbbk,\Bbbk)) \cong \operatorname{Ext}^*_\Lambda(\Bbbk,\Bbbk)$ agrees with that given by the characteristic morphism

$\varphi_{\underline{\Bbbk}}$. Now, consider a Jordan-Hölder filtration of Λ in $\operatorname{mod}\Lambda^e$

$$0 = B_{K+1} \subseteq B_K \subseteq \cdots \subseteq B_2 \subseteq B_1 \subseteq B_0 = \Lambda \tag{5.44}$$

with $B_i/B_{i+1} \in \operatorname{add}_{\Lambda^e}(\operatorname{Hom}_k(\underline{\Bbbk},\underline{\Bbbk}))$ for $0 \leq i \leq K$. The long exact sequence of $\mathrm{HH}^*(\Lambda,B)$ is compatible[1] with the module structure over $\mathrm{HH}^*(\Lambda,\Lambda)$ and so gives rise to commutative diagrams of the form (writing $\mathrm{HH}^n(B) := \mathrm{HH}^n(\Lambda,B)$ for short)

$$\begin{array}{ccccccccc}
\mathrm{HH}^{n-1}(B_i/B_{i+1}) & \to & \mathrm{HH}^n(B_{i+1}) & \to & \mathrm{HH}^n(B_i) & \to & \mathrm{HH}^n(B_i/B_{i+1}) & \to & \mathrm{HH}^{n+1}(B_{i+1}) \\
\downarrow{\scriptstyle \chi\smile -} & & \downarrow{\scriptstyle \chi\smile -} & & \downarrow{\scriptstyle \chi\smile -} & & \downarrow{\scriptstyle \chi\smile -} & & \downarrow{\scriptstyle \chi\smile -} \\
\mathrm{HH}^{n-1}(B_i/B_{i+1}) & \to & \mathrm{HH}^n(B_{i+1}) & \to & \mathrm{HH}^n(B_i) & \to & \mathrm{HH}^n(B_i/B_{i+1}) & \to & \mathrm{HH}^{n+1}(B_{i+1})
\end{array}$$

Since $B_i/B_{i+1} \in \operatorname{add}_{\Lambda^e}(\operatorname{Hom}_k(\underline{\Bbbk},\underline{\Bbbk}))$, from the above paragraph and Prop. 4.2 we see that the fourth column map is an isomorphism for $n \geq \dim\Lambda + 1$, and the first column map is an isomorphism for $n \geq \dim\Lambda + 2$ and an epimorphism for $n = \dim\Lambda + 1$. The 5-Lemma then shows that if the second and fifth column maps are isomorphism for $n \geq \dim\Lambda + 1$, then so is the third column map. The result for $B_0 = \Lambda$ then follows by induction from the base case $B_K = B_K/B_{K+1} \in \operatorname{add}_{\Lambda^e}(\operatorname{Hom}_k(\underline{\Bbbk},\underline{\Bbbk}))$, which follows from Prop. 4.2. □

Finally, recall that for Λ Gorenstein the enveloping algebra $\Lambda^{\mathsf{e}} = \Lambda^{op} \otimes_k \Lambda$ is also Gorenstein, then of dimension $\dim\Lambda^{\mathsf{e}} = 2\dim\Lambda$, see [16, Prop. 6.1]. Following Buchweitz, we define the Tate-Hochschild cohomology as the Ext algebra of the bimodule $\Lambda \in \mathrm{D}_{sg}(\Lambda^{\mathsf{e}})$ in the singularity category of Λ^{e}:

$$\widehat{\mathrm{HH}}^*(\Lambda,\Lambda) := \operatorname{Ext}^*_{\mathrm{D}_{sg}(\Lambda^{\mathsf{e}})}(\Lambda,\Lambda). \tag{5.45}$$

Corollary 5.3. *Let Λ be a Gorenstein monomial algebra. Then the Tate-Hochschild cohomology ring is given by periodic Hochschild cohomology:*

$$\widehat{\mathrm{HH}}^*(\Lambda,\Lambda) \cong \mathrm{HH}^*(\Lambda,\Lambda)[\chi^{-1}].$$

Proof. We first find a good model for $\widehat{\mathrm{HH}}^*(\Lambda,\Lambda)$ by constructing an appropriate complete resolution of the diagonal. Let $P_* \xrightarrow{\sim} {}_\Lambda\Lambda_\Lambda$ be a minimal projective resolution of the diagonal bimodule Λ. We may represent the class $\chi \in \mathrm{HH}^p(\Lambda,\Lambda)$ by a chain map

$$\chi : P_{*+p} \to P_* \tag{5.46}$$

[1] This follows from binaturality of $\operatorname{Ext}^*_{\Lambda^e}(\Lambda,M)$ making it a right module over $\operatorname{Ext}^*_{\Lambda^e}(\Lambda,\Lambda)$; note that a priori this only makes the map $- \smile \chi$ commute with the long exact sequence, but we use $\chi \smile -$ for consistency with the theorem's claim and use graded-commutativity.

The corresponding class $\varphi_{\underline{\Bbbk}}(\chi) = \chi \in \mathrm{Ext}^p_\Lambda(\underline{\Bbbk},\underline{\Bbbk})$ is then represented by

$$\underline{\Bbbk} \otimes_\Lambda \chi : \underline{\Bbbk} \otimes_\Lambda P_{*+p} \to \underline{\Bbbk} \otimes_\Lambda P_*. \tag{5.47}$$

Minimality of P_* means that $P_* \otimes_{\Lambda^e} \Lambda^e/\mathrm{rad}\,\Lambda^e \cong \underline{\Bbbk} \otimes_\Lambda P_* \otimes_\Lambda \underline{\Bbbk}$ has trivial differential, and so $\underline{\Bbbk} \otimes_\Lambda P_*$ is a minimal right resolution of $\underline{\Bbbk}$ over Λ. Right multiplication by χ then corresponds to

$$\begin{array}{ccc}
\mathrm{Ext}^n_\Lambda(\underline{\Bbbk},\underline{\Bbbk}) & \xrightarrow{-\cdot\chi} & \mathrm{Ext}^{n+p}_\Lambda(\underline{\Bbbk},\underline{\Bbbk}) \\
\| & & \| \\
\mathrm{Hom}_\Lambda(\underline{\Bbbk} \otimes_\Lambda P_n, \underline{\Bbbk}) & \xrightarrow{-\circ\chi} & \mathrm{Hom}_\Lambda(\underline{\Bbbk} \otimes_\Lambda P_{n+p}, \underline{\Bbbk}) \\
\| & & \| \\
\mathrm{Hom}_{\underline{\Bbbk}}(\underline{\Bbbk} \otimes_\Lambda P_n \otimes_\Lambda \underline{\Bbbk}, \underline{\Bbbk}) & \xrightarrow{-\circ\chi} & \mathrm{Hom}_{\underline{\Bbbk}}(\underline{\Bbbk} \otimes_\Lambda P_{n+p} \otimes_\Lambda \underline{\Bbbk}, \underline{\Bbbk}).
\end{array}$$

The first row is an isomorphism in degree $n \geq \dim\Lambda + 1$, and the third row is $\underline{\Bbbk}$-dual to the map χ tensored down

$$\chi \otimes_{\Lambda^e} \Lambda^e/\mathrm{rad}\,\Lambda^e : P_{n+p} \otimes_{\Lambda^e} \Lambda^e/\mathrm{rad}\,\Lambda^e \to P_n \otimes_{\Lambda^e} \Lambda^e/\mathrm{rad}\,\Lambda^e \tag{5.48}$$

which must then also be an isomorphism for $n \geq \dim\Lambda + 1$. By the Nakayama Lemma we conclude that $\chi : P_{n+p} \to P_n$ is an isomorphism in the same range. In particular Λ has an eventually periodic minimal projective bimodule resolution.

Let C_* be the unique periodic, unbounded, acyclic complex of finite Λ^e projectives which agrees with P_* in degree $n \geq \dim\Lambda + 1$; the complex C_* is then a complete resolution of (the Gorenstein-projective approximation of) Λ over Λ^e. By Buchweitz's theorem [35] we may compute the Tate-Hochschild cohomology algebra as

$$\widehat{\mathrm{HH}^*}(\Lambda,\Lambda) \cong \mathrm{H}^*(\mathrm{Hom}_{\Lambda^e}(C_*, C_*)) \tag{5.49}$$

The chain-map $\chi : P_{*+p} \to P_*$ extends by periodicity to $\chi : C_{*+p} \to C_*$, giving rise to a class $\chi \in \widehat{\mathrm{HH}^p}(\Lambda,\Lambda)$ which is the image of $\chi \in \mathrm{HH}^p(\Lambda,\Lambda)$ under the natural algebra map $\psi : \mathrm{HH}^*(\Lambda,\Lambda) \to \widehat{\mathrm{HH}^*}(\Lambda,\Lambda)$. More importantly, $\chi : C_{*+p} \to C_*$ represents the periodicity isomorphism by construction and therefore admits an inverse $\chi^{-1} : C_{*-p} \to C_*$, with $\chi^{-1} \in \widehat{\mathrm{HH}^{-p}}(\Lambda,\Lambda)$. The natural map then factors through the localisation as $\psi = \widetilde{\psi} \circ u$

$$\mathrm{HH}^*(\Lambda,\Lambda) \xrightarrow{u} \mathrm{HH}^*(\Lambda,\Lambda)[\chi^{-1}] \xrightarrow{\widetilde{\psi}} \widehat{\mathrm{HH}^*}(\Lambda,\Lambda). \tag{5.50}$$

By [35, 6.3.5] the map $\psi : \mathrm{HH}^n(\Lambda,\Lambda) \to \widehat{\mathrm{HH}^n}(\Lambda,\Lambda)$ is an isomorphism for all $n \geq \dim\Lambda^{\mathsf{e}} + 1 = 2\dim\Lambda + 1$, and so u is injective in the same degrees. Since multiplication by χ on $\mathrm{HH}^{\geq n}(\Lambda,\Lambda)$ acts by periodicity for $n \geq \dim\Lambda + 1$ by the last theorem, the map u is surjective in degree $n \geq \dim\Lambda + 1$.

Hence u, and therefore $\widetilde{\psi}$, is an isomorphism in degree $n \geq 2\dim\Lambda + 1$. Finally since $\widetilde{\psi}$ commutes with the action of the periodicity operators $\chi^{\pm 1}$ we see that $\widetilde{\psi}$ is an isomorphism in all degrees. $\square$

Chapter 6

Term rewriting and homotopical algebra

The chapter is organized as follows. In Section 1 we review the essential results coming from homotopy theory of associative algebras. In Section 2 we give our first homotopical re-formulation of uniqueness of normal forms as a general criterion in terms of perturbations of free resolutions (Theorem 2.1). Using that result, in Section 3 we prove another general criterion in terms of approximate solutions to the Maurer–Cartan equation in the tangent complex (Theorem 3.1). Using this latter result, we establish the main result of this chapter (Theorem 3.2): every model extending the Shafarevich complex of the given relations gives rise to its own Diamond Lemma, so that the resolvable ambiguities condition of the Diamond Lemma becomes the first non-trivial component of the Maurer–Cartan equation in the tangent complex of the model. Applying that latter result to some specific combinatorial models, we recover in Section 4 the classical Diamond Lemma in the case of the inclusion–exclusion model of the Dotsenko and Khoroshkin [52] and the Triangle Lemma in the case of our minimal model [171]. To conclude, we discuss some analogues and possible generalisations of our results for other types of algebras towards the end of the article. Already in the case of operads with monomial relations, explicit formulas for minimal models are not known in general, and only a much weaker version of the Triangle Lemma is available [31]; we hope that our work will attract due attention to this question.

1 Homotopy theory of associative algebras

Deformation theory of algebras borrows a lot of intuition from algebraic topology: one may say that algebraic tools for studying continuous deformations of spaces are very much amenable to the case of deformations of algebras. In algebraic topology, an important invariant of a space is its cohomology algebra. However, that algebra itself does not capture many important homotopy invariants. It turns out that to remedy that one should either consider cohomology together with certain higher

structures, or work with a bigger algebra of the same homotopy type, for example the differential graded algebra of singular cochains. This led Quillen to the general philosophy of homotopical algebra [153] suggesting that one should extend categories of algebras to better behaved "model categories" with a notion of weak equivalence, an abstraction of homotopy equivalence, and consider the homotopy category formed by equivalence classes. Alternatively, one may say that instead of studying an algebra it is often beneficial to study its model, which is a better behaved algebra of the same homotopy type in the model category. The construction in the next section can be thought of as providing an anchor towards constructing such models.

1.1 Shafarevich complexes. We shall now recall the notion of a Shafarevich complex of a system of elements in an algebra. Those complexes are very rarely models themselves, but they offer a good starting point for constructing a model, which will be one of the guiding principles for our main results.

Definition 19 (Shafarevich complex)**.** Let A be an associative (non-differential non-graded) algebra. The *Shafarevich complex* of a subset $S \subset A$ is the differential graded algebra

$$\mathsf{Sha}(S,A) = (A * T(U), d).$$

Here $A * T(U)$ is the free product of A and the tensor algebra, U is the graded vector space whose only non-zero component is $U_1 \cong \Bbbk S$, and d is the unique graded derivation satisfying $d(a) = 0$ for $a \in A$ and $d(e_s) = s \in A$ for each basis element $s \in U_1$.

It is an immediate consequence of the definition that the degree zero homology of $\mathsf{Sha}(S,A)$ is isomorphic to the quotient $\cong A/(S)$. In this chapter, we shall only consider the Shafarevich complexes $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$ for the free algebra $A = \Bbbk\langle X\rangle$; it is precisely those that were first defined in work of Golod and Shafarevich [70] and later called the Shafarevich complexes by Golod in [68]. In this particular case, the degree zero homology is the algebra $\Bbbk\langle X|S\rangle$. Of course, $\mathsf{Sha}(S,\Bbbk\langle X\rangle) = (T(V),d)$ where V is the graded vector space whose only non-zero components are $V_0 = \Bbbk X$ and $V_1 \cong \Bbbk S$, and d is the unique graded derivation satisfying $d(x) = 0$ for $x \in X$ and $d(e_s) = s \in \Bbbk\langle X\rangle \cong T(V_0)$ for each basis element $s \in V_1$. Thus, $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$ is a triangulated quasi-free algebra whose space of generators is concentrated in homological degrees zero and one corresponding to killing certain cycles in $T(V_0)$. In that capacity, it was rediscovered by Lemaire [124], and used by Anick in his work on Hilbert series of associative algebras [4, 5].
In general, the positive degree homology of $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$ does not vanish, so that differential graded algebra is not a model of $\Bbbk\langle X|S\rangle$. In fact, it is known that the homology of $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$ is generated by elements of homological degrees zero and one [69], but this remarkable result will not be immediately useful for us. We however will be mainly interested in models that extend Shafarevich complexes.

Definition 20 (Acyclic extension of the Shafarevich complex)**.** Let $A \cong \Bbbk\langle X|S\rangle$ be an associative algebra presented by generators and relations. We say that a model of the algebra A is an

acyclic extension of the Shafarevich complex $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$ if it is concentrated in non-negative homological degrees, and in homological degrees zero and one is isomorphic to the Shafarevich complex $\mathsf{Sha}(S,\Bbbk\langle X\rangle)$.

2 Rewriting systems and perturbation of models

Since any model of an associative algebra is quasi-free, it is not particularly interesting as an algebra: all the complexity is hidden in the differential. Thus, if we replace an algebra by a model, it is reasonable to expect that studying deformation theory is done via studying perturbations of the differential. The main result of this section confirms this expectation: essentially, it says that once we fix a filtration, deforming a monomial algebra with respect to that filtration is equivalent to perturbing the differential of its model.

Definition 21 (Word-homogeneous algebra)**.** Let us call a differential associative graded algebra (B,d) *word-homogeneous* if it has an extra grading by the free monoid $\langle X\rangle$, that is we have a decomposition into a direct sum of chain subcomplexes

$$(B,d)=\bigoplus_{u\in\langle X\rangle}(B_u,d),$$

such that $B_{u_1}B_{u_2}\subseteq B_{u_1u_2}$ for all $u_1,u_2\in\langle X\rangle$. For an element $b\in B_u$, we say the element b is word-homogeneous of *grade* u.

As above, let us consider a rewriting system $\Sigma=(X,W,f)$. Clearly, the non-differential non-graded algebra $(A_{\Sigma_0},0)$ is word-homogeneous. It is easy to establish that its minimal model is word-homogeneous; while we shall consider models that are not necessarily non-minimal, we shall restrict ourselves to word-homogeneous models when considering monomial algebras. Moreover, since the algebra A_{Σ_0} is concentrated in homological degree zero, it is enough to work with word-homogeneous models concentrated in non-negative homological degrees.
For such a model $(T(V),d)$, the differential graded subalgebra $(T(V_0),0)$ surjects onto A_{Σ_0}; in other words, the image of the vector space V_0 generates the algebra A_{Σ_0}. In the context of studying the rewriting system Σ, we are working with a distinguished set of generators X, so it makes sense to focus on models with $V_0\cong\Bbbk X$. Moreover, since $H_\bullet(T(V),d)\cong A_{\Sigma_0}$, the image of V_1 under d generates the two-sided ideal of $T(V_0)\cong\Bbbk\langle X\rangle$ generated by W; it makes sense to focus on models with $V_1\cong\Bbbk W$. Using the terminology we recalled earlier, this means that we work with word-homogeneous acyclic extensions of the Shafarevich complex $\mathsf{Sha}(W,\Bbbk\langle X\rangle)$. Finally, we make one slightly less trivial assumption: similar to many situations in rational homotopy theory, we shall consider models of finite type, that is require that $\dim V_n<\infty$ for all $n\geq 0$. This is true for the minimal model of a monomial algebra with finitely many relations, as established in [171], and for some other models of interest.

The following result is a mild generalisation of [171, Th. 5.1], which mimics [52, Th. 4.1], and is an adaptation to our case of the argument of Anick [2, Th. 1.4]. We say that a linear endomorphism of a word-homogeneous algebra is Σ-*filtered* if it sends every element of grade $u \in \langle X \rangle$ to a combination of elements of grades strictly less than u with respect to the ordering $<_\Sigma$.

Theorem 2.1. *Suppose that $(T(V), d)$ is a word-homogeneous acyclic extension of finite type of the Shafarevich complex $\mathsf{Sha}(W, \Bbbk\langle X \rangle)$. The rewriting system Σ is convergent if and only if there exists an acyclic extension $(T(V), d + d')$ of the Shafarevich complex $\mathsf{Sha}(R, \Bbbk\langle X \rangle)$ with a Σ-filtered perturbation d'.*

Note that being an acyclic extension of the Shafarevich complex $\mathsf{Sha}(R, \Bbbk\langle X \rangle)$ means in particular that $d'|_{V_0} = 0$ and that for each basis element $e_w \subset V_1$, we have

$$d'(e_w) = -f(w),$$

so d' is indeed Σ-filtered on elements of degree one.

Proof. Suppose first that such a model exists. Extend the partial ordering $<_\Sigma$ to a total well ordering arbitrarily, and consider the filtration of the graded vector space $T(V)$ associated to that ordering. Since we are dealing with a countable well-order that is not necessarily isomorphic to $\mathbb{N}$, one has to be careful, and either consider a sequence of spectral sequences, or consider generalised spectral sequences of countably filtered modules, such as the transfinite spectral sequences of Hu [95] or spectral sequences of transfinitely filtered modules of Rahmati [154]; either strategy works in our case, and allows one to construct a spectral sequence converging to $H_\bullet(T(V), d + d')$ with the zeroth term E^0 given by $(T(V), d)$. This forces our spectral sequence to collapse on the following page, since the homology $H_\bullet(T(V), d)$ is concentrated in homological degree zero. Examining the individual word-homogeneous components, we conclude that cosets of the monomial basis of A_{Σ_0} form a basis of the vector space A_Σ, and therefore Σ is convergent.

Suppose now that Σ is convergent. If we ignore the differential, there exist surjective homomorphisms from the algebra $T(V)$ to both the monomial algebra A_{Σ_0} and the algebra A_Σ: one may project it onto its part $T(V_0) \cong \Bbbk\langle X \rangle$ of homological degree 0, and the latter admits obvious projection maps to its respective quotients. Let us choose splittings for those projections; this amounts to exhibiting two idempotent endomorphisms $\bar{\pi}$ and π of $T(V)$ such that both of them annihilate all elements of positive homological degree, the former annihilates the ideal of $T(V_0) \cong \Bbbk\langle X \rangle$ generated by W, and the latter annihilates the ideal of $T(V_0) \cong \Bbbk\langle X \rangle$ generated by R. Since the model $T(V)$ is of finite type, there exists a word-homogeneous map $h\colon T(V) \longrightarrow T(V)$ such that $[d, h] = 1 - \bar{\pi}$.

We are going to define a derivation $d'\colon T(V) \longrightarrow T(V)$ of homological degree -1 satisfying the requested conditions; since d' is a derivation, it is enough to define it on the generators. We shall also define a map

$$h'\colon \ker(d + d') \to T(V)$$

of homological degree 1 such that

$$(d+d')(h+h')\big|_{\ker(d+d')} = 1-\pi.$$

This latter condition instantly implies that $(T(V), d+d')$ is a model of A_Σ.
We shall construct the maps and prove their properties inductively. More specifically, we shall prove by induction on $k \geq 0$ that one can define the values of the perturbation d' on generators of homological degree $k+1$ and the values of h' on elements of $\ker(d+d')$ of homological degree k so that the following conditions hold:

- *perturbation*: both maps d' (on generators of homological degree $k+1$) and h' (on elements of $\ker(d+d')$ of homological degree k) are Σ-filtered,
- *square-zero*: we have $(d+d')^2 = 0$ on generators of homological degree $k+1$,
- *homotopy*: we have $(d+d')(h+h') = 1-\pi$ on elements of $\ker(d+d')$ of homological degree k; of course, according to the definition of the map π, this condition reads $(d+d')(h+h') = 1$ on elements of positive homological degree.

As a basis of induction, we recall that the formula $d'(e_w) = -f(w)$ indicated in the statement of the theorem is Σ-filtered, and the square-zero condition is satisfied for degree reasons, as there are no elements of negative homological degree. Suppose that x is a degree zero element of grade u. We define

$$h'(x) = \begin{cases} 0, & \text{if } x \notin (W), \\ -(h+h')((1-\pi)(x) - (d+d')h(x)), & \text{if } x \in (W). \end{cases}$$

Since the map h is word-homogeneous and since we already know that d' is Σ-filtered on elements of homological degree 1, the map $d'h$ is Σ-filtered. Also, the map

$$(1-\pi) - dh = (1-\pi) - (1-\bar{\pi}) = (\bar{\pi} - \pi)$$

is Σ-filtered simply by definition of $<_\Sigma$. By induction with respect to the well-ordering $<_\Sigma$, we may assume the Σ-filtered condition for the map h' evaluated on sum of those elements, and the perturbation condition for h' evaluated on x follows. Finally, to show the homotopy property evaluated on elements of homological degree zero, we note that on elements $x \notin (W)$ this becomes $0 = 0$, and otherwise we have

$$(d+d')(h+h')(x) = (d+d')h(x) - (d+d')(h+h')((1-\pi)(x) - (d+d')h(x))$$

which, by induction on the well-ordering $<_\Sigma$, is equal to

$$(d+d')h(x) - (1-\pi)((1-\pi)(x) - (d+d')h(x)) = \\ (1-\pi)^2(x) - (1-\pi)(d+d')h(x) = (1-\pi)(x),$$

since π vanishes on the image of $d+d' = (R)$ and $1-\pi$ is a projector.
To make the step of induction, we proceed in a similar way. To define $d'(x)$ for a word homogeneous generator x of homological degree $k+1>1$, we put

$$d'(x) = -(h+h')(d+d')d(x).$$

The Σ-filtered property for d' now easily follows by induction. For the square-zero property, we note that

$$\begin{aligned}(d+d')^2(x) = (d+d')(d(x)-(h+h')(d+d')d(x)) = \\ (d+d')d(x)-(d+d')(h+h')((d+d')d(x)) = \\ (d+d')d(x)-(1-\pi)(d+d')d(x) = \pi((d+d')d(x)) = 0,\end{aligned}$$

since $(d+d')d(x) \in \ker(d+d')$, and since π vanishes on the image of $d+d'$. Suppose that x is word-homogeneous of grade u. Note that whenever $x \in \ker(d+d')$, we have $x-(d+d')h(x) \in \ker(d+d')$ as well. Since $\bar{\pi}$ vanishes on elements of positive homological degree, we have $dh(x) = (1-\bar{\pi})(x) = x$, and so, using the perturbation condition for d', we see that $x-(d+d')h(x)$ is a combination of elements of grades strictly less than u with respect to the ordering $<_\Sigma$. Consequently, we may define h' on elements of $\ker(d+d')$ of homological degree $k>0$ by the same inductive argument: we put

$$h'(x) = (h+h')(x-(d+d')h(x)).$$

Once again, a simple inductive argument shows that the Σ-filtered property for the map h' and the homotopy condition are satisfied.
We conclude that $(T(V), d+d')$ a differential graded algebra, with the homology $H_\bullet(T(V), d+d')$ isomorphic to A_Σ, and a Σ-filtered map d', as required. It remains to check that it is triangulated. For that, we first decompose the space of generators V according to the homological degree, and then for the fixed homological degree n, decompose V_n into word-homogeneous components. Since we work with a model of finite type, there are finitely many word-homogeneous components in each homological degree, and so once we linearly order them extending the ordering $<_\Sigma$, the total order on thus obtained components is either finite or isomorphic to $\mathbb{N}$. This total order gives a triangulation: by construction, the differential of each component is made of products of elements from the previous components. This completes the proof of the theorem. □

3 Differential graded Lie algebras and the Diamond Lemma

In this section, we present a deformation-theoretic version of the Diamond Lemma. This corresponds to the heuristics of deformation theory going back to Deligne, Drinfeld, and Feigin (and later formalised by Lurie [134, 135] and Pridham [149]) that, infinitesimally, any moduli space of

deformations in characteristic zero is controlled by a Maurer–Cartan equation in a suitable differential graded Lie algebra. We remark that while using this heuristics requires to restrict ourselves to a ground field $\Bbbk$ of characteristic zero, one can easily see that replacing, in Theorem 3.1, the Maurer–Cartan equation $[d+F,d+F]=0$ by the square-zero condition $(d+F)^2=0$ leads to a result that is valid without any assumption on the ground field.

3.1 Differential graded Lie algebras. We begin with a brief recollection of differential graded Lie algebras and their Maurer–Cartan elements.

Definition 22 (Differential graded Lie algebra)**.** A *differential graded Lie algebra* is a pair (L,d), where L is a graded Lie algebra, that is a graded vector space equipped with a degree zero operation $L\otimes L\to L$, $a\otimes b\mapsto[a,b]$, called the *Lie bracket*, which satisfies the graded skew-symmetry property $[a,b]=-(-1)^{|a||b|}[b,a]$ for homogeneous elements $a,b\in L$ and the graded Jacobi identity

$$[a,[b,c]]=[[a,b],c]+(-1)^{|a||b|}[b,[a,c]]$$

for homogeneous elements $a,b,c\in L$, and $d\colon L\to L$ is a degree -1 map satisfying the condition $d^2=0$ as well as the graded derivation property

$$d[a,b]=[d(a),b]+(-1)^{|a|}[a,d(b)].$$

From the homotopical algebra point of view, one of the protagonist in the homotopy theory for differential graded Lie algebras is the set of Maurer–Cartan elements of such algebra.

Definition 23. Let (L,d) be a differential graded Lie algebra over a field $\Bbbk$ of characteristic different from two. The condition

$$d(x)+\frac{1}{2}[x,x]=0$$

is called the *Maurer–Cartan equation* of L, and any solution $x\in L_{-1}$ is called a *Maurer–Cartan element*.

Let L be a graded Lie algebra. Note that for an element $x\in L$ of degree -1 and any $a\in L$, the Jacobi identity implies

$$[x,[x,a]]=[[x,x],a]+(-1)^{|x||x|}[x,[x,a]]=[[x,x],a]-[x,[x,a]],$$

and therefore

$$[x,[x,a]]=\frac{1}{2}[[x,x],a].$$

Thus, if we suppose that x satisfies the condition $[x,x]=0$ (note that for elements of odd degree the graded antisymmetry property is actually symmetry, so this condition is non-empty), the degree -1 endomorphism $[x,-]$ squares to zero; the Jacobi identity implies that it also satisfies the derivation property. Thus, $(L,[x,-])$ is a differential graded Lie algebra.

3.2 Tangent complexes and deformations. Our Lie algebras of interest arise as Lie algebras of derivations of models of associative algebras. Suppose that A is an associative algebra, and that (B,d) is a model of A. The graded vector space $\mathrm{Der}(B)$ of all derivations of the graded associative algebra B has a natural structure of a graded Lie algebra: the graded bracket

$$[D_1,D_2] = D_1 \circ D_2 - (-1)^{|D_1||D_2|} D_2 \circ D_1$$

of two derivations is again a derivation. Clearly, $d \in \mathrm{Der}(B)$ and

$$[d,d] = d \circ d - (-1)^{|d||d|} d \circ d = 2d^2 = 0;$$

as we saw above, the map $[d,-]\colon \mathrm{Der}(B) \to \mathrm{Der}(B)$ makes $\mathrm{Der}(B)$ a differential graded Lie algebra. We note that in the previous section, d denoted the differential of the Lie algebra, and now our context forces us to consider differential graded Lie algebras whose differentials are of the form $[d,-]$ with d being an element of the algebra, the differential of the model; we hope that this does not lead to a confusion. This algebra is called the *tangent complex* associated to the model B. Using the work [160] of Schlessinger and Stasheff as an inspiration, Hinich showed in [91] that the homotopy type of this Lie algebra does not depend on the choice of a model.
In our context, the Maurer–Cartan heuristics of deformation theory acquires very precise meaning. Since the differential of the tangent complex of a model is of the form $[d,-]$, the Maurer–Cartan equation in that differential graded Lie algebra is equivalent to the equation $[d+x,d+x] = 0$, which in turn is equivalent to the square-zero condition for the derivation $d+x$. Thus, the Maurer–Cartan equation for the differential graded Lie algebra $(\mathrm{Der}(B),[d,-])$ describes perturbations of the differential of B.

3.3 Perturbing solutions to the Maurer–Cartan equation. In this section, we shall prove what is essentially the Diamond Lemma in disguise. It claims that having the Maurer–Cartan equation satisfied on generators of homological degree at most two of a model is sufficient to ensure existence of a perturbation. As above, we consider a rewriting system $\Sigma = (X,W,f)$.

Theorem 3.1. *Suppose that $(T(V),d)$ is a word-homogeneous acyclic extension of finite type of the Shafarevich complex $\mathsf{Sha}(W,\Bbbk\langle X\rangle)$. The rewriting system Σ is convergent if and only if there exists an degree -1 element $F \in \mathrm{Der}(T(V))$ such that*

- *for each $e_w \in V_1$, we have $F(e_w) = -f(w)$,*
- *$F|_{V_2}$ is Σ-filtered,*
- *the Maurer–Cartan equation for the element F holds when evaluated on generators of homological degree at most two.*

Proof. In view of Theorem 2.1, we have to prove that finding a perturbation that only satisfies the Maurer–Cartan equation on generators of homological degree at most two is enough to ensure that there exists a perturbation that is a Maurer–Cartan element. We shall show by induction on k that

we can find a Σ-filtered derivation F for which the Maurer–Cartan equation holds on generators of homological degree at most k, the basis of induction being $k = 2$.

Note that $[d+F, d+F]$ is the Lie bracket of derivations, thus itself a derivation, so since it vanishes on generators of homological degree at most k, it vanishes on all elements of $T(V)$ of homological degree at most k. Let us consider a word-homogeneous element $c \in V_{k+1}$. We note that the element

$$\nabla(c) = \left(\partial(F) + \frac{1}{2}[F,F]\right)(c) = d(F(c)) + F(d(c)) + F(F(c)) \in T(V)$$

is in the kernel of $d+F$. Indeed, we have

$$d(\nabla(c)) = d(d(F(c)) + F(d(c)) + F(F(c))) = (dF)((d+F)(c)),$$

and since the element $(d+F)(c)$ is of homological degree at most k, the Maurer–Cartan equation for F holds when evaluated on that element. This means that

$$\begin{aligned} d(\nabla(c)) &= -(Fd + F^2)((d+F)(c)) \\ &= -F(d(F(c))) - F(F(d(c))) - F(F(F(c))) = -F(\nabla(c)), \end{aligned}$$

as required. The element $\nabla(c)$ is of homological degree $k+1-2 = k-1$, and we know that the Maurer–Cartan equation means that the derivation $d+F$ squares to zero on elements of degree at most k. Thus, it is not unreasonable to ask whether $\nabla(c)$ is in the image of $d+F$. Let us show that it is indeed the case. We write $\nabla(c) = x+y$, where x is the word-homogeneous component of maximal grade u. Since the differential d is word-homogeneous, and the derivation F is Σ-filtered on elements of degree at most k, it follows that

$$\begin{cases} d(x) = 0, \\ d(y) + F(x+y) = 0. \end{cases}$$

Since $|x| = |\nabla(c)| = k-1 \geq 1$ and the model $(T(V), d)$ is acyclic in positive degrees, it follows that we can write $x = da$ for some $a \in T(V)$. The element

$$y - F(a) = \nabla(c) - (d+F)(a)$$

represents the same homology class for $d+F$, but its component of grade u has been replaced by a combination of elements of grades strictly less than u with respect to the ordering $<_\Sigma$. By induction with respect to the well-ordering $<_\Sigma$, we see that $\nabla(c)$ is a boundary of $d+F$, so we can write $\nabla(c) = (d+F)(b)$. Let us now consider a new derivation F' which coincides with F on elements of homological degree at most k, but satisfies

$$F'(c) = F(c) - b.$$

We note that since ∇ is Σ-filtered, each element a above (and hence the resulting element b) is a linear combination of elements of grades strictly less than the grade of c with respect to the ordering $<_\Sigma$, so F' is still Σ-filtered. Moreover, since the elements $d(c)$ and $F'(c)$ are of homological degree k, we have

$$\begin{aligned} d(F'(c)) + F'(d(c)) + F'(F'(c)) \\ = d(F(c)-b) + F(d(c)) + F(F(c)-b) = \nabla(c) - d(b) - F(b) = 0. \end{aligned}$$

Making such adjustments for all elements of homological degree $k+1$, we obtain a perturbation that satisfies the Maurer–Cartan equation up to degree $k+1$, which allows us to proceed by induction. Since the correction does not change values of F on generators of low homological degrees, we may consider the limiting value of F; it is a *bona fide* Maurer–Cartan element extending the original perturbation of the differential. □

3.4 Maurer–Cartan equation and the Diamond Lemma. In this section, we are finally rewarded for going through all the previous arguments involving models and the Maurer–Cartan equation; the reward is a general result stating that each word-homogeneous acyclic extension of $\mathsf{Sha}(W, \Bbbk\langle X\rangle)$ leads to its own Diamond Lemma criterion for convergence of Σ. At the core of such result is the following definition of an S-polynomial associated to a generator of homological degree two of such model.

Definition 24 (Obstruction)**.** Let $(T(V), d)$ be a word-homogeneous acyclic extension of the Shafarevich complex $\mathsf{Sha}(W, \Bbbk\langle X\rangle)$. For each element $c \in V_2$, the element $d(c)$ is of degree 1, and therefore is of the form

$$d(c) = \sum_i a_i e_{w_i} b_i \in T(V_0) \otimes V_1 \otimes T(V_0).$$

We define the *obstruction* $S_c \in \Bbbk\langle X\rangle$ associated to the element c by the formula

$$S_c = \sum_i a_i f(w_i) b_i.$$

We remark that our usage of the term "obstruction" is different from that of Anick in [2]: for us, as it will become apparent below, S_c is an actual obstruction to convergence of Σ.

We are now ready to state and prove our model-specific Diamond Lemma. As above, we consider a rewriting system $\Sigma = (X, W, f)$.

Theorem 3.2. *Suppose that $(T(V), d)$ is a word-homogeneous acyclic extension of finite type of the Shafarevich complex $\mathsf{Sha}(W, \Bbbk\langle X\rangle)$. The rewriting system Σ is convergent if and only if for each word-homogeneous element $c \in V_2$, the corresponding obstruction S_c is mapped to zero by some reduction with respect to Σ.*

Proof. According to Theorem 3.1, Σ is convergent if and only if there exists a degree -1 element $F \in \mathrm{Der}(T(V))$ such that

- for each $e_w \subset V_1$, we have $(d+F)(e_w) = w - f(w)$,
- $F|_{V_2}$ is Σ-filtered,
- the Maurer–Cartan condition $[d+F, d+F] = 0$ for the element F holds when evaluated on generators of homological degree at most two.

Since the Maurer–Cartan equation evaluated on an element of homological degree k is an element of homological degree $k-2$, we may replace "at most two" by "exactly two" without changing the statement.

Suppose $c \in V_2$ is word-homogeneous of grade u. We note that the formula

$$S_c = \sum_i a_i f(w_i) b_i$$

for the obstruction S_c means that S_c is a combination of terms of grade less than u, so existence of its reduction equal to zero is clearly equivalent to existence of a representation of S_c as a two-sided linear combination

$$S_c = \sum_i a_i'(w_i' - f(w_i'))b_i',$$

where all the terms in that sum are of grade less than u. Indeed, a sequence of basic reductions produces such a combination, and *vice versa*, existence each such combination immediately suggests a sequence of basic reductions.

Recall that the Maurer–Cartan equation for F is exactly the square-zero condition for $d+F$. Let us evaluate it on an element $c \in V_2$ of grade u. We obtain

$$d(d(c)) + F(d(c)) + d(F(c)) + F(F(c)) = 0.$$

We already remarked that the element $d(c) \in T(V)$ is of homological degree one, and therefore can be written as

$$d(c) = \sum_i a_i e_{w_i} b_i \in T(V_0) \otimes V_1 \otimes T(V_0).$$

The first term in the square-zero condition vanishes, while the second is

$$F(d(c)) = F\left(\sum_i a_i e_{w_i} b_i\right) = \sum_i a_i F(e_{w_i}) b_i = -\sum_i a_i f(w_i) b_i = -S_c.$$

The remaining terms in the square-zero condition are

$$d(F(c)) + F(F(c)) = (d+F)(F(c)).$$

Since $F|_{V_2}$ is Σ-filtered, the element $F(c)$ is of the form $\sum_i a'_i e_{w'_i} b'_i$, where all the terms in the sum are of grades less than u. Finally, we obtain

$$d(F(c)) + F(F(c)) = (d+F)\left(\sum_i a'_i e_{w'_i} b'_i\right) = \sum_i a'_i (w'_i - f(w'_i)) b'_i.$$

We conclude that the validity of the Maurer–Cartan equation on $c \in V_2$ of grade u is equivalent to the equation

$$S_c = \sum_i a'_i (w'_i - f(w'_i)) b'_i,$$

where all the terms in that sum are of grade less than u, which, according to our observation above, completes the proof. □

4 Specific cases of models and their Diamond Lemmas

In this section, we discuss two particular examples of models of monomial algebras that allow us to recover the Diamond Lemma and the Triangle Lemma for classical term rewriting.

4.1 The inclusion-exclusion model. In [52], Dotsenko and Khoroshkin constructed a combinatorial (usually non-minimal) model for a shuffle operad with monomial relations; specialising to shuffle operads generated by elements of arity one, one obtains a model for an associative algebra with monomial relations. For the benefit of the reader who does not wish to go through the operad construction, we describe that model here. As above, we denote $A_{\Sigma_0} = \Bbbk\langle X \mid W\rangle$.

Let $u \in \langle X\rangle$. Suppose that there are exactly m different divisors of the word u that are among the monomial relations W. We introduce formal symbols $D_{u,1}, \dots, D_{u,m}$ in one-to-one correspondence with those divisors, and denote by A_u the Grassmann algebra $\Lambda(D_{u,1}, \dots, D_{u,m})$. The direct sum

$$A_W := \bigoplus_{u \in \langle X\rangle} A_u$$

has an associative algebra structure defined as follows. For an element $u' \in \langle X\rangle$ which is a divisor of u, there is a natural inclusion $A_{u'} \hookrightarrow A_u$ sending each generator $D_{u',i}$ to the corresponding $D_{u,j}$, where j is the position of the i-th divisor of u' (viewed now as a divisor of u) in the list of divisors of u. The product of two elements $a_1 \in A_{u_1}$ and $a_2 \in A_{u_2}$ is the product in $A_{u_1 u_2}$ of their images with respect to the inclusions $A_{u_1}, A_{u_2} \hookrightarrow A_{u_1 u_2}$. If we set the homological degree of each $D_{u,i}$ to be equal to one, the algebra A_W is graded. Moreover, it has a differential graded algebra structure, where the differential d, when restricted to each Grassmann algebra A_u, is the unique derivation of that algebra sending all generators $D_{u,i}$ to 1.

Let us call a Grassmann monomial $D_{u,i_1} \wedge \cdots \wedge D_{u,i_k} \in A_u$ *indecomposable* if there does not exist a factorisation $u = u_1 u_2$ into a product of two non-empty monomials u_1 and u_2 for which each of

the divisors of u corresponding to S_{u,i_p} is either a divisor of u_1 or a divisor of u_2. It is easy to see that as an associative algebra, the algebra A_W is freely generated by its indecomposable elements. Moreover, according to [52, Th. 2.2], the differential graded algebra (A_W, d) is a model of the algebra A_{Σ_0}. We call this model the *inclusion-exclusion model*. It is immediately seen to be a word-homogeneous acyclic extension of finite type of the Shafarevich complex $\mathsf{Sha}(W, \Bbbk\langle X\rangle)$. Moreover, all homological degree two word-homogeneous generators c of that model of grade $u \in \langle X\rangle$ are of two possible forms:

- *inclusion*: $c = S_{u,i} \wedge S_{u,j}$ where the i-th divisor of u is its proper divisor u' and the j-th divisor of u is equal to u,
- *overlap*: $c = S_{u,i} \wedge S_{u,j}$ where the i-th divisor of u is a left divisor u' of u, the j-th divisor of u is a right divisor u'' of u, and these two subwords have a non-trivial common divisor.

For each generator of the first type, we have $u = au'b$ for some $a, b \in \langle X\rangle$, therefore such generators are in one-to-one correspondence with the inclusion ambiguities of [21]. We note that for such generator, we have

$$d(c) = e_u - ae_{u'}b,$$

and the obstruction associated to this generator is the basic reduction associated to the triple (a, u', b) applied to the relation $u - f(u)$. Similarly, for each generator of the second type, we have $u = u'a = bu''$ for some $a, b \in \langle X\rangle$, therefore such generators are in one-to-one correspondence with the overlap ambiguities of *op. cit.*. Moreover, for such generator, we have

$$d(c) = be_{u''} - e_{u'}a,$$

and the obstruction associated to this generators is the element usually called the S-polynomial corresponding to the overlap of u' and u'' [31]. This proves the following result.

Proposition 4.1 (Bergman's Diamond Lemma [21, Th. 1.2])**.** *The rewriting system Σ is convergent if and only if each basic reduction of each relation from R and each S-polynomial between two relations from R is mapped to zero by some reduction with respect to Σ.*

As an immediate consequence, we see that the rewriting system

$$xyz \mapsto x^3 + y^3 + z^3$$

on $\Bbbk\langle x, y, z\rangle$ discussed in Section 2.2.1 is convergent, since we already established its termination, and the word xyz does not have self-overlaps.

4.2 The minimal model. In [171], we constructed, for each monomial algebra $A_{\Sigma_0} = \Bbbk\langle X|W\rangle$, the minimal model which we shall now recall and use. In this section, the rewriting system Σ is assumed minimal: W contains no words of length one and no words from W divide one another.

We begin with recalling the definition of (right) *Anick chains*, a notion that was to some extent present implicitly in work of Backelin [9], then used by Green, Happel and Zacharia [73], and became known to a wider community from the work of Anick [2].

Definition 25 (Anick chain)**.** We say that an *Anick* 0*-chain* is an element $x \in X$, and define the *tail* of such element to be equal to it. For $n > 0$, we say that an *Anick n-chain* is a monomial $c \in \langle X \rangle$ such that

(1) we can write $c = c't$ where c' is an Anick $(n-1)$-chain,
(2) if t' is the tail of c', then $t't$ has a right divisor which is a relation from W,
(3) no proper left divisor of c satisfies the first two conditions.

The *tail* of c is the element $t_c := t$.

In particular, the Anick 1-chains are the defining relations W, with the tail of each monomial relation given by the monomial obtained by deleting its first letter, and the Anick 2-chains are precisely the monomials $c \in \langle X \rangle$ which can be written as $c = u'a = bu''$ for two relations $u', u'' \in W$, and which do not have other divisors from W.
It is shown in [2] that a monomial c admits at most one structure of a chain: if c is an n-chain with tail t_c, then both n and t_c are uniquely determined. We define the homological degree of an Anick n-chain c to be equal to n, and consider the graded associative algebra B freely generated by Anick chains. This algebra has a unique derivation d whose value $d(c)$ on an Anick n-chain c is the sum over all ways to represent c, viewed as a monomial, as a concatenation $c_1 \cdots c_k$ of an Anick n_1-chains c_1, ..., an Anick n_k-chain c_k with $n_1 + \cdots + n_k = n - 1$, of the terms

$$-(-1)^{\binom{n+1}{2}+n_1} c_1 \otimes \cdots \otimes c_k.$$

The main theorem of [171] states that for a minimal rewriting system Σ, the quasi-free algebra (B, d) is the minimal model of A_{Σ_0}. It is a word-homogeneous acyclic extension of finite type of the Shafarevich complex $\mathsf{Sha}(W, \Bbbk\langle X \rangle)$.
The description of Anick 2-chains mentioned above indicates that they are among the overlaps from the previous section; in fact, they are *minimal overlaps* that have only two divisors from W. For the differential of the model, we have $d(c) = be_{u''} - e_{u'}a$, as before. This proves the following result.

Proposition 4.2 (Triangle Lemma [31, Sec. 2.4.3])**.** *A minimal rewriting system* Σ *is convergent if and only if each S- polynomial corresponding to a minimal overlap is mapped to zero by some reduction with respect to* Σ*.*

5 Analogues and generalisations

In words of Kontsevich and Soibelman [112], "the deformation theory of associative algebras is a guide for developing the deformation theory of many algebraic structures; conversely, all the

concepts of what should be the "deformation theory of everything" must be tested in the case of associative algebras". We believe that our work constitutes a successful test of putting the Diamond Lemma in the homotopical context, and it is natural to conclude this chapter with a discussion of analogues and possible generalisations of our results.

5.1 Multiplicative algebraic structures. We begin with outlining a context in which an immediate generalisation of our approach is available.

Definition 26 (Multiplicative algebraic structure)**.** We say that an algebraic structure P is *multiplicative* if it is described by a colored operad obtained as a $\Bbbk$-linear span of a set-theoretic operad. In classical terms, we require that the free $\Bbbk$-linear P-algebra generated by a vector space V is equal to the $\Bbbk$-linear span of the free P-algebra in the category of sets generated by a basis of V.

The key consequence of this definition is that free $\Bbbk$-linear P-algebras are naturally equipped with monomial bases, so one can talk about rewriting systems and orderings of monomials, and most of our results apply *mutatis mutandis* if one can talk about models of algebras in the same way as above. Work of Hinich [91] mentioned earlier allows one to verify that condition easily enough.
Some remarks are in order. First, talking about colored operads means that this formalism includes, for instance, quotients of path algebras, since the latter are obtained as $\Bbbk$-linear spans of small categories and are not just algebras in a very classical sense. Second, there are situations where the deviation from the set-theoretic property is "moderate", i.e. the value of any structure operation on basis elements is a basis element up to a non-zero constant; it is easy to extend our work to such algebras. Finally, it is also possible to include algebras that are complete in a suitable sense; we already indicated in the introduction that complete commutative associative algebras were one of the central examples at the moment of inception of both topics discussed in this chapter, through the theory of standard bases of Hironaka [93, Sec. III.1] and the deformation theory of analytic spaces of Palamodov [145].

Commutative associative algebras. The free commutative associative algebra coincides with the linear span of the free commutative monoid, and the homotopy theory for commutative associative algebras suggests that our results work over a field of characteristic zero. An analogue of the inclusion-exclusion model exists, and generators of homological degree at most two of the minimal model of a given monomial algebra can be determined directly (for further information on models of monomial algebras, we refer the reader to [18]). If the partial order $<_\Sigma$ comes from a total multiplicative order of monomials, the result corresponding to the inclusion-exclusion model is the Buchberger criterion [34], and the result corresponding to the minimal model is known as the Chain Criterion [33, 110].

Non-associative algebras. If one considers free non-associative algebras, better known as magmatic algebras for one binary operation and as absolutely free algebras in the more general case, the free algebra is the linear span of the free set- theoretic magma (consisting of appropriate decorated

planar trees), and models are surprisingly manageable. In particular, the inclusion- exclusion model of a given monomial algebra is readily available, and the minimal model for a monomial algebra whose set of generators and relations are already chosen minimal is simply the non-associative Shafarevich complex, since there are no non-associative overlaps. As a consequence, for a general rewriting system convergence is equivalent to the fact that all basic reductions between the relations R are mapped to zero by some reduction with respect to Σ, and, in particular, a rewriting system where pairwise reductions are impossible is convergent. This result goes back to [180], see also [29, Appendix 3] and a more recent paper [64].

Shuffle algebras. Shuffle algebras [53, 158], known also as permutads [127], are associative algebras for the monoidal category of non-symmetric sequences with respect to the monoidal structure given by Cauchy tensor product. The corresponding definition is also available for sets; elements of the free monoid are decorated permutations and their suitable generalisations. All results of this chapter, including a generalisation of the inclusion-exclusion model and our minimal model, can be adapted; the corresponding results are the Diamond Lemma for shuffle algebras [31, Th. 4.5.1.4] and the Triangle Lemma for shuffle algebras [31, Prop. 4.5.3.2].

Non-symmetric operads and shuffle operads. When one considers operads, whether non-symmetric or shuffle ones, each free operad is the linear span of the corresponding free set operad consisting of appropriate decorated planar trees. It is important to note that the algebraic structure on the space of decorated trees in this case is much richer than the one considered in the case of magmatic algebras above, so the corresponding Diamond Lemmas (and especially their applications) are sufficiently non-trivial. Models are once again available under relatively mild assumptions [138]. In particular, the inclusion-exclusion model can be defined for every shuffle operad with monomial relations [52, Sec. 2]. This way one obtains the Diamond Lemma for non-symmetric operads and for shuffle operads [31].

Finding an explicit description of the minimal model of the given operad with monomial relations is an open problem. Partial results in the spirit of the Triangle Lemma are available for operads [31], but even in homological degree two, the computation of the spaces of generators of minimal models of monomial operads still has to be completed in full generality. One calculation hinting at the noticeable complexity of this question was made by Sköldberg several years ago (unpublished), and we shall now describe the result of that calculation; another discussion of that phenomenon can be found in the recent preprint of Iyudu and Vlassopoulos [100].

The minimal model of a monomial operad is homogeneous: one can separate generators according to their underlying tree monomials. In the case of associative algebras, minimal models of monomial algebras satisfy the homological purity condition: for each monomial, the corresponding component of the space of generators is concentrated in one homological degree. Somewhat surprisingly, it turns out that in the case of operads, this homological purity condition does not hold for some underlying tree monomials. The simplest possible example is obtained as follows. Consider the free non-symmetric operad generated by one binary operation, and consider the set of

monomial relations W consisting of all tree monomials of arity four in that operad. Then for the tree monomial

For example, for the rewriting system $x^3 \mapsto xyz - y^3 - z^3$ arising in the example discussed in Section 2.2.1 for a total multiplicative ordering, the monomial x^3 has two self-overlaps, $x^4 = x^3 \cdot x = x \cdot x^3$ and $x^5 = x^3 \cdot x^2 = x^2 \cdot x^3$. Only the first of them is minimal. This means that there is no need to consider the S-polynomial corresponding to the overlap x^5 when checking the Diamond Lemma criterion. That said, in this case already the S-polynomial corresponding to the overlap x^4 cannot be reduced to zero, so the rewriting system is not convergent, as we indicated when discussing that example.

of arity seven, as well as the monomials obtained from it by reflections at internal vertices, the corresponding component of the space generators of the minimal model is two-dimensional, with one generator in degree three and one generator in degree four. This example is a particular case of the remarkable construction from the very recent preprint of Qi, Xu, Zhang and Zhao [151, Sec. 5]: it is the "operadization" of the algebra $\Bbbk\langle x, y\rangle/\mathfrak{m}^3$. We feel that explicitly computing minimal models for monomial operads obtained by that construction is likely to exhibit some of the most interesting phenomena behind minimal models of general monomial operads.

Non-multiplicative algebraic structures. For an algebraic structure that is not multiplicative, our approach does not adapt immediately. Upon examining the existing literature on Gröbner–Shirshov bases, see, for example, [22–24] and references therein, we feel that the most obvious way to deal with a non-multiplicative algebraic structure is to embed it into a bigger multiplicative one. For example, Lie algebras can be embedded in their universal envelopes, and so one can work with a rewriting system within a multiplicative algebraic structure. Equivalence of the associative algebra methods applied to the universal envelopes and the Lie-algebraic methods of Shirshov was established by Bokut and Malcolmson [26]. Alternatively, a Lie algebra can be regarded as an anti-commutative magmatic algebra; the corresponding study of Gröbner–Shirshov bases was undertaken in [25]. Recent papers on Gröbner–Shirshov bases for pre-Lie algebras [28], Novikov algebras [174], and Sabinin algebras [125] follow the same logic. This raises a question of computing not only models of monomial algebras for multiplicative algebraic structures, but also models of meaningful classes of algebras obtained when extending non-multiplicative structures to multiplicative ones. We hope to address this in more detail elsewhere.

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